Socio-Cognitive Engineering · SCE 2026

Morning Care

Pepper as a tooth-brushing aid for care home residents with dementia

Group 04

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Chapter 1

Foundation

1.1 Operational Demands

1.1.1 Situated Activities

SA-05 Brushing teeth

Evidence basis	Provisional. Literature and group reasoning only. Rogers et al. (1999) describe morning care routines, including oral hygiene, of nursing-home residents with dementia and the graduated prompting staff use. Mihailidis et al. (2008) break a comparable stepwise ADL (handwashing) into steps for prompting people with dementia. Longitudinal study about recall of everyday task `https://pubmed.ncbi.nlm.nih.gov/11997490/` Analysis of logical memory recall with Alzheimer's `https://pubmed.ncbi.nlm.nih.gov/12597076/`
Goal	Maintain oral hygiene as part of the resident's normal morning routine, with as much independence as possible
Performing Actor(s)	ST-08: Resident with dementia (performs the activity) ST-09: Care home staff member (supports and prompts when needed)
Trigger	Morning routine, after waking, washing and/or breakfast
Constituent Tasks	1. Recognize that is time to brush teeth 2. Go to the bathroom/sink 3. Locate toothbrush and toothpaste 4. Pick up toothbrush 5. Open toothpaste 6. Put toothpaste on toothbrush 7. Brush all teeth 8. Rinse/spit 9. Put toothbrush and toothpaste away
Context	Resident's bathroom in a dementia care home usually during a morning/evening care routine. Care staff may be nearby, the resident's cognitive abilities and need for assistance can vary.
Frequency and duration	Twice daily approximately 2-5 minutes per episode.
Tools and artifacts	Toothbrush, toothpaste, sink, cup, mirror, towels, possible visual prompts/hints used by the care home
Existing workarounds	• Verbally reminding the resident to brush their teeth • Placing toothbrush in front of them • Putting toothpaste onto the toothbrush • Demonstrating the brushing action • Breaking the activity into verbal instructions • Physically assisting the resident
Observed breakdowns	• Forgetting to start tooth brushing • Losing track of the sequence • Performing a single step over and over again • Stopping before completing the whole activity • An unexpected medical issue happens

SA-06 Taking prescribed medication

Evidence basis

Provisional. Literature and group reasoning only.

While, Duane, Beanland & Koch (2013), "Medication management: the perspectives of people with dementia and family carers."

Longitudinal study about recall of everyday task `https://pubmed.ncbi.nlm.nih.gov/11997490/`

Analysis of logical memory recall with Alzheimer's `https://pubmed.ncbi.nlm.nih.gov/12597076/`

Goal

Take the right medication at the right time; have information about the medication and why it needs to be taken

Performing Actor(s)

ST-08: Resident with dementia (performs the activity)

ST-09: Care home employee (prepares the medication and reminds the resident)

Trigger

Morning routine, after waking, washing or breakfast: depending on the needs of the patient and the type of medication

Constituent Tasks

1. Recognize that it is time to take medication
2. Identify which medicine should be taken
3. Go to where the medicines are stored
4. Locate the medicine container
5. Open the medicine container
6. Put a designated amount of pills in the mouth
7. Locate cup of water
8. Swallow pills with the help of the water
9. Confirm that task was completed

Context

Care home, morning, staff present

Frequency and duration

Daily, or as often as the prescription says. Duration 2-5 minutes per episode.

Tools and artifacts

Glass, water, medication: pills, syringe, suppository, cream/ointment, food (when needs to be taken with food or to mask the taste)

Existing workarounds

Post-it note with instructions, pill organizer, reminder app, reminder by caretaker in person, placing the medication in a visible spot, delegate an employee whose task is to assist with taking medication.

Observed breakdowns

- The resident has forgotten what type of medication they need to take, or has forgotten that they need to take medication entirely.
- The post-it note with instructions is missed.
- Medication has run out and was not refilled on time.
- The pill organizer (which medication on which day at which time) was not refilled or was refilled incorrectly by accident.
- The care home staff member is overloaded and did not check in with the resident on time.

SA-07 Morning care round

Evidence basis

Provisional. Literature and group reasoning only.

Constructed from the care home worker's part in PS-05 and PS-06 and from HP-05. Rogers et al. (1999) describe nursing-home morning care routines for residents with dementia, and found that graduated prompting took considerably more caregiver time than usual care.

Goal

Get every resident on the round through their morning care safely and in time, giving each the support they actually need, and know afterwards what each resident needed so care can continue at handover

Performing Actor(s)

ST-09: Care home staff member

Trigger

Start of the morning shift

Constituent Tasks

1. Read the handover notes from the previous shift

2. Go from resident to resident: wake them, and assist with washing and dressing where needed
3. Prompt each resident through personal-care activities such as tooth brushing (SA-05)
4. Hand out medication and remind residents to take it (SA-06)
5. Return to residents who stalled or did not finish
6. Record the care given and hand over at the end of the shift

Context One care worker is responsible for several residents during the busy morning period, under time pressure and with frequent interruptions. Residents' needs for support vary from day to day.

Frequency and duration Every morning shift. 5 - 8 residents per care worker, 30 - 60 minutes per round.

Tools and artifacts Handover notes and care record, medication list or trolley, residents' own personal-care items

Existing workarounds
- Preparing items in advance, such as putting toothpaste on the toothbrush (SA-05)
- Verbal reminders given in passing while attending to another resident
- Delegating one employee to handle medication for the whole round (SA-06)

Observed breakdowns
- Having to return repeatedly to the same resident to prompt the next step (PS-05, PS-06)
- Being unable to attend to two residents at once, so one of them waits
- Prompting step by step takes considerable caregiver time (Rogers et al., 1999)
- Checking in on a resident too late because the care worker is overloaded (SA-06)

1.1.2 Stakeholders

ST-08 Resident with dementia

Evidence basis Provisional. Literature and group reasoning only. The stakeholder set is derived from SA-05 to SA-07. Values and priorities are the group's reasoning.

Type Direct - interacts with the System during the activity

Values and motivations
- HV-10: Dignity (high)
- HV-09: Autonomy (high)
- HV-13: Well-being (high)
- HV-12: Competence (medium)
- HV-11: Privacy (medium)

Relationship to other Stakeholders Receives day-to-day support from ST-09 care workers. ST-10 family members may visit and be involved in care decisions, including consent. Receives medical care from ST-12, and lives in a home run by ST-11.

ST-09 Care home staff member

Evidence basis Provisional. Literature and group reasoning only. The stakeholder set is derived from SA-05 to SA-07. Values and priorities are the group's reasoning.

Type Direct - receives handover requests and reviews the session record

Values and motivations
- HV-13: Well-being (high)
- HV-14: Attentive care (high)
- HV-10: Dignity (medium)
- HV-09: Autonomy (medium) - endorses it for residents, but weighs it against the time available
- HV-12: Competence (medium)
- HV-11: Privacy (medium) - for residents, and against surveillance of their own work

Relationship to other Stakeholders Supports several ST-08 residents per shift. Employed by ST-11. Keeps ST-10 informed about their relative, and coordinates with ST-12 on medication and health.

ST-10 Family members of Resident

Evidence basis Provisional. Literature and group reasoning only. The stakeholder set is derived from SA-05 to SA-07. Values and priorities are the group's reasoning.

Type	Indirect - affected by outcomes, does not interact with the System and does not receive its records
Values and motivations	• HV-13: Well-being (high) • HV-10: Dignity (high) • HV-09: Autonomy (medium) • HV-14: Attentive care (medium) • HV-11: Privacy (low) - may want more information about their relative than the resident would choose to share
Relationship to other Stakeholders	Relatives of ST-08. In contact with ST-09 about their relative's care, and may be involved in consent decisions on ST-08's behalf.

ST-11 Management or the Owners of the care home

Evidence basis	Provisional. Literature and group reasoning only. The stakeholder set is derived from SA-05 to SA-07. Values and priorities are the group's reasoning.
Type	Indirect - decides on adopting the System, does not interact with it
Values and motivations	• HV-14: Attentive care (high) - staffing and workload • HV-13: Well-being (high) - quality of care • HV-11: Privacy (medium) - data-protection obligations for residents and staff • HV-10: Dignity (medium)
Relationship to other Stakeholders	Employs ST-09. Responsible for the home's quality of care, staffing and data protection, and for the living environment of ST-08.

ST-12 Medical staff: GP, Nurse

Evidence basis	Provisional. Literature and group reasoning only. The stakeholder set is derived from SA-05 to SA-07. Values and priorities are the group's reasoning.
Type	Indirect - responsible for care plans and medication, does not interact with the System
Values and motivations	• HV-13: Well-being (high) • HV-11: Privacy (medium) - medical confidentiality • HV-09: Autonomy (medium)
Relationship to other Stakeholders	Prescribes and reviews ST-08's medication and health. Relies on ST-09 to carry out care plans day to day.

1.1.3 Problem Scenarios

PS-05 Losing track of the tooth brushing routine

Evidence basis	Provisional. Constructed from SA-05 and the ST-08/ST-09 records, which rest on literature and group reasoning only. The pattern of stalling mid-sequence and needing repeated prompts matches Rogers et al. (1999) and the COACH study (Mihailidis et al., 2008). No resident or care worker has confirmed it.
Narrative	It is morning in the dementia care home. The resident has finished their breakfast and returns to their room to perform their normal morning routine. A care worker reminds the resident that it is time to brush their teeth. The resident goes into the bathroom and stands in front of the sink. The toothbrush and toothpaste are visible next to the sink. The resident picks up the toothbrush but then pauses and appears unsure of what to do next. After some thought, they put the toothbrush back down. The care worker notices this and verbally prompts the resident to pick up the toothpaste. However, after putting the toothpaste on the toothbrush they again pause and are unsure of what to do next. The resident requires another verbal prompt from the care worker to eventually successfully finish this activity.
Stakeholders involved	• ST-08: Resident with dementia • ST-09: Care worker
Activities involved	SA-05: Brushing teeth; SA-07: Morning care round (the care worker's side)
Breakdown point	The activity breaks down when the resident is unable to determine their next action independently in the tooth brushing sequence without the help of verbal prompts from for example the care worker

Causes

Root causes:

- Difficulty remembering the correct sequence of actions
- Difficulty recognizing what an object is for

Contributing causes:

- Instructions being too complex
- Environmental distractions

Sub-problems

1. Initiation: Resident may have difficulty with recognizing that it is time to start brushing their teeth
2. Sequencing: Resident may begin the activity but lose track of which action comes next
3. Completion: Resident may stop too early with brushing their teeth

Out of scope

- Medication routines - a separate activity (SA-06) with its own Problem Scenario (PS-06) and Use Case (UC05), deferred to a later cycle so this cycle can specify one activity properly.
- Other personal-care activities such as dressing, washing and toileting - they share the initiation, sequencing and completion sub-problems, but involve more intimate exposure and physical assistance. Extending the approach to them is a later decision, if it works for tooth brushing.
- Physically assisting the resident (for example guiding their hand) - the robot cannot do this safely; it stays with the care worker, who is called in via handover.
- Dental health assessment - a clinical judgement for ST-12 or a dental professional, not something the System should infer.

PS-06 Difficulty completing the medication routine

Evidence basis

Provisional. Constructed from SA-06 and the ST-08/ST-09 records, which rest on literature (While et al., 2013) and group reasoning only. No resident or care worker has confirmed it.

Narrative

A resident with dementia needs to take prescribed medication in the morning. At the right time, a care worker approaches the resident with their medication and explains that it is time to take it. At first the resident agrees to take the medicine.

The care worker places the medication and a glass of water in front of the resident. However, the resident becomes distracted by activity elsewhere in the room and does not take the medication. After some time, the resident appears to have forgotten why the medication and water are in front of them.

The care worker returns their attention to the medication and reminds them that they need to take it. The resident is not sure and asks what the tablets are for. The care worker explains again and provides reassurance. Following some more prompts, the resident takes the medication.

Stakeholders involved

- ST-08: Resident with dementia
- ST-09: Care worker

Activities involved

SA-06: Taking prescribed medication; SA-07: Morning care round (the care worker's side)

Breakdown point

The activity breaks down when the resident does not independently continue with taking the medication after it has been presented and requires repeated prompting and reassurance from the care worker.

Causes

Root causes:

- Difficulty remembering the purpose of the medication after it has been presented
- Difficulty remembering and doing the care worker's instruction
- Difficulty independently starting the next action in the medication-taking sequence.

Contributing causes:

- Instructions being too complex
- Environmental distractions

Sub-problems

1. Initiation - Resident may have difficulty with recognizing that it is time to take their medication

2. Sequencing - Resident may begin the activity but lose track of which action comes next or what the purpose is of the action/medicine
3. Completion - The Resident only takes some of the medication but not all.
4. Communication - The resident fails to confirm that they have taken the medication, risking taking the same medication a second time.

Out of scope

- Medication prescribing and dosage decisions, changing medication - decided by the primary care practitioner.
- Determining whether a medication is appropriate - primary care practitioner.
- Diagnosing medical conditions - not handled by the robot (such as when the resident does not talk for an extended period of time).
- Physically administering medication - would require a dedicated product involving computer vision and is very error-prone and time-consuming.
- Emergency medical treatment - out of scope. Notifying a care worker when there is an immediate risk of serious harm during the activity is in scope treating or assessing the emergency is not.
- Residents who have to take critical medication such as heart medication - the care home blacklists these residents to require a care home employee to be present.
- Residents who only have to take vitamin C daily or anti-congestion meds preventively are permitted - they may get confused or suddenly not know where they can find the medication, but little harm follows.
- General monitoring of the resident - we only report a failure to complete the task, or an immediate risk of harm during it, and avoid surveilling the patient in general. This excludes fall detection outside the activity and general failure to respond to a query.

1.2 Human Values and Human Factors

1.2.1 Human Values

HV-09 Autonomy

Evidence basis Provisional. Literature and group reasoning only.

Situated meaning Autonomy means being able to participate in everyday activities and make choices for themselves, rather than having tasks automatically taken over by caregivers or technology. In daily activities such as brushing teeth or taking medication, this means allowing the person to initiate and perform the steps they are still capable of doing, while providing assistance or prompting only when needed.

Held by ST-08 (high); ST-09 (medium - endorses it for residents, but weighs it against the time available); ST-10 (medium); ST-12 (medium)

Intrinsic or instrumental It is an intrinsic value because having choice and control over one's own everyday activities is valuable in itself. It can also instrumentally support related values such as dignity and wellbeing.

Recognition criteria Served: the person is given meaningful opportunities to make choices, initiate actions and perform steps independently. assistance or prompting is provided when needed rather than automatically taking over.

Violated: choices are unnecessarily overridden, the resident is not given an opportunity to act independently, or caregivers/technology routinely take over activities the resident remains capable of performing.

Source Self-determination theory: Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist*, 55(1), 68-78.

HV-10 Dignity

Evidence basis Provisional. Literature and group reasoning only.

Situated meaning Dignity means treating people as individuals with individual preferences, abilities, and personal histories, rather than reducing them to their diagnosis or level of dependency. During everyday care activities, this means communicating respectfully, involving the

	resident in what is happening, avoiding unnecessary exposure or embarrassment, and providing assistance in a way that preserves the person's sense of self-worth.
Held by	ST-08 (high); ST-10 (high); ST-09 (medium); ST-11 (medium)
Intrinsic or instrumental	It is an intrinsic value because being treated with respect and as a person of equal worth is valuable in itself. It can also instrumentally support related values such as autonomy and well-being.
Recognition criteria	Served: the resident is addressed respectfully, included in decisions concerning them, and supported in ways that preserve their sense of identity and self-worth. Care is provided discretely and without unnecessary infantilisation, embarrassment, or exposure. Violated: the resident is spoken about or treated as though they are not present, unnecessarily infantilised, embarrassed, exposed, ignored, or treated primarily as a task or diagnosis rather than as a person.
Source	Value Sensitive Design: Friedman, B., Kahn, P. H., Jr., & Borning, A. (2008). Value Sensitive Design and Information Systems. In K. E. Himma & H. T. Tavani (Eds.), *The Handbook of Information and Computer Ethics*. Personhood in dementia care: Kitwood, T. (1997). *Dementia Reconsidered: The Person Comes First*. Open University Press.

HV-11 Privacy

Evidence basis	Provisional. Literature and group reasoning only.
Situated meaning	Privacy means that people retain appropriate control over access to their body, personal space, belongings, activities, and personal information, even when they require assistance or monitoring. In everyday care, this means that observation, reminders, sensors, or caregiver involvement should not be more intrusive than necessary and that sensitive activities such as dressing, toileting, medication use, or personal hygiene should be handled discretely. The care home employee (ST-09) should not face surveillance at their workplace besides what is necessary.
Held by	ST-08 (medium); ST-09 (medium - for residents, and against surveillance of their own work); ST-11 (medium - data-protection obligations); ST-12 (medium - medical confidentiality); ST-10 (low - may want more information than the resident would share)
Intrinsic or instrumental	It is primarily an intrinsic value because having personal boundaries and control over access to oneself is valuable in itself. It also instrumentally supports dignity, autonomy, and trust.
Recognition criteria	Served: personal information, spaces, and care activities are accessed only when necessary and by appropriate people. Residents are given private space where possible, and monitoring or assistance is limited to what is required for safe and effective care. Violated: residents are unnecessarily watched, exposed, interrupted, or monitored; personal information is accessible to people who do not need it; or technology collects or communicates more information than is necessary for the care activity.
Source	Value Sensitive Design (Friedman et al., 2008). Surveillance in residential dementia care: Niemeijer et al. (2010), *International Psychogeriatrics*, 22(7), 1129-1142, PMID 20199699.

HV-12 Competence

Evidence basis	Provisional. Literature and group reasoning only.
Situated meaning	Competence means enabling residents to experience that they are still capable of successfully carrying out meaningful parts of everyday activities despite cognitive decline. In activities such as brushing teeth, dressing, preparing for bed, or taking medication, this means adapting tasks and providing suitable prompts so that residents can use the abilities they still have instead of immediately having the activity completed for them.
Held by	ST-08 (medium); ST-09 (medium)
Intrinsic or instrumental	It is primarily an instrumental value because maintaining and supporting the resident's ability to perform everyday activities helps preserve autonomy, dignity, and well-being. Experiencing oneself as capable may also have intrinsic value for the resident.
Recognition criteria	Served: activities are matched to the resident's remaining abilities, instructions or prompts are understandable, and the resident is given opportunities to successfully complete steps themselves. Assistance increases only when the resident needs it.

	Violated: tasks are made unnecessarily difficult or confusing, the resident repeatedly experiences avoidable failure, or caregivers or technology take over tasks that the resident could still perform with appropriate support.
Source	Self-determination theory, competence need (Ryan & Deci, 2000).

HV-13 Well-being

Evidence basis	Provisional. Literature and group reasoning only.
Situated meaning	Well-being means supporting the resident's emotional and physical comfort and helping everyday life remain calm, meaningful, and pleasant. In the context of dementia care, this includes reducing avoidable stress, confusion, frustration, discomfort, and anxiety while supporting positive experiences such as familiarity, confidence, social connection, enjoyment, and a sense of security during daily activities.
Held by	ST-08 (high); ST-09 (high); ST-10 (high); ST-11 (high); ST-12 (high)
Intrinsic or instrumental	It is an intrinsic value because the resident's physical and emotional quality of life is valuable in itself. It can also be supported instrumentally by values such as autonomy, dignity, competence, privacy, and social connection.
Recognition criteria	Served: the resident is generally comfortable and calm during the activity, assistance reduces rather than increases confusion or distress, and the activity supports positive experiences such as satisfaction, familiarity, enjoyment, or social connection. Violated: an activity, care practice, or technology causes avoidable anxiety, frustration, discomfort, confusion, isolation, or distress, particularly when the negative effect persists or could reasonably have been prevented through a different form of support.
Source	Value Sensitive Design (Friedman et al., 2008).

HV-14 Attentive care

Evidence basis	Provisional. Literature and group reasoning only.
Situated meaning	Attentive care means care staff having enough time and attention to give each resident the support they actually need during the morning round, without being pulled back to the same resident repeatedly or having to rush other residents. For the care worker it is about being able to do the job well; for management it is about staffing and the quality of care the home provides.
Held by	ST-09 (high); ST-11 (high); ST-10 (medium - wants their relative to be attended to)
Intrinsic or instrumental	Instrumental. It serves the residents' Well-being (HV-13) and Dignity (HV-10), because rushed staff tend to take over activities or skip them, and it serves the care worker's own well-being at work. Because it is instrumental, it should not be traded against the values it serves: saving staff time by taking over activities the resident could do would defeat its purpose.
Recognition criteria	Served: a care worker can attend to another resident while one completes an activity; they return to the same resident less often; prompting time per resident falls without completion falling. Violated: residents are left waiting; staff take over activities to save time; the System adds work, for example through frequent low-value alerts that have to be checked.
Source	Value Sensitive Design (Friedman et al., 2008). Caregiver time in graduated prompting: Rogers et al. (1999), *Journal of the American Geriatrics Society*, 47(9), 1049-1057. Derived from SA-07 and HP-05.

1.2.2 Value Tensions

VT-07 Autonomy vs Well-being

Evidence basis	Provisional. Literature and group reasoning only.
Values in tension	HV-09 Autonomy: pulls toward allowing the resident to initiate and perform everyday activities independently. HV-13 Well-being: pulls toward intervening when independent performance may lead to distress, confusion, discomfort, or unsafe outcomes.
Domain area	safety/physical-environmental

Nature of the conflict	Giving residents more freedom to perform activities independently supports autonomy, but too little assistance may result in mistakes, confusion, frustration, or unsafe situations.
Non-negotiable floor	None identified. No regulation or hard safety standard has been found that bounds this trade-off; recorded explicitly so it reads as a genuine trade-off. Notifying a care worker on immediate risk of serious harm is a group design decision and is part of the current resolution.
Current resolution	The situated objective is to support the resident in completing everyday activities as independently as possible while preventing avoidable distress or harm. The conditional function is: IF the resident is able to continue the activity independently, THEN no assistance is provided; IF difficulty is detected, THEN increasingly supportive prompting is provided; IF a safety-critical condition occurs (an immediate risk of serious harm during the activity), THEN a care worker is notified at once, pre-empting any prompting.
Alternatives considered	Immediate assistance whenever difficulty is detected: not preferred because it unnecessarily reduces autonomy. No intervention unless explicitly requested: not preferred because a resident with dementia may not always be able to recognise or communicate when assistance is required.
Status of the tension	Provisionally resolved

VT-08 Privacy vs Well-being

Evidence basis	Provisional. Literature and group reasoning only.
Values in tension	HV-11 Privacy: pulls toward allowing the resident to maintain their privacy and personal space. HV-13 Well-being: pulls toward intervening when independent performance may lead to distress, confusion, discomfort, or unsafe outcomes.
Domain area	safety and privacy
Nature of the conflict	The System can only offer timely support if it knows how the activity is going, and the more it senses and records, the better it can judge. But every extra sensor, and every detail recorded or shared, reduces the resident's privacy during a personal-care activity in their own bathroom - and cameras in particular make support feel like being watched.
Non-negotiable floor	none
Current resolution	Monitor only the information necessary to determine whether support is needed, and avoid sharing sensitive information unless the agreed conditions for caregiver involvement are met.
Alternatives considered	Constant monitoring of all activities: not preferred because it unnecessarily reduces privacy and dignity.
Status of the tension	Provisionally resolved

VT-09 Competence vs Well-being

Evidence basis	Provisional. Literature and group reasoning only.
Values in tension	HV-12 Competence: pulls toward giving the resident opportunities to perform activities and individual steps themselves, so that they can use and maintain their remaining abilities and experience themselves as capable. HV-13 Well-being: pulls toward preventing unnecessary frustration, confusion, anxiety, or distress when an activity becomes too difficult for the resident.
Domain area	safety
Nature of the conflict	Giving the resident time and opportunity to attempt an activity independently can support competence by allowing them to use their remaining abilities and experience successful task completion. However, if the activity or individual step is beyond the resident's current ability, continued attempts may lead to repeated failure, confusion, frustration, or distress. Providing assistance earlier may protect well-being, but can also remove opportunities for the resident to demonstrate and maintain competence.

Non-negotiable floor	none
Current resolution	The situated objective is to allow the resident to use and maintain their remaining abilities while avoiding unnecessary frustration or distress. The conditional function is: IF the resident is able to continue the activity independently, THEN allow them sufficient time to complete the step themselves; IF repeated difficulty is detected, THEN provide a supportive prompt; IF the resident remains unable to continue or shows signs of distress, THEN increase the level of assistance.
Alternatives considered	Immediate assistance when the resident hesitates: not preferred because hesitation does not necessarily indicate inability and may remove opportunities to exercise competence. Allowing the resident to continue attempting the task without assistance: not preferred because repeated unsuccessful attempts may cause unnecessary frustration, confusion, or distress.
Status of the tension	Provisionally resolved

VT-10 Dignity vs Competence

Evidence basis	Provisional. Literature and group reasoning only.
Values in tension	HV-12 Competence: pulls toward prompting the resident through the activity so they can still do it themselves. HV-10 Dignity: pulls toward not being instructed through a basic personal-care activity at all, least of all by a machine, because being talked through tooth brushing can be experienced as being treated like a child.
Domain area	safety (psychological) - this tension is not about physical harm but about the resident's standing as a person
Nature of the conflict	The prompting that lets the resident complete more steps themselves is the same prompting that can feel infantilising. Kitwood (1997) names infantilisation and outpacing as detractors that erode personhood in dementia care, and an embodied robot's gaze, timing and tone can enact the same thing (Furlanis et al., 2026). The more explicit and frequent the prompts, the more competence is supported and the greater the dignity risk.
Non-negotiable floor	None identified.
Current resolution	Prompts are phrased as offers rather than instructions, address the resident by name in an adult register, and stop as soon as the resident resumes (F8). The robot is framed as a tool rather than a carer or supervisor, and its expressivity is kept low (see §1.3 TECH-10 and HFC-13).
Alternatives considered	No prompting at all, care worker only: not preferred, because it removes the competence benefit and leaves PS-05 as it is. Continuous step-by-step narration: not preferred, because it maximises the infantilisation risk and removes opportunities to act unprompted.
Status of the tension	Open. Tested by UC04-F8-CL14 (adverse Claim). Strained by TECH-10 (Pepper), whose humanoid form raises the risk.

VT-11 Autonomy vs Well-being (detection reliability)

Evidence basis	Provisional. Literature and group reasoning only.
Values in tension	HV-09 Autonomy: pulls toward prompting only when the resident is genuinely stuck, so that a false stall detection does not interrupt someone who was managing on their own. HV-13 Well-being: pulls toward detecting every stall, so that a resident who is stuck is not left confused or distressed.
Domain area	reliability
Nature of the conflict	Stall detection has to trade missed stalls against false alarms. A sensitive detector catches more real stalls but prompts residents who were only pausing; a conservative one leaves stuck residents waiting. In the COACH study, mistimed and false-alarm prompts produced frustration and confusion (Mihailidis et al., 2008), so false prompts are not merely an annoyance - they cause the harm the System is meant to prevent.

Non-negotiable floor	None identified.
Current resolution	Favour fewer false prompts: F8 waits a configured interval before any prompt, and the lowest prompt level is the least intrusive, so a false detection costs little. The false-prompt rate is verified in UC04-F8-PR8 (M-11).
Alternatives considered	Maximum sensitivity with immediate prompting: not preferred, because false prompts interrupt the resident and COACH found they cause frustration.
Status of the tension	Provisionally resolved. Open until EM-05 shows an acceptable false-prompt rate, and UC04-F8-CL13 tests whether the wait itself causes frustration.

VT-12 Autonomy vs Well-being (medication timing)

Evidence basis	Provisional. Literature and group reasoning only.
Values in tension	HV-09 Autonomy: pulls toward letting the resident take their medication when they are ready. HV-13 Well-being: pulls toward the medication being taken on time, as prescribed.
Domain area	time-criticality
Nature of the conflict	The longer the resident is left to take medication in their own time, the greater the risk it is taken late or not at all; the more firmly timing is enforced, the less the resident decides.
Non-negotiable floor	For time-critical medication, yes: PS-06 records that the care home requires a care worker to be present for residents on critical medication such as heart medication, so those residents are excluded. Ramos et al. (2026), a scoping review of 33 studies on medication management in Alzheimer's disease, report polypharmacy, potentially inappropriate medications and transitions of care as recurring safety risks, which is consistent with such a floor. It does not supply the floor itself: its regulatory discussion maps onto US Joint Commission hospital goals, not Dutch care-home rules. It also reports that technology-based supports such as automated dispensers and apps have mostly been evaluated in small usability or pilot studies, which matters for TECH-15. TODO: cite the care home's protocol or the applicable Dutch guideline. For the remaining medication, none identified.
Current resolution	Deferred with UC05 (later cycle). DS-03 sketches a resolution: the resident may take the medication when ready, with a gentle reminder after a few minutes and a care worker notified if reminders go unanswered.
Alternatives considered	Not yet considered - to be done when UC05 is specified.
Status of the tension	Open, deferred to UC05.

VT-13 Competence vs Attentive care

Evidence basis	Provisional. Literature and group reasoning only.
Values in tension	HV-12 Competence: pulls toward giving the resident time and graduated prompting, so they do as much as possible themselves. HV-14 Attentive care: pulls toward getting each resident through their morning care quickly, so the care worker can attend to the others.
Domain area	time-criticality
Nature of the conflict	Graduated prompting preserves competence, but Rogers et al. (1999) found it took considerably more caregiver time than usual care. Under morning time pressure (SA-07), care workers therefore tend to take over activities residents could still do themselves, which is quicker but costs competence.
Non-negotiable floor	None identified.
Current resolution	The System carries the waiting and the lower prompt levels, so the extra time graduated prompting takes is spent by the robot rather than the care worker, who is called only when prompting is exhausted, the resident is distressed, or there is a safety risk (F9).

Alternatives considered	Care worker prompts as now: not preferred, because it is the SA-07 breakdown this design addresses. System completes the activity quickly for the resident: not possible for tooth brushing, and would sacrifice competence anyway.
Status of the tension	Open. Tested by UC04-F9-CL18, which asks whether care-worker time actually falls (M-17), and by UC04-F9-CL20, which asks whether brushing quality falls while the care worker is away.

VT-14 Privacy vs Well-being (brushing quality)

Evidence basis	Provisional. Literature and group reasoning only. Raised by the midterm feedback, which asked how well the teeth were actually brushed.
Values in tension	HV-11 Privacy: pulls toward the System knowing only where the resident is in the sequence, and keeping nothing about how they brushed their teeth, as HP-04 asks. HV-13 Well-being: pulls toward knowing whether the teeth were actually cleaned, since oral hygiene is the Goal of SA-05 and step completion alone does not show it.
Domain area	privacy and health
Nature of the conflict	The more precisely the System measures how well a resident brushes, the more closely it observes a personal-care activity. A camera or smart mirror would show coverage best, but that is the being-watched experience UC04-F7-CL16 says must not happen, and the reason TECH-13 was rejected. A position-sensing toothbrush (TECH-16) gives coverage without images, but produces a record of how each part of the mouth was brushed. Brushing time is the coarsest signal and reveals least, and it is only a proxy for plaque.
Non-negotiable floor	None identified.
Current resolution	The System uses brushing time only, within the session, to decide whether to offer once that the resident keep brushing (F8 early-stop rule, §2.3 Escalation specification), and stores none of it: F10 keeps its three items. Whether the teeth are actually clean is judged by people. During evaluation a trained assessor scores plaque (M-18); outside evaluation the home's existing oral-care and dental routines judge it (ST-12). The System does not, because dental health assessment is out of scope for it (PS-05).
Alternatives considered	Smart mirror or camera analysis of brushing (TECH-17): rejected, because it would make UC04-F7-CL16 fail by design. Position-sensing toothbrush with coverage feedback (TECH-16): deferred until EM-06 shows whether brushing time is enough (UC04-F8-CL19, UC04-F9-CL20). Coverage feedback would also invite more specific prompts about which teeth were missed, which raises the infantilisation risk tested by UC04-F8-CL14. Adding brushing time to the F10 record: not preferred, because it conflicts with HP-04's constraint and with UC04-F10-PR10. To be reconsidered only if care staff say in EM-07 that they need it and residents agree.
Status of the tension	Open. Tested by UC04-F7-CL16 (whether toothbrush sensing is experienced as being watched) and UC04-F9-CL20 (whether keeping quality data out of the System costs oral health).

1.2.3 Human Factors Concepts

HFC-09 Perceived Autonomy and Technology Acceptance

Evidence basis	Human-factors literature only. No interviews or observations have been carried out by this project. Peek et al. (2014), a systematic review of factors influencing acceptance of technology for ageing in place, identifies a framework of 27 pre-implementation factors including perceived usefulness, perceived need, and concerns about the technology. Ryan & Deci (2000) supplies the autonomy mechanism.
Concept statement	Assistive technology that preserves the resident's sense of choice and control is more likely to be accepted and continuously used than technology that automatically takes over activities or makes decisions on the resident's behalf.

Human factor addressed	Perceived autonomy; motivation to engage with and continue using assistive technology.
Proposed mechanism	When assistance is provided automatically or without giving the resident an opportunity to act, the interaction may be experienced as controlling. This can reduce the resident's sense of autonomy and willingness to engage with the support. Giving the resident a meaningful opportunity to initiate an activity, choose, or respond before assistance is increased preserves perceived control and may increase acceptance of the support.
Source	Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist*, 55(1), 68-78. doi:10.1037/0003-066X.55.1.68. Peek, S. T. M., Wouters, E. J. M., van Hoof, J., Luijkx, K. G., Boeije, H. R., & Vrijhoef, H. J. M. (2014). Factors influencing acceptance of technology for aging in place: a systematic review. *International Journal of Medical Informatics*, 83(4), 235-248. PMID 24529817. Mihailidis, A., Boger, J. N., Craig, T., & Hoey, J. (2008). The COACH prompting system to assist older adults with dementia through handwashing: an efficacy study. *BMC Geriatrics*, 8, 28. PMID 18992135. - the concrete case of a prompting system in this population.
Conditions and population of validity	The mechanism is established more generally and in older-adult populations. Transfer to people with moderate dementia must be treated cautiously because it assumes that the resident can perceive and interpret the choice being offered. The level and presentation of choice may therefore need to be adapted to cognitive ability.
Predicted Effect type	Higher perceived autonomy; greater willingness to engage with assistance; sustained use; reduced avoidance or rejection of the technology.
Associated Measures	M-15: Perceived autonomy / felt control during the interaction. Use the episode-specific items from Custers, A. F. J., Westerhof, G. J., Kuin, Y., Gerritsen, D. L., & Riksen-Walraven, J. M. (2013). Need fulfillment in the nursing home: resident and observer perspectives in relation to resident well-being. *European Journal of Ageing*, 10(3), 201-209. doi:10.1007/s10433-013-0263-y. Caution: in that study residents rated their own need fulfilment *higher* than trained observers did, with only weak-to-moderate agreement. A self-report measure cannot stand alone here - pair it with the observer rating. See §2.3 Open item O-17.

HFC-10 Perceived surveillance and technology acceptance

Evidence basis	Literature only. No interviews or observations have been carried out by this project. Niemeijer et al. (2010) is a systematic review of 79 papers on the ethical and practical acceptability of surveillance technology in residential care for people with dementia or intellectual disabilities. Niemeijer et al. (2015) then interviewed residents themselves.
Concept statement	Monitoring that is perceived as continuous, intrusive, or unnecessarily detailed can reduce a person's sense of privacy and may lead to discomfort, avoidance, or rejection of the technology.
Human factor addressed	Perceived surveillance; privacy perception; technology acceptance.
Proposed mechanism	Increasing the amount or visibility of monitoring can provide caregivers or technology with more information about whether assistance is needed. However, when the resident becomes aware of being continuously observed or believes that more personal information is being collected than necessary, the support may be experienced as surveillance. This may reduce perceived privacy and willingness to accept the system.
Source	Niemeijer, A. R., Frederiks, B. J. M., Riphagen, I. I., Legemaate, J., Eefsting, J. A., & Hertogh, C. M. P. M. (2010). Ethical and practical concerns of surveillance technologies in residential care for people with dementia or intellectual disabilities: an overview of the literature. *International Psychogeriatrics*, 22(7), 1129-1142. doi:10.1017/S1041610210000037. PMID 20199699. Niemeijer, A. R., Depla, M. F. I. A., Frederiks, B. J. M., & Hertogh, C. M. P. M. (2015). The experiences of people with dementia and intellectual disabilities with surveillance technologies in residential care. *Nursing Ethics*, 22(3), 307-320. PMID 24913545. - the

	key finding for us: surveillance technology can support autonomy, but only when embedded in a person-centred approach. Also relevant: Niemeijer et al. (2011), *Journal of Medical Ethics* 37(5), 303-310, which used concept mapping to find an inherent duality in professional views, rooted in the moral conflict between safety and freedom - i.e. our VT-07 and VT-08, found independently.
Conditions and population of validity	Must be established specifically for older adults and, where possible, people with dementia. Cognitive impairment may affect whether and how monitoring is perceived, so findings from general technology users should not automatically be transferred to this population.
Predicted Effect type	Perceived privacy; discomfort with monitoring; willingness to accept monitoring; avoidance or rejection behaviour.
Associated Measures	M-16: Perceived privacy / monitoring acceptability. Interview coding for expressions of being watched or monitored, plus care-staff judgement of what the session record exposes. Bespoke, but the concern set is drawn from Niemeijer et al. (2010, 2015) rather than invented.

HFC-11 Adaptive assistance, competence and distress

Evidence basis	Literature only. No interviews or observations have been carried out by this project. Rogers et al. (1999) is the strongest support: systematic graduated prompting of nursing-home residents with dementia produced a 26% increase in self-dressing, significantly fewer physical assists, and significantly *fewer* disruptive behaviours than usual care - so matching assistance to ability improved performance and reduced distress together, in a severely impaired cohort (mean MMSE 6.07). Caveat on direction. This concept says assistance should match ability, but it does not settle *which way* to approach that match. The dementia ADL literature largely favours errorless learning with vanishing cues - prompt early to prevent the error, then fade support (de Werd et al., 2013) - whereas §2.3's F8 waits and escalates. The REDALI-DEM trial (Voigt-Radloff et al., 2017; n = 161) found structured relearning improved ADL performance and held for six months, but that errorless learning gave no additional effect over trial-and-error learning - so neither direction is settled. See §2.3 Open item O-16.
Concept statement	Assistance should be matched to the person's current ability: providing too much assistance can remove opportunities to exercise competence, while providing too little assistance when a task exceeds the person's current capability can result in repeated failure, confusion, or distress.
Human factor addressed	Task performance; cognitive capability; perceived competence; frustration and distress.
Proposed mechanism	When an activity remains within the resident's current capabilities, allowing sufficient time to perform the activity supports successful task completion and perceived competence. When task demands exceed those capabilities, repeated unsuccessful attempts can increase cognitive difficulty and emotional distress. Increasing assistance only when difficulty persists can therefore preserve opportunities for competence while limiting unnecessary frustration.
Source	Rogers, J. C., Holm, M. B., Burgio, L. D., Granieri, E., Hsu, C., Hardin, J. M., & McDowell, B. J. (1999). Improving morning care routines of nursing home residents with dementia. *Journal of the American Geriatrics Society*, 47(9), 1049-1057. PMID 10484245. de Werd, M. M. E., Boelen, D., Olde Rikkert, M. G. M., & Kessels, R. P. C. (2013). Errorless learning of everyday tasks in people with dementia. *Clinical Interventions in Aging*, 8, 1177-1190. doi:10.2147/CIA.S46809. PMID 24049443. Voigt-Radloff, S., de Werd, M. M. E., Leonhart, R., Boelen, D. H. E., Olde Rikkert, M. G. M., Fliessbach, K., Klöppel, S., Heimbach, B., Fellgiebel, A., Dodel, R., Eschweiler, G. W., Hausner, L., Kessels, R. P. C., & Hüll, M. (2017). Structured relearning of activities of daily living in dementia: the randomized controlled REDALI-DEM trial on errorless learning. *Alzheimer's Research & Therapy*, 9, 22. doi:10.1186/s13195-017-0247-9. - participants lived at home rather than in care homes, so this transfers to our population only cautiously.

	Mihailidis, A., Boger, J. N., Craig, T., & Hoey, J. (2008). The COACH prompting system to assist older adults with dementia through handwashing: an efficacy study. *BMC Geriatrics*, 8, 28. PMID 18992135. - mistimed and false-alarm prompts here produced observed frustration and confusion, which is the failure mode this concept predicts.
Conditions and population of validity	The appropriate level and timing of assistance is likely to vary with the resident's cognitive abilities, the familiarity and complexity of the activity, and their momentary condition. Evidence should therefore be selected specifically for people with dementia or closely comparable older-adult populations rather than assumed from general users.
Predicted Effect type	Successful independent task steps; perceived competence; frequency of assistance; frustration, confusion, or distress during the activity.
Associated Measures	M-12: Independent task-step completion - the scoring used in Mihailidis et al. (2008), directly reusable. M-13: Prompt level required - bespoke ordinal coding, levels matching SA-05's recorded workarounds. M-14: Observed frustration/distress - the Resistiveness to Care Scale (Mahoney et al., 1999), 13 observed behaviours, α .82-.87, built for ADL care episodes in advanced dementia, with later reliability and validity support (Galik et al., 2017). Validated, not invented. Optionally paired with the Observed Emotion Rating Scale for affect (psychometrics: Lawton, Van Haitsma & Klapper, 1996, *J Gerontol B*, 51(1), P3-P14).

HFC-12 Personhood and malignant social psychology

Evidence basis	Literature only. No interviews or observations have been carried out by this project. Kitwood (1997) is the founding text of person-centred dementia care.
Concept statement	How people with dementia are treated in everyday interactions can erode or sustain their sense of being a person. Infantilising them, outpacing them (going faster than they can follow), or treating them as a task rather than a person erodes personhood, even when the help itself is well-meant and effective.
Human factor addressed	Personhood; experienced dignity; stigma.
Proposed mechanism	Each interaction carries an implicit message about the person's standing. Being addressed like a child, rushed, or talked through a basic activity signals that the person is no longer competent, which undermines their sense of self and can show up as withdrawal, distress or resistance. The effect comes from how help is delivered - tone, pacing, wording - not only from whether help is given, so an assistive System can enact it through its prompting style.
Source	Kitwood, T. (1997). *Dementia Reconsidered: The Person Comes First*. Buckingham: Open University Press. The list of detractors (including infantilisation and outpacing) is reported consistently in secondary sources; check page numbers against the book before quoting.
Conditions and population of validity	Developed for people with dementia in care settings, so it transfers directly to our population. It is a conceptual framework drawn from care practice rather than an experimentally tested mechanism.
Predicted Effect type	Experienced infantilisation; resistance or withdrawal during care; resident and staff descriptions of the System as talking down to the resident.
Associated Measures	M-14 (Resistiveness to Care Scale, for resistance during the activity); M-16 (interview coding, extended to code descriptions of being talked down to - see UC04-F8-CL14).

HFC-13 Embodiment and perceived role of assistive robots

Evidence basis	Literature only. No interviews or observations have been carried out by this project. Pemberton et al. (2026) is a qualitative interview study (n = 13); Furlanis et al. (2026) is explicitly conceptual. Both are summarised in §2.3 Open item O-19.
Concept statement	The physical form and behaviour of an assistive robot shape the role older adults assign to it. A humanoid, expressive robot tends to be read as a supervisor ("Warden"), while a more mechanical, tool-like robot is read as an aid; and a robot's gaze, timing and tone can come across as condescending even when meant to be gentle.
Human factor addressed	Perceived role of the robot; perceived surveillance; experienced dignity; technology acceptance.

Proposed mechanism	People interpret robots as social actors through posture, timing, tone and spatial behaviour. A human-like body invites human-like role attributions, and oversight by a human-like figure without a relationship is experienced as being watched. Exaggerated slowness or gentleness toward older adults is read as condescension. A tool-like framing and restrained expressivity preserve emotional distance, which Pemberton et al. (2026) report older adults prefer in private contexts. The original source for the private-context point is mixed: for a personal-care task, Prakash & Rogers (2015) found a robotic face rated more likeable than a human face, but not more trusted or more useful, and their interviewees were split between wanting a human-looking carer and an impersonal one. Older adults also call robots "patronizing" when they are toy-like or imitate human companionship (Coghlan et al., 2021).
Source	Pemberton, L., Mehandjiev, N., & Carter, M. (2026). The Tool, the Pet, the Carer & the Warden: A Conceptual Framework of How Older Adults Perceive Socially Assistive Robots. *Information Systems Frontiers*, 28, 1039-1059. doi:10.1007/s10796-026-10699-2. Furlanis, T., Brščić, D., & Kanda, T. (2026). Robotic Bias and Its Dimensions. *Proceedings of HRI '26*, 562-571. doi:10.1145/3757279.3785613. Lee, H. R., Tan, H., & Šabanović, S. (2016). That robot is not for me: Addressing stereotypes of aging in assistive robot design. *RO-MAN '16*, 312-317. doi:10.1109/ROMAN.2016.7745148. Read: interviews with 7 independently living older adults (57 to 72), who rejected robots designed around a disability stereotype of ageing. Prakash, A., & Rogers, W. A. (2015). Why some humanoid faces are perceived more positively than others: effects of human-likeness and task. *International Journal of Social Robotics*, 7(2), 309-331. doi:10.1007/s12369-014-0269-4. Coghlan, S., Waycott, J., Lazar, A., & Barbosa Neves, B. (2021). Dignity, autonomy, and style of company: dimensions older adults consider for robot companions. *Proceedings of the ACM on Human-Computer Interaction*, 5(CSCW1). doi:10.1145/3449178.
Conditions and population of validity	Pemberton et al. studied cognitively intact older adults ageing in place, in a VR home setup - not care-home residents with dementia. Whether residents with moderate dementia attribute roles to a robot in the same way is uncertain and must not be assumed. Furlanis et al. provide a framework, not measurements.
Predicted Effect type	Perceived surveillance; experienced infantilisation; acceptance or rejection of the robot, depending on form and behaviour.
Associated Measures	M-16 (perceived privacy and monitoring - including descriptions of the robot as watching or supervising); M-14 (resistance during the activity).

HFC-14 Alert fatigue

Evidence basis	Literature only. No interviews or observations have been carried out by this project. Added to close §2.3 Open item O-31. Cvach (2012) has been read. It is an integrative review of 72 articles on physiologic monitor alarms published from 2000 to 2011, appraised with the Johns Hopkins Nursing Evidence-Based Practice model. It reports that 80% to 99% of monitor alarms are false or clinically insignificant, and that a single patient can generate up to 700 alarms a day. Citing Bliss and colleagues, it describes nurses matching their response rate to how reliable they believe an alarm source to be: a source perceived as 10% reliable gets a response about 10% of the time. Response also worsens as workload and task complexity rise. Its recommendations include sending alarms only at actionable limits, giving them high, medium and low priorities, and routing them through notification systems that can delay and escalate. Cvach also notes that the evidence is mostly observational and single-site, with few randomised trials. Sendelbach & Funk (2013): only the abstract has been read, because the full text could not be obtained. It defines alarm fatigue as sensory overload from excessive alarms, which leads to desensitisation and missed alarms, and reports that 72% to 99% of clinical alarms are false. Cite it for the definition only.
Concept statement	When staff receive many alerts, and many of them need no action, they become desensitised: they respond more slowly, ignore alerts or switch them off, and so may miss the ones that matter.

Human factor addressed	Attention and vigilance of care staff; workload; trust in an alerting system.
Proposed mechanism	Each alert interrupts the care worker and has to be judged. If most alerts turn out to need nothing, responding stops paying off, and the care worker learns to treat alerts from that source as low value. Response slows for all of them, including the rare urgent one. Fewer alerts, clear priorities, and alerts sent only when the recipient can and should act keep each alert informative. Cvach (2012) reports the learning step as nurses matching their response rate to the perceived reliability of the alert source.
Source	Cvach, M. (2012). Monitor alarm fatigue: an integrative review. *Biomedical Instrumentation & Technology*, 46(4), 268-277. doi:10.2345/0899-8205-46.4.268. PMID 22839984. Sendelbach, S., & Funk, M. (2013). Alarm fatigue: a patient safety concern. *AACN Advanced Critical Care*, 24(4), 378-386. doi:10.1097/NCI.0b013e3182a903f9. PMID 24153215.
Conditions and population of validity	Established for nurses in hospital and critical-care monitoring, where alarm volumes are far higher than one System per resident would produce (up to 700 a patient a day; Cvach, 2012). Transfer to care-home staff on a morning round is plausible but unconfirmed. HP-05's need for calls to be rare and meaningful is the same concern, stated by the group without data.
Predicted Effect type	Slower acknowledgement of handover requests; care workers ignoring or muting the notification channel; time saved on prompting being spent on notifications instead.
Associated Measures	M-17 (care-worker load, extended to count F9 notifications and those judged unnecessary); M-11 (handover timing and re-escalation in EM-05).

1.2.4 Measures

M-11 Step-tracking and stall-detection performance

Evidence basis	The protocol follows the system-performance reporting of the COACH prompting system (Mihailidis et al., 2008), the closest published precedent for prompting people with dementia through a stepwise ADL.
Construct measured	Whether the System correctly identifies which SA-05 step the resident is on, how quickly it detects a stall, how often it prompts when no stall has occurred, and how quickly handover requests arrive.
Type	Quantitative
Instrument or procedure	Staged-event protocol against a ground-truth log: scripted steps, stalls, resumptions and confounders at a mock sink, with each event timestamped by the experimenter.
Scoring or coding scheme	Proportion of steps correctly identified; stall-detection latency (s); spurious-prompt rate (prompts per session with no stall); time from triggering condition to handover request (s). Proposed pass thresholds are in UC04-F7-PR7 to UC04-F9-PR9 (§2.3), alongside the escalation timing they check. They are to be confirmed in a first EM-05 pilot with the selected item sensors (TECH-11); see §2.3 Open item O-29.
Validity and reliability	Validity rests on the accuracy of the ground-truth log, not on prior psychometric evidence.
Serves	Verification
Burden	No human subjects - staged events only. Requires a mock sink set-up and the prototype.

M-12 Independent step completion

Evidence basis	Step-level scoring as used in COACH (Mihailidis et al., 2008), which scored older adults with moderate-to-severe dementia on handwashing steps completed without human assistance. Directly reused, not adapted.
Construct measured	How many steps of the SA-05 sequence the resident completes with no prompt, per session.
Type	Quantitative
Instrument or procedure	Observer scoring of each SA-05 step during the real morning routine.

Scoring or coding scheme	One point per SA-05 step completed without any prompt or human assistance; summed per session (0-9 for SA-05's nine steps). Compared against the same resident's pre-System baseline.
Validity and reliability	Used as the primary outcome in Mihailidis et al. (2008). Inter-rater reliability for our own observers, per step (completed unprompted: yes/no): Two observers independently code at least 20% of sessions; agreement is reported as Cohen's kappa, with $\varkappa \geq .60$ treated as acceptable ("substantial" from .61, Landis & Koch, 1977). Coders are trained on pilot sessions until they reach that level.
Serves	Validation
Burden	Human subjects - residents with dementia, supported consent. An observer present during a personal-care activity, which is itself a privacy load (see HV-11, VT-08).

M-13 Prompt level required

Evidence basis	Bespoke ordinal coding. The levels correspond to SA-05's own recorded existing workarounds and to the assistance levels used in graduated prompting of nursing-home residents with dementia (Rogers et al., 1999).
Construct measured	The highest level of assistance needed to get the resident through each step.
Type	Quantitative (ordinal)
Instrument or procedure	Observer coding per SA-05 step, during the real morning routine.
Scoring or coding scheme	Per step, code the highest level reached: 0 none, 1 orienting, 2 naming the next action, 3 demonstration, 4 care-worker takeover. Report the distribution across steps and sessions.
Validity and reliability	No prior validity evidence for this exact scale. Inter-rater reliability: as for M-12, but using weighted kappa because the levels are ordinal. Two observers independently code at least 20% of sessions; agreement is reported as Cohen's kappa, with $\varkappa \geq .60$ treated as acceptable ("substantial" from .61, Landis & Koch, 1977). Coders are trained on pilot sessions until they reach that level.
Serves	Validation
Burden	Human subjects - as M-12, collected in the same observation.

M-14 Observed distress and resistance

Evidence basis	Resistiveness to Care Scale (Mahoney et al., 1999), developed specifically for ADL care episodes in advanced dementia, with later reliability and validity support (Galik et al., 2017). Optionally paired with the Observed Emotion Rating Scale for affect (Lawton, Van Haitsma & Klapper, 1996).
Construct measured	Frustration, distress and resistance shown by the resident during the activity.
Type	Quantitative (observational, coded)
Instrument or procedure	Resistiveness to Care Scale - 13 observed behaviours - rated during or from the care episode. Optionally the Observed Emotion Rating Scale alongside it.
Scoring or coding scheme	Each of the 13 behaviours is rated for duration (0 absent, 1 less than 16 s, 2 16-59 s, 3 1-2 min, 4 more than 2 min) and intensity (1 mild, 2 moderate, 3 extreme). Duration and intensity are multiplied per item and summed to a total score from 0 (no resistiveness) to 156 (Mahoney et al., 1999, as described in "Revising the Resistiveness to Care Scale", PMC5897769). Compared against the resident's pre-System baseline.
Validity and reliability	Validated, not invented: internal consistency α .82-.87 (Mahoney et al., 1999), with further reliability and validity support in cognitively impaired older adults (Galik et al., 2017). For the Observed Emotion Rating Scale, psychometrics in Lawton et al. (1996); Lawton et al. (1999) found reliability "less than desirable" when rated by certified nursing assistants rather than trained research staff.
Serves	Validation
Burden	Human subjects - residents with dementia, supported consent. If care staff rather than researchers do the rating, rater-training time has to be budgeted (Lawton et al., 1999).

M-15 Perceived autonomy and felt control

Evidence basis	Episode-specific need-fulfilment items from Custers et al. (2013), which measured Dutch nursing-home residents' fulfilment of the needs for autonomy, competence and relatedness during a specific videotaped caregiving episode, with a trained-observer counterpart. Provisional: that sample was physically frail residents (n = 36, mean age 80), not specifically residents with dementia.
Construct measured	The resident's felt sense of doing the activity themselves rather than being led through it, during a specific session.
Type	Quantitative, with an observer counterpart
Instrument or procedure	Episode-specific autonomy items from Custers et al. (2013), asked after the session, paired with an observer rating of the same episode.
Scoring or coding scheme	Self-report: the three autonomy items used for the videotaped caregiving episode in Custers et al. (2013), for example "Do you feel that you could make your own choices concerning, for example, your clothes?", adapted to tooth brushing. Each item is answered on a 5-point scale (1 no, absolutely not; 2 no; 3 partly; 4 yes; 5 yes, absolutely) in a short semi-structured interview straight after the session. Score: mean of the three items. Observer: the 7-point "support of autonomy" scale from the same study, rating how far the prompting respects the resident's perspective and supports their choices, using the study's behavioural definitions for each point. Report self-report and observer scores separately; do not combine them.
Validity and reliability	In Custers et al. (2013) the need-fulfilment scale for the videotaped episode had $\alpha = .69$, and observers agreed within one scale point on 79% of episodes for support of autonomy. Residents rated their own need fulfilment higher than observers did, with only weak-to-moderate agreement, so the self-report has a documented ceiling and must not be used alone. Our adaptation to tooth brushing is untested. Observer agreement will be checked as in M-12, using weighted kappa.
Serves	Validation
Burden	Human subjects - residents with dementia, supported consent. Self-report may be difficult for residents with more advanced dementia; the observer rating also needs a trained observer present.

M-16 Perceived privacy and monitoring acceptability

Evidence basis	The concern set is drawn from Niemeijer et al. (2010), a systematic review of surveillance technology in residential dementia care, and Niemeijer et al. (2015), which interviewed residents themselves.
Construct measured	Whether residents experience the System as watching them or talking down to them, and whether care staff judge what the session record exposes as proportionate.
Type	Mixed (qualitative coding, plus a check of record content)
Instrument or procedure	Validation component, residents: a short, verbally administered post-session interview in EM-06 (bespoke, three open questions: how it felt to have the robot there while brushing their teeth; whether it felt like being watched; whether they liked how it spoke to them). Validation component, care staff: the EM-07 interview, including their judgement of what the session record exposes. Verification component (record content): inspection of stored session records against the three specified items.
Scoring or coding scheme	Code each statement into one of: W - the System watches, checks on, or supervises the resident; R - the System records or reports on the resident; T - the System talks down to the resident or treats them like a child (for UC04-F8-CL14); N - neutral or positive about the System's presence. A statement can carry more than one code. Code each record item named by care staff as P (proportionate) or I (intrusive). Report counts per code per resident; any W, R or T code is reported with its quote. Record-content check: pass if the record holds exactly the three specified items and no step-level trace.

Validity and reliability	No prior validity evidence; the code set follows the concerns in Niemeijer et al. (2010, 2015) and Kitwood (1997). Inter-coder reliability: two coders code all interviews independently, kappa per code, $\varkappa \geq .60$ acceptable (Landis & Koch, 1977); disagreements are resolved by discussion and the resolution recorded.
Serves	Both - Verification (record content) and Validation (perceived privacy)
Burden	Human subjects for the validation component - residents with dementia (supported consent) and care staff. The record-content check needs no human subjects.

M-17 Care-worker prompting load

Evidence basis	Time-sampled observation as used in Rogers et al. (1999), which reported the proportion of a morning care session spent on physical assists.
Construct measured	How much of the care worker's time a resident's activity takes.
Type	Quantitative
Instrument or procedure	Time-sampled observation of care-worker prompting during the activity.
Scoring or coding scheme	Count and total duration of care-worker prompting episodes per session. Compared against the pre-System baseline. In the System phase, also the number of F9 notifications per care worker per round, by priority, and (from EM-07) how many the care worker judged unnecessary, so that time saved on prompting is not simply moved to handling notifications (HFC-14).
Validity and reliability	Method follows Rogers et al. (1999); our specific coding is bespoke. Inter-rater reliability: two observers time the same prompting episodes in at least 20% of sessions; agreement on episode counts as Cohen's kappa and on durations as an intraclass correlation, with $\varkappa$ or ICC $\geq .60$ treated as acceptable.
Serves	Validation
Burden	Human subjects - care staff (observed at work, which is itself a workplace-surveillance concern for ST-09) and residents.

M-18 Brushing quality

Evidence basis	Added in response to the midterm feedback, which asked how well the teeth were actually brushed. M-12 cannot answer that, because a resident can complete SA-05 task 7 and still brush badly. Part (a): two candidate instruments were read, both designed for assessing oral hygiene in older people in residential care. Proposed choice: the Plaque Score of the Mucosal-Plaque Score (MPS; Henriksen et al., 1999). Group to confirm (§2.3 Open item O-32). The Oral Health Assessment Tool (OHAT; Chalmers et al., 2005) was set aside for this purpose. It is an eight-category screening tool that carers complete to decide on dental referral, and plaque is covered only by its "oral cleanliness" category. Against a dentist's Plaque Index on 21 residents, that category agreed in 42.9% of cases (r = 0.15, not significant), and carers rated plaque much lower than the dentist did. OHAT would suit a general oral-health screen, but here it would measure the one thing M-18 needs least well. MPS adds a four-point Plaque Score (PS) to a four-point Mucosal Score (MS). Henriksen et al. describe the mucosal part as showing the sequelae of long-standing oral-care deficits and the plaque part as showing current hygiene, so PS is the part that can respond within EM-06. MS is recorded alongside as a secondary score. Part (a) limits to state wherever it is cited: MPS was developed to assess groups of residents, a ward or an institution, not individuals; its reliability samples were small (8 to 24 residents) at one Oslo institution for older people with mental disabilities; and validity against a reference standard was not tested. Part (b): brushing time from the toothbrush motion sensor that TECH-11 already includes. Bespoke. No primary data collected by this project.
Construct measured	How well the resident's teeth were cleaned: (a) plaque left after the morning routine, the outcome; (b) total time the toothbrush was in brushing motion per session, a process

proxy.

Type

Quantitative

Instrument or procedure

(a) MPS scoring by a trained assessor (a dental hygienist, or a researcher trained by one) shortly after the morning routine, on a fixed schedule in both phases of EM-06, for example twice a week per resident. Only with consent, and only if the resident is willing on the day. As in Henriksen et al. (1999), the resident sits in a chair or lies in bed; the assessor uses daylight plus a head torch or flashlight and two dental mirrors, and removes dentures to inspect the mucosa. Henriksen et al. report two to four minutes per resident. Assessors calibrate beforehand on the criteria and a pictorial manual (one to two hours in the source). The source's 25-photograph manual is in Norwegian and available from the authors on request; obtaining it, or building an equivalent with a dental hygienist, is part of O-32.

(b) Brushing time per session, logged from the toothbrush motion sensor in both EM-06 phases (the sensor can run without Pepper in the baseline phase). This log exists only for the evaluation, under research consent, kept apart from the F10 session record, and deleted after analysis. It is verified against the ground-truth log in EM-05.

Scoring or coding scheme

(a) Henriksen et al. (1999), scored on teeth and dentures:

- Plaque Score (PS): 1 no easily visible plaque; 2 small amounts of hardly visible plaque; 3 moderate amounts of plaque; 4 abundant confluent plaque. Food remnants are not counted as plaque.
- Mucosal Score (MS): 1 normal gingiva and oral mucosa; 2 mild inflammation; 3 moderate inflammation, including gingiva that bleeds easily under pressure or denture ulcers; 4 severe inflammation, including spontaneous gingival bleeding.
- MPS = MS + PS, range 2 to 8: 2 to 4 good or acceptable, 5 to 6 unacceptable, 7 to 8 poor.
- Tie rule from the source: when unsure between 1 and 2, or 2 and 3, score the lower; when unsure between 3 and 4, score 4.

PS is the primary score, compared against the same residents' baseline-phase scores. Because MPS was built for groups, the result is reported as the change across residents, and no conclusion is drawn about a single resident's oral health from it.

(b) Seconds of brushing motion per session, and the proportion of sessions that reach the resident's B-target (§2.3 Escalation specification). Compared against baseline.

Validity and reliability

(a) Henriksen et al. (1999) report reliability only. Weighted kappa for PS was 0.62 for one dentist scoring 24 residents a week apart, 0.71 between two dentists on 20 residents, and 0.80 among a dentist, two dental hygienists and a nurse on 8 residents after training. MS was weaker between the two dentists (0.47). Validity against a reference standard was not tested. Inter-rater reliability in this project as for M-12: two assessors independently score at least 20% of assessments, weighted kappa or ICC ≥ .60 acceptable (Landis & Koch, 1977). (b) Bespoke. Validity depends on the sensor telling brushing apart from other handling of the brush, which EM-05 checks against staged brushing. Brushing time says nothing about which teeth were brushed, so (b) never stands in for (a).

Serves

Both: Validation (plaque, brushing time in EM-06) and Verification (brushing-time accuracy in EM-05)

Burden

Human subjects for (a): residents with dementia, supported consent. An oral check is a physical intrusion into a personal-care area, so it is kept short and skipped when the resident is unwilling or distressed. (b) is data about how a personal-care activity was performed, which HP-04's constraints exclude from the System. It is justified only as time-limited research data under consent (VT-14).

1.2.5 Evaluation Methods

EM-05 Staged-session verification

Evidence basis

Staged-event testing against a ground-truth log follows the system-performance evaluation of the COACH prompting system (Mihailidis et al., 2008), the closest published precedent for prompting people with dementia through a stepwise ADL.

Procedure summary	Scripted steps, stalls, resumptions and confounders are performed at a mock sink against a timestamped ground-truth log. The System's step identification, stall detection, brushing time, prompting and handover requests are compared with the log, including unacknowledged handovers to check re-sending and escalation. Stored session records are inspected for their content. The first run is a pilot that sets the proposed escalation values (§2.3 Open item O-29).
Purpose	Verification
Human subjects required	No
Measures used	M-11; M-16 (record-content component only); M-18 (brushing-time component only)
Cost and feasibility	Low. Needs a mock sink set-up, the prototype and experimenter time; no ethics approval for human subjects. Can run as soon as a prototype exists.
Known limitations	Cannot establish real-world accuracy with an actual resident in an actual bathroom. Scripted stalls may not resemble real hesitation.

EM-06 Within-resident baseline comparison

Evidence basis	Design follows the single-subject alternating baseline-intervention structure used in COACH (Mihailidis et al., 2008) with older adults with moderate-to-severe dementia.
Procedure summary	Residents are observed during the real morning routine, first without the System to establish each resident's own baseline, then with it, alternating between the two phases (A-B-A-B). In the baseline phases Pepper stands at the doorway on standby, so that the phases differ only in whether the Functions run. Trained observers score the activity during each session, and the toothbrush sensor logs brushing time in both phases. A trained assessor, who is not told the phase, scores plaque shortly after the routine on a fixed schedule in both phases (M-18). After each session in both phases the resident is asked the M-15 autonomy items, and after System sessions also M-16's three short, verbally administered interview questions, only if they are willing and not distressed. The full study design is ER-02 (§3).
Purpose	Validation
Human subjects required	Yes - residents with dementia (supported consent) and the care staff working with them.
Measures used	M-12, M-13, M-14, M-15, M-16 (resident post-session interview), M-17, M-18
Cost and feasibility	High. Needs access to a care home, ethics approval, supported consent, trained observers present during a personal-care activity, and enough sessions per resident for both phases.
Known limitations	Small n and no control group. Cannot separate novelty effects from the design. The observer's presence during a personal-care activity may itself change behaviour. Post-session interview answers from residents with dementia may be brief or unreliable (see M-15's ceiling effect), so they are read alongside the observer measures, never alone.

EM-07 Care-staff interview

Evidence basis	The interview's concern set is drawn from Niemeijer et al. (2010, 2015) on the acceptability of surveillance technology in residential dementia care.
Procedure summary	Care staff are interviewed about the handover record the System produces and about whether they perceive the System as intrusive, for residents or for themselves. Interviews are coded with M-16's coding scheme, including whether staff describe the prompting as talking down to residents.
Purpose	Validation
Human subjects required	Yes - care staff.
Measures used	M-16, M-17
Cost and feasibility	Moderate. Needs informed consent and staff time during working hours; lower ethics load than EM-06 because residents are not participants.
Known limitations	Cannot establish the resident's own privacy experience - only how staff read it. Staff accounts of their own prompting load are self-reports, not observations.

1.3 Technology

TECH-10 Pepper humanoid robot (selected)

Evidence basis
Vendor documentation (SoftBank Robotics / Aldebaran); the course's own design-pattern examples use Pepper. No hands-on testing by this project yet.

Category
Social Robot

Capabilities
Humanoid robot about 1.2 m tall on a wheeled base, with a tablet on its chest, microphones, speakers, head and body cameras, a depth sensor, touch sensors, and LEDs in its eyes and shoulders. Relevant here: spoken prompts that address the resident by name (F8); the chest tablet can show a picture or short video of the next step, which gives F8 its demonstration level without the robot miming the action; speech recognition for simple replies; touch sensors as a non-verbal way to respond.

Limitations
- Size: its height and wheeled base may not fit in a care-home bathroom, and it must never block the resident's path. It will probably have to stand at the bathroom doorway or just outside, within earshot and sight of the sink. To be checked on site.
- Form: a humanoid, expressive robot is more likely to be read as a supervisor ("Warden") and to come across as condescending (HFC-13). This is the main drawback.
- Cameras: its cameras are physically present even when unused, so the resident may still feel watched.
- Speech recognition of older adults with dementia in a noisy bathroom is likely to be unreliable, so F8 must not depend on understanding what the resident says.
- It cannot detect the resident's position in the tooth-brushing sequence without its cameras - that is TECH-11's job.

Maturity and feasibility
Commercially available and supplied by the course, with a documented software toolkit. Feasible within the project for spoken prompts, tablet content and touch input. Not waterproof, so it has to be kept away from the sink.

Failure and fallback behaviour
If Pepper is unavailable (battery, crash, network), TECH-11 still tracks the activity and the System reports a degraded state to the care worker through TECH-12; the activity falls back to the existing SA-05 workarounds. This is what UC04-F7-PR7's robustness clause verifies. Pepper has to be charged between sessions.

Benefits
Serves the Goal on SA-05 by prompting the resident through stalls (F8); HV-12 Competence by letting the resident do as much as possible themselves; HV-14 Attentive care by taking over the waiting and low-level prompting from the care worker (SA-07).

Drawbacks
Strains VT-10 (Dignity vs Competence): its humanoid form raises the infantilisation risk. Mitigations: tool-like framing, low expressivity (no gaze-following during the activity, no exaggerated gentleness), adult-register prompts phrased as offers, demonstration on the tablet rather than by gesture.

Strains VT-08 (Privacy vs Well-being): cameras are present. Mitigations: cameras are not used by any Function; the camera streams are disabled in software and the lenses are visibly covered, so residents and staff can see they are not in use; no audio is stored.

If UC04-F8-CL14 or UC04-F7-CL16 hold for Pepper, TECH-14 is the fallback.

Selected or rejected
Selected - the project's robot platform. Chosen over TECH-14 despite the embodiment evidence, which is why the mitigations above are part of the design and not optional.

TECH-11 Sensors on the tooth-brushing items (selected)

Evidence basis
Group reasoning. Connected toothbrushes with built-in motion sensors are commercially available; weight and flow sensors are standard, off-the-shelf components. COACH (Mihailidis et al., 2008) tracked a comparable stepwise ADL, but with a camera (TECH-13).

Category
Sensing / AI-ICT

Capabilities
Tracks the SA-05 sequence from the items themselves rather than the person: a motion sensor in or clipped to the toothbrush (picked up, brushing motion, put down), a weight pad under the toothbrush cup and toothpaste (lifted, returned), and a flow sensor on the tap (rinsing). From these, F7 infers the current step and detects a stall when no expected change follows within the configured interval. The toothbrush sensor also gives

	the total time spent brushing, which F8 uses within the session to offer once to keep brushing (§2.3 Escalation specification). Requires nothing from the resident, and observes the activity, not the resident.
Limitations	• It cannot sense steps that do not involve a sensed item - recognising that it is time (step 1) and walking to the bathroom (step 2). Initiation can only be inferred from the time window running down with no toothbrush pick-up. • It cannot see distress or a collapse. Distress is only detected if the resident says so or touches Pepper. A possible emergency is inferred when there is no sensor change and no response to prompts for a configured time, and is then treated as safety-critical (F9) - the same rule the manual's example uses for non-response. • It cannot tell whether all teeth were brushed well, only that brushing motion occurred and for how long. Coverage would need TECH-16; plaque is measured by people (M-18). • Sensors can be moved, get wet, or be confused by a staff member handling the items.
Maturity and feasibility	Prototype level: needs to be built from off-the-shelf sensors and a microcontroller. Feasible within the project for a mock sink (EM-05); robustness in a real bathroom is unproven.
Failure and fallback behaviour	If a sensor stops reporting, the System reports a degraded state to the care worker (TECH-12) rather than prompting on bad data; the activity falls back to the existing SA-05 workarounds.
Benefits	Serves the Goal on SA-05 (knowing where the resident is in the sequence at all). Serves HV-11 Privacy: no camera, and nothing is worn.
Drawbacks	Strains VT-11 (detection reliability): tracking from item events is coarser than video, so stall detection will produce some false and some missed stalls. Accepted in order to stay within VT-08.
Selected or rejected	Selected - enables F7 without a camera.

TECH-12 Care-staff notification and session record (selected)

Evidence basis	Group reasoning; care homes commonly already have a nurse-call system and work phones for staff. HP-05 requires alerts to be rare and meaningful.
Category	AI/ICT
Capabilities	Sends a handover request to the care worker's work phone - room, activity, step reached, reason (F9) - and stores the three-item session record (F10) on a local server for the care record.
Limitations	Integration with the home's existing nurse-call system depends on the vendor and is probably not feasible in this project, so the prototype uses a separate phone app. A second channel could add to staff workload if it alerts too often, and frequent low-value alerts are what HFC-14 (alert fatigue) warns against.
Maturity and feasibility	High - standard messaging and storage.
Failure and fallback behaviour	Alerting follows the handover rules in the §2.3 Escalation specification: four message types (routine, priority, urgent, information), each with its own acknowledgement time, one re-send, and escalation to a second care worker; urgent messages also go through the nurse-call system where it can be connected. Failures are logged. If the record cannot be stored, the session is flagged as unrecorded rather than silently lost.
Benefits	Serves HV-14 Attentive care (the care worker is only called when needed) and HV-13 Well-being (help arrives when prompting fails or there is a safety risk).
Drawbacks	Strains VT-08: the session record is data about a personal-care activity, so what it holds is limited to three items (UC04-F10-PR10) and it is not shared with ST-10.
Selected or rejected	Selected.

TECH-13 Camera-based activity recognition (rejected)

Evidence basis	COACH (Mihailidis et al., 2008) tracked handwashing with a camera over the sink; it is the closest published precedent.

Category	Sensing / AI-ICT
Capabilities	Could track every step, including ones that do not involve a sensed item, and could see distress or a collapse.
Limitations	Continuous video of a resident in their bathroom during a personal-care activity.
Maturity and feasibility	Research prototypes exist; Pepper's own cameras could in principle be used.
Failure and fallback behaviour	Not applicable - rejected.
Benefits	Would serve the Goal on SA-05 with better detection than TECH-11, which would ease VT-11.
Drawbacks	Violates HV-11 Privacy and strains VT-08 more than anything else considered. Continuous monitoring is the concern HFC-10 describes (Niemeijer et al., 2010, 2015), and it would make UC04-F7-CL16 fail by design.
Selected or rejected	Rejected - the privacy cost in a bathroom outweighs the detection gain. Retained because it is the reason F7 works from items instead, and it would be the first option to revisit if TECH-11 proves too unreliable.

TECH-14 Tool-like sink-side prompting device (rejected)

Evidence basis	HFC-13: older adults prefer tool-like framings that keep emotional distance (Pemberton et al., 2026). For personal care specifically the evidence is mixed: Prakash & Rogers (2015) found a robotic face rated more likeable than a human one, but not more trusted or more useful, and interviewees were split. COACH delivered prompts through a screen and speaker at the sink (Mihailidis et al., 2008).
Category	AI/ICT
Capabilities	A small speaker, light and screen mounted by the sink that gives the same F8 prompts, without a humanoid body.
Limitations	No social presence; cannot be the course's robot platform.
Maturity and feasibility	High - simple hardware.
Failure and fallback behaviour	Same as TECH-10.
Benefits	Would serve HV-10 Dignity and HV-11 Privacy. Best supported by the evidence for VT-10 (Dignity) and VT-08 (Privacy): less risk of being read as a supervisor or of talking down to the resident. Fits inside a bathroom.
Drawbacks	Does not use the project's robot platform.
Selected or rejected	Rejected - project constraint: the project uses Pepper. Retained because the evidence favours it for personal care, and it is the fallback if UC04-F8-CL14 or UC04-F7-CL16 hold for Pepper.

TECH-15 Medication tray or dispenser with sensors (deferred)

Evidence basis	Previously the embodiment in RP-02 and DS-03 (§2.1, §2.2).
Category	Sensing / AI-ICT
Capabilities	A tray with a weight sensor, or a lockable dispenser with an open-state sensor, which detects whether medication has been picked up.
Limitations	Cannot track a tooth-brushing sequence, so it cannot enable F7 (§2.3 Open item O-22).
Maturity and feasibility	High - simple sensors.
Failure and fallback behaviour	To be specified with UC05.
Benefits	Would serve the Goal on SA-06.
Drawbacks	To be assessed with UC05, including VT-12.
Selected or rejected	Deferred - fits UC05 (medication), which is deferred to a later cycle.

TECH-16 Position-sensing connected toothbrush (deferred)

Evidence basis	Group knowledge of commercially available connected electric toothbrushes, which report brushing coverage in a companion app. Vendor claims only; no accuracy study has been read, and none with older adults or people with dementia is known to the group (§2.3 Open item O-32).
Category	Sensing / AI-ICT
Capabilities	Estimates from motion and orientation sensors which region of the mouth is being brushed, and for how long, so it could report coverage without any image.
Limitations	• Requires an electric toothbrush. Replacing a resident's familiar manual brush may itself disrupt a routine they still manage (group reasoning). • Region estimates assume a fairly standard brushing posture and grip, which may not hold for residents with dementia. • Consumer apps send data to the vendor's cloud. That link would have to be cut, and the data read locally.
Maturity and feasibility	Commercially available, but reading its data outside the vendor app is uncertain. Not needed for F7, which only needs brushing motion.
Failure and fallback behaviour	Same as TECH-11.
Benefits	Would serve HV-13 by showing whether all of the mouth was brushed, which brushing time cannot.
Drawbacks	Strains VT-14: it produces a per-region record of how a personal-care activity was performed. Coverage feedback would invite more specific prompts ("the left side"), which raises the infantilisation risk tested by UC04-F8-CL14.
Selected or rejected	Deferred. Revisit if UC04-F9-CL20 holds or UC04-F8-CL19 fails, that is, if brushing time turns out not to be enough.

TECH-17 Smart mirror or camera-based brushing analysis (rejected)

Evidence basis	Group knowledge that research prototypes and consumer products of this kind exist; none has been reviewed. Rejected on the same grounds as TECH-13, so the comparison does not depend on their accuracy.
Category	Sensing / AI-ICT
Capabilities	A camera in or behind the bathroom mirror tracks the brush against the face, giving coverage and technique.
Limitations	Continuous video of the resident's face during a personal-care activity in their bathroom.
Maturity and feasibility	Prototypes and consumer products exist.
Failure and fallback behaviour	Not applicable, rejected.
Benefits	Would serve HV-13 Well-being, with the most complete picture of how well the resident brushed.
Drawbacks	Violates HV-11 Privacy and strains VT-08 and VT-14 more than any other brushing-quality option. It is the continuous monitoring HFC-10 describes, and it would make UC04-F7-CL16 fail by design. It also conflicts with covering Pepper's cameras: the System would hide one camera and add another.
Selected or rejected	Rejected. Retained because it is the option the midterm feedback points toward, and the reason brushing quality is measured by people (M-18) instead.

1.4 Foundation Coherence

1.4.1 Edge Inventory

Edges are listed by source ID, with the relation named as it appears on the source page. Targets outside §1.1 to §1.3 (Functions F7 to F10, OBJ, DS, HP, RP, and the UC04 Premises and Claims) are forward edges into the Specification and are included so the inventory is complete.

Table 1.1: Edges of the Situated Activities

Source	Relation	Target(s)
SA-05	performed-by	ST-08 (supported by ST-09)
SA-05	illustrated-by	PS-05
SA-06	performed-by	ST-08 (supported by ST-09)
SA-06	illustrated-by	PS-06
SA-07	performed-by	ST-09
SA-07	illustrated-by	PS-05, PS-06

Table 1.2: Edges of the Stakeholders

Source	Relation	Target(s)
ST-08	performs	SA-05, SA-06
ST-08	holds	HV-09, HV-10, HV-11, HV-12, HV-13
ST-08	appears-in	PS-05, PS-06
ST-08	elaborated-as	HP-04 (§2.2)
ST-09	performs	SA-07
ST-09	supports	SA-05, SA-06
ST-09	holds	HV-09 to HV-14
ST-09	appears-in	PS-05, PS-06
ST-09	elaborated-as	HP-05 (§2.2)
ST-10	holds	HV-09, HV-10, HV-11, HV-13, HV-14
ST-11	holds	HV-10, HV-11, HV-13, HV-14
ST-12	holds	HV-09, HV-11, HV-13

Table 1.3: Edges of the Problem Scenarios

Source	Relation	Target(s)
PS-05	involves	ST-08, ST-09, SA-05, SA-07
PS-05	motivates	OBJ-10
PS-05	reimagined-as	DS-02 (§2.1)
PS-06	involves	ST-08, ST-09, SA-06, SA-07
PS-06	motivates	UC05 (later cycle)
PS-06	reimagined-as	DS-03 (§2.1)

Table 1.4: Edges of the Human Values

Source	Relation	Target(s)
HV-09	held-by	ST-08, ST-09, ST-10, ST-12
HV-09	party-to	VT-07, VT-11, VT-12
HV-09	operationalized-by	M-12, M-15
HV-09	explained-by	HFC-09
HV-10	held-by	ST-08, ST-09, ST-10, ST-11
HV-10	party-to	VT-10
HV-10	operationalized-by	M-14, M-16
HV-10	explained-by	HFC-12, HFC-13
HV-11	held-by	ST-08, ST-09, ST-10, ST-11, ST-12
HV-11	party-to	VT-08, VT-14
HV-11	operationalized-by	M-16

HV-11	explained-by	HFC-10, HFC-13
HV-12	held-by	ST-08, ST-09
HV-12	party-to	VT-09, VT-10, VT-13
HV-12	operationalized-by	M-12, M-13
HV-12	explained-by	HFC-11
HV-13	held-by	ST-08 to ST-12
HV-13	party-to	VT-07, VT-08, VT-09, VT-11, VT-12, VT-14
HV-13	operationalized-by	M-14, M-18
HV-13	explained-by	HFC-11
HV-14	held-by	ST-09, ST-10, ST-11
HV-14	party-to	VT-13
HV-14	operationalized-by	M-17
HV-14	explained-by	HFC-14

The "held-by" edges agree with the "holds" edges on the Stakeholders page for all six values.

Table 1.5: Edges of the Value Tensions

Source	Relation	Target(s)
VT-07	involves	HV-09, HV-13
VT-07	bounds	F8, F9
VT-07	tested-by	UC04-F8-CL13
VT-08	involves	HV-11, HV-13
VT-08	bounds	F7, F9, F10
VT-08	strained-by	TECH-10, TECH-12 (TECH-13 rejected for straining it most; TECH-17 rejected)
VT-08	tested-by	UC04-F7-CL16, UC04-F10-CL17
VT-09	involves	HV-12, HV-13
VT-09	bounds	F8
VT-09	tested-by	UC04-F8-CL10, UC04-F8-CL13
VT-10	involves	HV-10, HV-12
VT-10	bounds	F8
VT-10	strained-by	TECH-10
VT-10	tested-by	UC04-F8-CL14
VT-11	involves	HV-09, HV-13
VT-11	bounds	F7, F8
VT-11	strained-by	TECH-11
VT-11	tested-by	UC04-F8-PR8, UC04-F8-CL13
VT-12	involves	HV-09, HV-13
VT-12	bounds	UC05 (later cycle)
VT-13	involves	HV-12, HV-14
VT-13	bounds	F8, F9
VT-13	tested-by	UC04-F9-CL18, UC04-F9-CL20
VT-14	involves	HV-11, HV-13
VT-14	bounds	F7, F8, F10
VT-14	strained-by	TECH-16 (deferred), TECH-17 (rejected for straining it most)
VT-14	tested-by	UC04-F7-CL16, UC04-F9-CL20

Table 1.6: Edges of the Human Factors Concepts

Source	Relation	Target(s)
HFC-09	explains	HV-09
HFC-09	associated-measure	M-15
HFC-09	grounds	UC04-F8-CL12
HFC-10	explains	HV-11
HFC-10	associated-measure	M-16
HFC-10	grounds	UC04-F7-CL16
HFC-11	explains	HV-12, HV-13
HFC-11	associated-measure	M-12, M-13, M-14
HFC-11	grounds	UC04-F8-CL10, UC04-F8-CL11, UC04-F8-CL13, UC04-F9-CL15, UC04-F9-CL18, UC04-F8-CL19
HFC-12	explains	HV-10
HFC-12	associated-measure	M-14, M-16
HFC-12	grounds	UC04-F8-CL14
HFC-13	explains	HV-10, HV-11
HFC-13	associated-measure	M-16, M-14
HFC-13	informs	TECH-10 selection rationale
HFC-13	grounds	UC04-F8-CL14, UC04-F7-CL16
HFC-14	explains	HV-14
HFC-14	associated-measure	M-17, M-11
HFC-14	grounds	UC04 handover rules (F9), TECH-12 alerting policy, UC04-F9-CL18

Table 1.7: Edges of the Measures

Source	Relation	Target(s)
M-11	used-by	EM-05
M-11	operationalizes	UC04-F7-PR7, UC04-F8-PR8 (timing component), UC04-F9-PR9
M-11	associated with	HFC-14
M-12	operationalizes	HV-12, HV-09; UC04-F8-CL10, UC04-F8-CL11, UC04-F9-CL15, UC04-F9-CL20
M-12	used-by	EM-06
M-12	associated with	HFC-11
M-13	operationalizes	HV-12; UC04-F8-CL11, UC04-F8-CL13
M-13	used-by	EM-06
M-13	associated with	HFC-11

M-14	operationalizes	HV-13, HV-10; UC04-F8-CL10, UC04-F8-CL13, UC04-F8-CL14, UC04-F8-CL19
M-14	used-by	EM-06
M-14	associated with	HFC-11, HFC-12, HFC-13
M-15	operationalizes	HV-09; UC04-F8-CL12
M-15	used-by	EM-06
M-15	associated with	HFC-09
M-16	operationalizes	HV-11, HV-10; UC04-F10-PR10 (record-content component), UC04-F8-CL14, UC04-F7-CL16, UC04-F10-CL17, UC04-F8-CL19
M-16	used-by	EM-05, EM-06 (resident interview), EM-07
M-16	associated with	HFC-10, HFC-12, HFC-13
M-17	operationalizes	HV-14; UC04-F9-CL18
M-17	used-by	EM-06, EM-07
M-17	associated with	HFC-14
M-18	operationalizes	HV-13; UC04-F8-CL19, UC04-F9-CL20, UC04-F7-PR7 (brushing-time component)
M-18	used-by	EM-05 (part b), EM-06
M-18	associated with	VT-14

Table 1.8: Edges of the Evaluation Methods

Source	Relation	Target(s)
EM-05	applies	M-11, M-16, M-18
EM-05	produces Evidence for	UC04-F7-PR7, UC04-F8-PR8, UC04-F9-PR9, UC04-F10-PR10
EM-06	applies	M-12, M-13, M-14, M-15, M-16, M-17, M-18
EM-06	produces Evidence for	UC04-F8-CL10, UC04-F8-CL11, UC04-F8-CL12, UC04-F8-CL13, UC04-F8-CL14, UC04-F9-CL15, UC04-F7-CL16, UC04-F9-CL18, UC04-F8-CL19, UC04-F9-CL20
EM-07	applies	M-16, M-17
EM-07	produces Evidence for	UC04-F8-CL14, UC04-F7-CL16, UC04-F10-CL17, UC04-F9-CL18

The "used-by" edges on the Measures page and the "applies" edges on the Evaluation Methods page agree for all eight Measures.

Table 1.9: Edges of the Technology Options

Source	Relation	Target(s)
TECH-10 (selected)	enables	F8, F9 (resident-facing part)
TECH-10	addresses	Goal (SA-05), HV-12, HV-14
TECH-10	strains	VT-10, VT-08
TECH-10	realized-in	RP-02 (§2.2)
TECH-11 (selected)	enables	F7, F8 (early-stop rule), F10 (session data)

TECH-11	addresses	Goal (SA-05), HV-11
TECH-11	strains	VT-11
TECH-12 (selected)	enables	F9, F10
TECH-12	addresses	HV-13, HV-14
TECH-12	strains	VT-08
TECH-12	alerting policy grounded on	HFC-14
TECH-13 (rejected)	would strain	VT-08 (no outgoing edges to the Specification)
TECH-14 (rejected)	would ease; fallback for	VT-10, VT-08; TECH-10 if UC04-F8-CL14 or UC04-F7-CL16 hold, or if O-30 fails
TECH-15 (deferred)	none	No edges this cycle, deferred with UC05
TECH-16 (deferred)	would strain	VT-14 (no outgoing edges to the Specification this cycle)
TECH-17 (rejected)	would strain	VT-08, VT-14 (no outgoing edges to the Specification)

1.4.2 Open Tensions

No Value Tension has been tested yet, since no Evaluation Method has been run. The table gives each tension's status as recorded in §1.2.2, the design stance taken pending evidence, and what is meant to test it.

Tension	**Values**	**Recorded status**	**Stance taken pending evidence**	**Tested by**
VT-07 Autonomy vs Well-being	HV-09, HV-13	Provisionally resolved	Graduated support: no assistance while the resident can continue, increasingly supportive prompting when difficulty is detected, and immediate notification of a care worker on an immediate risk of serious harm (group decision, §2.3 Open item O-11). No regulatory floor identified.	UC04-F8-CL13
VT-08 Privacy vs Well-being	HV-11, HV-13	Provisionally resolved	Monitor only what is needed to decide whether support is needed; share nothing sensitive unless the agreed conditions for caregiver involvement are met. No camera (TECH-13 rejected); Pepper's cameras disabled and covered; the session record is limited to three items and not shared with ST-10.	UC04-F7-CL16, UC04-F10-CL17; record content verified by UC04-F10-PR10 (M-16)
VT-09 Competence vs Well-being	HV-12, HV-13	Provisionally resolved	Allow time to complete a step, give a supportive prompt on repeated difficulty, and increase assistance if the resident still cannot continue or shows distress.	UC04-F8-CL10, UC04-F8-CL13
VT-10 Dignity vs Competence	HV-10, HV-12	Open	Prompts phrased as offers in an adult register, addressing the resident by name, stopping as soon as the resident resumes; the robot is framed as a tool with low expressivity. Strained by the choice of Pepper (TECH-10) over the tool-like device (TECH-14), which HFC-13 favours; TECH-14 is the named fallback (§2.3 Open items O-19, O-20).	UC04-F8-CL14 (adverse Claim)
VT-11 Autonomy vs Well-being (detection reliability)	HV-09, HV-13	Provisionally resolved, open until EM-05 shows an acceptable false-prompt rate	Favour fewer false prompts: F8 waits a configured interval and starts at the least intrusive prompt level. Accepted as the cost of item sensing (TECH-11) without a camera. Timing values are proposals awaiting the first EM-05 pilot (§2.3 Open item O-29).	UC04-F8-PR8 (M-11), UC04-F8-CL13
VT-12 Autonomy vs Well-being (medication timing)	HV-09, HV-13	Open, deferred to UC05	Deferred with UC05. A floor exists for time-critical medication: residents on such medication are excluded (PS-06), and the source for that floor is still a TODO in §1.2.2. DS-03 sketches a resolution for the remaining medication.	No Claim yet; to be written with UC05

VT-13 Competence vs Attentive care	HV-12, HV-14	Open	The System carries the waiting and the lower prompt levels; the care worker is called only when prompting is exhausted, the resident is distressed, or there is a safety risk (F9). Alert volume is limited by the handover rules grounded on HFC-14 (§2.3 Open item O-31).	UC04-F9-CL18 (M-17), UC04-F9-CL20
VT-14 Privacy vs Well-being (brushing quality)	HV-11, HV-13	Open	Brushing time is used within the session only and stored nowhere; whether teeth are clean is judged by people (M-18 during evaluation, the home's oral-care routines and ST-12 otherwise). TECH-17 rejected; TECH-16 deferred until EM-06 shows whether brushing time is enough. Part of §2.3 Open item O-32 is still open (per-resident targets, confirming the plaque instrument).	UC04-F7-CL16, UC04-F9-CL20; TECH-16 decision also depends on UC04-F8-CL19

1.4.3 Completeness Check

Table 1.10: Checks the manual names

Check	Result	Stance
Any Situated Activity with no Goal stated	None. SA-05, SA-06 and SA-07 each state a Goal.	SA-06's Goal is addressed only by TECH-15, which is deferred with UC05 (§2.3 Open item O-22). Recorded as a scope decision for this cycle.
Any Human Value with no recognition criteria	None. HV-09 to HV-14 each state served and violated criteria.	No action.
Any Human Factors Concept with no associated Measure	None. HFC-09 to HFC-14 each name at least one Measure.	No action. The reverse links on the Measures page have been completed (see *One-sided and inconsistent edges*).
Any Technology Option addressing nothing	The three selected options (TECH-10 to TECH-12) each address a Goal or Human Value. The five rejected or deferred options (TECH-13 to TECH-17) have no "addresses" edge, which follows from their status. In their Benefits fields, TECH-13 names Goal (SA-05), TECH-15 names Goal (SA-06) and TECH-16 names HV-13; TECH-14 now names HV-10 and HV-11, and TECH-17 names HV-13.	Fixed in §1.3, so that every option, rejected or not, states what it would have served.

Table 1.11: Further gaps

Gap	Stance
SA-06, PS-06, VT-12 and TECH-15 form one cluster with no Claim, no Measure and no Evaluation Method.	Deferred with UC05. New Measures for it start at M-24 and new Evaluation Methods at EM-09, since M-19 to M-23 and EM-08 belong to the main concept's Kinnect study.
VT-12 has no testing Claim.	Deferred with UC05, as above. Every other tension has at least one testing Claim or Premise.
HV-09 Autonomy and HV-10 Dignity are not addressed by any selected Technology Option.	Both are carried by Function behaviour: F8's wait and offer-style prompts follow the VT-07, VT-09 and VT-10 resolutions. They are tested by UC04-F8-CL12 (HV-09) and UC04-F8-CL14 (HV-10).
No Technology Option addresses the Goal on SA-07 directly.	SA-07's Goal reaches the design through HV-14 (addressed by TECH-10 and TECH-12) and OBJ-15. Recorded.
ST-10, ST-11 and ST-12 have no Persona and perform no Situated Activity.	All three are Indirect Stakeholders. A Persona is built for Direct Stakeholders only, and both Direct Stakeholders have one (ST-08 → HP-04, ST-09 → HP-05). No action.
M-18 Brushing quality has no associated Human Factors Concept.	M-18 is justified through VT-14 and HV-13, and was added for the midterm feedback. Recorded; no HFC is proposed for this cycle.
Every Foundation entity is provisional: no resident, care worker or manager has been observed or interviewed.	Recorded on every Evidence basis field. The edges on this page inherit that status.
Premise thresholds and escalation timing are proposals.	Pending the first EM-05 pilot. Affects VT-11's stance.
Whether Pepper (TECH-10) fits at a care-home bathroom doorway is unchecked.	fallback is TECH-14. Affects TECH-10's feasibility and the VT-10 stance.

Chapter 2

Specification

2.1 Design Scenarios

DS-02

It is morning in the dementia care home. The resident has finished breakfast and returns to their room. The care worker, Myra, reminds them that it is time to brush their teeth, and moves on to help the next resident on the round.

The resident goes into the bathroom and stands in front of the sink. Pepper stands quietly just outside the bathroom door, its camera lenses covered and its eye lights dimmed. The resident picks up the toothbrush. The sensor in the toothbrush registers the pick-up, and the System notes that the resident has reached that step. Pepper says nothing.

The resident pauses, toothbrush in hand, and seems unsure what comes next. Pepper waits. After the configured interval nothing has changed, so Pepper gives the lightest prompt, an orienting one, calmly and by name: "Mary, you are brushing your teeth this morning." Nothing changes, so after a short pause Pepper names the next action as an offer: "Would you like some toothpaste on your brush?" The resident picks up the toothpaste, and the weight pad under it registers the lift. Pepper goes quiet again.

The resident puts toothpaste on the brush but then stops again. This time neither the orienting prompt nor the naming prompt ("You could brush your teeth now.") is enough, so Pepper repeats the sentence once and its chest tablet shows a short, simple picture of someone brushing. The resident starts brushing; the toothbrush sensor registers the motion and Pepper stops prompting at once.

After about half a minute the resident stops brushing and lowers the brush. Pepper waits, then offers once: "Would you like to brush a little longer?" The resident brushes for another minute. Had they put the brush down instead, Pepper would have let it go and nobody would have been called.

The resident rinses and puts the toothbrush back in the cup. Pepper says "All done," once, in an ordinary adult tone - no praise that would suit a child - and nothing more.

Meanwhile Myra has been helping another resident get dressed. They were not called back, because the resident finished with Pepper's prompts. On their phone the session record shows three things only: tooth brushing completed, highest prompt level "demonstration", no care worker needed. Nothing about how the resident brushed their teeth is stored.

If the resident had stopped responding and nothing had moved for several minutes, Pepper would have told the resident that someone was coming, and Myra's phone would have received a short message: the resident's room, tooth brushing, step reached, reason for the call.

DS-03

Medication - kept for UC05, deferred to a later cycle (§2.3 Open item O-22). The tray-based device described here corresponds to TECH-15 (§1.3), not to the Pepper-based System selected for UC04.

A resident with dementia needs to take prescribed medication in the morning. At the right time, a care worker approaches the resident with their medication and explains that it's time to take it. At first, the resident agrees to take the medicine.

The care worker places the medication and a glass of water on the tray in front of the helper robot, and the device gives a brief, friendly acknowledgement for the patient, letting them know that they can take the medication whenever they are ready. The helper robot then goes quiet.

The resident becomes distracted by activity somewhere else in the room and does not take the medication. After a few minutes in which the tray weight has not changed, the helper robot notices that the patient did

not take the medication. The device then glows warm light and gently reminds the patient about the medication.

The resident returns their attention to the tray and the robot and, unsure of why the tablets are there, asks the robot for clarification. The device responds simply, friendly, and gently that these are the patient's morning tablets, that help with their condition, and that should be taken with the water on the tray. In case the resident does have more questions about the medication, the robot calmly repeats its purpose and reassures the patient to take it. Because of the reassuring but still matter-of-fact, adamant tone, the resident does not need the care worker to re-explain everything from scratch, and is convinced into taking the medication.

The resident therefore picks up the medication and water, and the helper robot detects the weight change. The robot also senses that the medication dispenser/container has been opened, which indicates that the patient indeed took the medication. The device then asks the resident to confirm that they took the tablets. The helper robot gives a short, friendly acknowledgement, and saves to the record that the patient has taken the medication, avoiding this way the risk of it being offered again later by mistake.

The care worker was free to attend to another resident during the distraction, instead of having to repeatedly prompt. They are only pulled back if the robot's requests go unanswered several times, in which case the device sends a quiet notification that a check-in with the patient is needed.

2.2 Personas and Profiles

HP-04 Mary, resident with dementia

Evidence basis	Literature and group reasoning only - no residents or care staff have been interviewed or observed by this project, so this Persona is provisional. Built on ST-08/ST-09 (§1.1), PS-05 and PS-06, and the literature cited in §1.2.
Elaborates	ST-08
Role in Scenario	The person whose morning routine it is (DS-02). Instead of being a passive recipient, they are given time to do each step themselves, and are prompted only when they stall, at the lowest level that gets them going again.
Task Demands	1. Recognizing that it is time to brush their teeth, and starting. 2. Keeping track of which step comes next once they have started, without the care worker coming back each time. 3. Finishing the whole routine rather than stopping part-way.
Specific Capabilities and Needs	Can still carry out each step of tooth brushing and respond to a short spoken cue, and can say yes or no. Cannot reliably remember the next step once distracted, may not recognise an object for what it is, and may repeat a step. Needs calm, short, simple cues, time to act before being prompted, and nothing said to them in a way they would find childish.
Values and Motivations	• HV-09: Autonomy - The resident wants to feel that the routine is within their own control, and that they can carry it out themselves. • HV-10: Dignity - The resident wants to be spoken to as a capable adult, not talked through a basic activity like a child. • HV-11: Privacy - The resident does not want to be watched or recorded in their bathroom. • HV-12: Competence - The resident wants to experience that they can still brush their own teeth once reminded, rather than having it done for them. • HV-13: Well-being - The resident wants their mornings to be calm, and to avoid the frustration of getting stuck and not knowing what comes next.
Social Network and Relationships	Receives support from ST-09 (the care worker) and may have ST-10 (family) involved in their care. DS-02 does not remove their reliance on the care worker; it reduces how often they need them during the routine, because the System holds the sequence steady when the care worker is with another resident.
History and Experiences	No prior technology rejection recorded. Previous experiences include stalling part-way through tooth brushing (PS-05) and needing repeated reminders and redirection to complete tasks (PS-06). Being approached repeatedly and having everything re-explained can be confusing and tiring, which is why smaller, gentler cues are desirable.

Constraints and Privacy Considerations	Will not accept being filmed in their bathroom. Monitoring should go no further than knowing where they are in the routine, and nothing about how they brushed their teeth should be kept.

HP-05 Myra, care home staff member

Evidence basis	Literature and group reasoning only - no residents or care staff have been interviewed or observed by this project, so this Persona is provisional. Built on ST-08/ST-09 (§1.1), PS-05 and PS-06, and the literature cited in §1.2.
Elaborates	ST-09
Role in Scenario	Responsible for several residents on the morning round (SA-07). In DS-02 their role shifts from walking each resident through their routine step by step to being called in only when prompting has run out, the resident is distressed, or there is a safety risk.
Task Demands	1. Being able to help another resident while one completes their routine. 2. Receiving a short, meaningful message when they are actually needed, and responding to it. 3. Knowing afterwards what support each resident needed, without constant double-checking.
Specific Capabilities and Needs	Can talk to the resident, reassure them, demonstrate and physically help. Can intervene when something is wrong. Needs calls from the System to be rare and meaningful rather than many low-value alerts, and needs the session record to be short enough to read on their phone during the round.
Values and Motivations	• HV-09: Autonomy - Wants the resident to complete the routine themselves with the help of the device, rather than having to walk them through it every time. • HV-10: Dignity - Wants the resident to be treated respectfully even when they cannot be there to assist in person. • HV-11: Privacy - Wants to limit surveillance of residents, and of their own work. • HV-12: Competence - Wants the resident to keep using the abilities they still have, so prefers gentle prompts that let the resident do it themselves over stepping in and doing it for them. • HV-13: Well-being - Has the resident's best interest in mind, and wants the routine to go calmly. • HV-14: Attentive care - Wants the time to give each resident the attention they need, instead of being pulled back to the same resident again and again.
Social Network and Relationships	Supports several residents per shift, is employed by ST-11, keeps ST-10 informed, and coordinates with ST-12 on health and medication. In DS-02 they remain the person who explains, reassures and physically helps; the System only takes over the waiting and the low-level prompting.
History and Experiences	No prior technology rejection recorded. Previous experiences include the burden of repeatedly returning to one resident during the routine (SA-07), and having to choose between waiting for a resident and moving on to the next.
Constraints and Privacy Considerations	Will not accept a device that collects and stores data in a way that could function as workplace surveillance. Will not accept a system that cannot flag genuine problems, or that sends too many non-critical alerts, as this would defeat the purpose of not having to check on residents constantly.

RP-02 Pepper as tooth-brushing aid

Evidence basis	Vendor documentation for Pepper (TECH-10); design choices follow HFC-13 and VT-10 (tool-like framing, low expressivity) and VT-08 (no cameras). Not yet tested with residents.
Role in Scenario	Stands at the bathroom doorway during the tooth-brushing routine (DS-02). Stays quiet while the resident manages, prompts at the lowest sufficient level when they stall (F8), and calls the care worker when prompting runs out, the resident is distressed, or there is a safety risk (F9).
Social Competencies	Slow, non-interruptive turn-taking. Waits and holds silence before prompting, and stops as soon as the resident resumes. Responds to a simple yes/no or a touch on its head or hand, but does not depend on understanding speech. Does not initiate conversation beyond the routine.

Embodiment	Pepper humanoid robot (TECH-10), about 1.2 m tall, on a wheeled base, with a chest tablet, speakers, microphones, touch sensors and LED eyes. Positioned at the bathroom doorway, never blocking the resident's path, kept away from water. Cameras disabled and lenses visibly covered; activity tracking comes from sensors on the tooth-brushing items (TECH-11), not from the robot. Arm gestures and head movement kept minimal during the activity, and no gaze-following of the resident.
Narrative and Identity Framing	Presented to residents and staff as a helper for the morning routine - a tool, not a carer, companion or supervisor. Avoids anything that suggests it is watching or checking on the resident, and avoids the exaggerated gentleness and praise that would read as treating the resident like a child (HFC-12, HFC-13).
Communication Strategies	Short spoken sentences, addressing the resident by name, in an ordinary adult register, phrased as offers ("Would you like...") rather than instructions. Prompt levels: orienting, then naming the next action, then a picture or short video of the step on the chest tablet (demonstration). Soft eye lights instead of alarms or beeping. Tells the resident when a care worker has been called. Sends handover requests and the session record to the care worker's phone (TECH-12).
Privacy and Data Handling	Processes only the item-sensor events (TECH-11) and the resident's simple responses. No video. No audio stored beyond the moment of the interaction. Keeps a three-item session record (completed or not, highest prompt level, whether a care worker was called) for the care record; no step-by-step trace, and nothing is shared with family (ST-10).
Context-Aware Capabilities	Adapts when it prompts to the time since the last sensor event, and escalates according to the sensor state and the resident's responses. The wait interval is configured per resident (a per-resident adaptive interval is the planned response if UC04-F8-CL18 holds). Does not build a behavioural profile of the resident over time.

2.3 Objectives, Use Cases, Functions, Premises and Claims

2.3.1 Objectives

OBJ-10 Activity reaches completion without care-worker prompting carrying it

Evidence basis	Derived from the Goal recorded on SA-05, and from PS-05's account of how that Goal currently fails. Literature: Rogers et al. (1999) found that systematic graduated prompting of nursing-home residents with dementia produced a 26% increase in self-dressing and significantly fewer physical assists than usual care, in a severely impaired cohort (mean MMSE 6.07). That establishes the effect direction is achievable. No primary data collected by this project. SA-05's own Evidence basis rests on literature and group reasoning only.
Statement	The resident reaches the last step of the tooth-brushing activity, and the verbal prompting a care worker currently has to supply is no longer what carries them there.
Type	Operational
Derived from	Goal on SA-05 ("maintain oral hygiene as part of the resident's normal morning routine, with as much independence as possible")
Priority	Must. This is the Goal SA-05 states, and the point at which PS-05 breaks down.

OBJ-11 A genuine opportunity to act before assistance arrives

Evidence basis	Derived from HV-09 and HV-12, and from VT-09's current resolution. Literature: Self-determination theory (Ryan & Deci, 2000) for the value itself. For why this matters specifically in dementia care, Kitwood (1997) names *disempowerment* - "not allowing or enabling a person to use the abilities they still have" - as one of the detractors that erode personhood. No primary data collected by this project.
Statement	At each step of the activity, the resident has had a real opportunity to act on their own before any assistance arrives.
Type	Value-derived

Derived from	HV-09 Autonomy and HV-12 Competence. Informed by VT-09's resolution, which already states the trade-off this Objective sits on.
Priority	Must. Without it, OBJ-10 could be satisfied by a System that simply narrates every step - which is what the existing workaround recorded on SA-05 already does, and what PS-05 identifies as the problem rather than the solution.

OBJ-12 Difficulty resolved where it appears, before it becomes distress

Evidence basis	Derived from HFC-11. Literature: de Werd et al. (2013) review error-prevention techniques for teaching everyday tasks to people with dementia; Rogers et al. (1999) found disruptive behaviours *decreased* significantly under systematic prompting. Both support the principle that assistance timing matters to distress. Contested on direction - the same literature favours preventing the error rather than allowing it, which is not what F8 does. See Open item O-16. No primary data collected by this project.
Statement	Where the resident cannot continue unaided, the difficulty is resolved at the step where it appears, rather than accumulating into confusion or distress.
Type	Human-factors
Derived from	HFC-11 Adaptive assistance, competence and distress
Priority	Must. This is the other side of OBJ-11; VT-09 is the trade-off between the two, and a design that serves only one of them has not resolved that tension.

OBJ-13 Support without the feeling of being watched

Evidence basis	Derived from HV-11, and from the scope decision recorded in PS-06's out-of-scope field ("we only report a failure in completing the task, we avoid surveilling the patient in general"). Literature: Niemeijer et al. (2010), a systematic review of 79 papers on the ethical and practical acceptability of surveillance technology in residential care for people with dementia. Niemeijer et al. (2015) then asked residents themselves, and concluded such technology can contribute to autonomy *only* when embedded in a person-centred approach - which is the condition this Objective encodes. No primary data collected by this project.
Statement	The resident is supported through the activity without experiencing it as being watched, and without detail about a personal-care activity reaching anyone who does not need it.
Type	Value-derived
Derived from	HV-11 Privacy. Bounded by VT-08's resolution, and explained by HFC-10.
Priority	Should. It constrains how F7 and F10 may be built rather than adding a capability of its own - but the constraint is hard, because tooth brushing is a personal-care activity.

OBJ-14 Prompting that does not talk down to the resident

Evidence basis	Derived from HV-10's recognition criteria, which names infantilisation explicitly. Literature: Kitwood (1997) lists *infantilisation* - "treating the person like a child; using condescending language or gestures toward an adult" - among the 17 detractors of personhood that make up what he called malignant social psychology in dementia care. *Outpacing* (giving information or choices too quickly) is a second detractor directly relevant to how F8 paces a prompt. Robot-specific support: Lee et al. (2016) on ageing stereotypes embedded in assistive robot design; Coghlan et al. (2021), whose older participants called companion robots "patronizing" when they were toy-like or imitated human companionship; Furlanis et al. (2026) for why an embodied system carries this risk differently from a screen, via their "embodied treatment" dimension - gaze, timing, tone. See Open item O-19. No primary data collected by this project.
Statement	The resident is prompted through the activity without being addressed as less capable than they are.
Type	Value-derived

Derived from	HV-10 Dignity
Priority	Should. Distinct from OBJ-11: OBJ-11 is about whether the opportunity to act exists, this is about how the prompt reads when it does arrive. A System could serve one and violate the other.

OBJ-15 The care worker's time goes where it is needed, and care continues at handover

Evidence basis	Derived from the Goal recorded on SA-07 (the care worker's morning round), added to close Open item O-12 and O-23. Literature: Rogers et al. (1999) found that graduated prompting of nursing-home residents with dementia took considerably more caregiver time than usual care - the time cost this Objective asks the System to absorb. No primary data collected by this project.
Statement	The care worker is only pulled back to a resident when prompting has run out, the resident is distressed, or there is a safety risk, and afterwards knows what support each resident needed.
Type	Operational
Derived from	Goal on SA-07 ("get every resident on the round through their morning care safely and in time, giving each the support they actually need, and know afterwards what each resident needed"). Serves HV-14 Attentive care; bounded by VT-13.
Priority	Should. It is the project's stated aim of relieving care staff, but it must not be met at the expense of OBJ-10 to OBJ-12: a System that saves staff time by letting activities go unfinished fails.

2.3.2 Use Cases

UC04 Support the resident through the morning tooth-brushing routine

Evidence basis	Formalizes SA-05 and PS-05. Closest prior system: COACH (Mihailidis et al., 2008) prompted six older adults with moderate-to-severe dementia through handwashing using audio and audio-video prompts, and scored what M-12 scores here - one point per step completed without human assistance. It is the nearest published precedent for this Use Case and should be cited wherever this design is defended. No primary data collected by this project. No resident or care worker has been observed or interviewed; SA-05 and PS-05 rest on literature plus the group's own reasoning.
Objective(s) served	OBJ-10, OBJ-11, OBJ-12, OBJ-13, OBJ-14, OBJ-15
TDP(s) and IDP(s) used	TDP-02 Graduated activity support with human handover (§2.4) allocates the tasks between resident, System and care worker; IDP-04 Respectful graduated prompting (§2.5) shapes F8's prompts. Both are drafted from UC04 alone. Confirming that they recur needs UC05 (see UC05's scope decision).
Actors	ST-08 Resident with dementia (Direct) - the only Actor who interacts with the System in real time. ST-09 Care home staff member (Direct) - receives handover from F9 and reviews the F10 record; does not operate the System during the activity. The System - Pepper (TECH-10) for prompts and interaction, sensors on the tooth-brushing items (TECH-11) for tracking, and a staff notification channel and session record (TECH-12). See §1.3. ST-10 Family members (Indirect) - affected by outcomes, not an Actor in this Use Case, and not a recipient of anything F10 records.
Trigger	The resident's tooth-brushing window opens - SA-05's trigger, after waking, washing and/or breakfast - and either the resident is at the sink, or the window is running down with the activity not started. Both are in scope, because PS-05's first sub-problem is initiation.
Pre-condition	The resident's SA-05 step sequence is configured, together with the assistance baseline they currently need.

The item sensors (TECH-11) are fitted and reporting, and Pepper (TECH-10) is charged and positioned at the bathroom doorway, within earshot and sight of the sink, without blocking the resident's path. Pepper's cameras are disabled and visibly covered.

A care worker is reachable within the home for F9.

Consent arrangements for ST-08 are in place, on the terms VT-08's resolution sets.

Post-condition

One of the following holds:

- the resident has completed SA-05 step 9, unaided or after prompting at some level;
- a care worker has taken over following F9;
- the window has closed with the activity incomplete, and that fact is recorded.

In every case F10 holds a session record, and no step-by-step trace of the activity is retained.

Action Sequence

1. The tooth-brushing window opens. The System begins tracking the activity (F7). **Deliberate addition** - SA-05 task 1 is the resident recognising that it is time, which PS-05 shows failing.
2. The resident enters the bathroom and begins (SA-05 tasks 2-3).

2-alt. The resident has not begun by the end of the window. F8 issues an initiation prompt. If there is no response, step 9.

3. The resident performs a step; the System tracks progress (F7). SA-05 tasks 4-8.

3-alt. During SA-05 task 7, brushing stops before the resident's target brushing time, or the same item event repeats without the next step following. Either is treated as a stall on the current step; go to step 4. The early-stop case has its own limits, set out in the *Escalation specification*.

4. A stall is detected on the current step. The System waits the configured interval and does nothing. **Deliberate addition** - the wait is the design, not an implementation delay. See OS-04, and Open item O-16 for the literature that contests it.
5. The resident resumes unaided. Return to step 3.
6. The stall persists past the interval. F8 issues the lowest-level prompt available for that step.
7. The resident resumes. F8 stops prompting; return to step 3.
8. The stall persists. F8 re-prompts one level more specific, up to the demonstration level.

8-alt. Signs of distress at any point. Remaining prompt levels are skipped; go to step 9.

9. Prompting is exhausted without progress. F9 requests care-worker assistance, passing the room, the activity, the step reached, and the reason.
10. A safety-critical condition occurs at any point - an immediate risk of serious harm, inferred from no sensor change and no response to prompts for a configured time (TECH-11 limitation). F9 immediately, pre-empting any prompting, per VT-07's current resolution. **Deliberate addition** - in scope by group decision (Open item O-11).
11. The resident completes SA-05 step 9, or a care worker takes over, or the window closes. F10 records the session. **Deliberate addition** relative to SA-05; it corresponds to the care worker's own recording and handover task (SA-07 task 6).

Exception handling. If the System cannot track the activity - sensor loss, power, or an unrecognised sequence - it reports a degraded state to the care worker rather than failing silently, and the activity falls back to the existing SA-05 workarounds. Verified by UC04-F7-PR7.

Function(s) realized

F7, F8, F9, F10 - specified in full below.

Functions

Function F7 (track position in the activity sequence): the System determines which step of the SA-05 sequence the resident is currently on, and whether progress to the next step has stalled, from the resident's observable actions on the activity's own tools and fixtures - requiring the resident neither to report anything nor to operate the System. During SA-05 task 7 it also sums the time the toothbrush is in brushing motion, which it uses within the session and does not keep (VT-14). It observes the activity only, not the resident in general. Serves OBJ-10, OBJ-13. Bounded by VT-08, VT-11. Enabled by TECH-11 (item sensors) - deliberately not by Pepper's cameras (TECH-13 rejected).

Function F8 (prompt at the lowest sufficient level, after a wait): on a detected stall, the System waits a configured interval, then issues the least specific prompt that could unblock the current step. If the stall persists it re-prompts one level more specific - orienting the resident to the activity, then naming the next action, then demonstrating it - and stops as soon as the resident resumes. If brushing stops before the resident's target brushing time, F8 offers once to continue; this offer never leads to a handover on its own. Prompts address the resident by name and in the second person, phrased as an offer rather than an instruction. Serves OBJ-10, OBJ-11, OBJ-12, OBJ-14. Bounded by VT-07, VT-09, VT-10, VT-11. Enabled by TECH-10 (Pepper): spoken prompts for the orienting and naming levels, and a picture or short video on its chest tablet for the demonstration level, rather than miming with its arms (VT-10 mitigation).

Where F8's ladder comes from. It is not invented: it is the escalation SA-05 already records as existing workarounds - verbal reminder, then naming, then demonstrating the brushing action, then physical assistance - made systematic, with physical assistance deliberately left to the care worker via F9 rather than taken on by the System. Rogers et al. (1999) is the published precedent for graduated prompting of this kind in nursing-home dementia care.

Where it is contested. F8 is a least-to-most ladder. The dementia ADL literature largely runs the other way, favouring errorless learning and vanishing cues - prompt early to prevent the error, then fade support across sessions (de Werd et al., 2013). The largest trial of that approach, REDALI-DEM, then found structured relearning improved ADL performance but errorless learning gave no advantage over trial-and-error learning. So the evidence neither endorses F8's direction nor rules it out.

Design bet (group decision, Open item O-16). *F8 keeps its least-to-most direction as a deliberate bet, not an established finding: we expect the wait-then-escalate ladder to preserve competence (OBJ-11) without causing avoidable frustration. UC04-F8-CL13 is the adverse Claim designed to settle it. If CL13 holds, F8 is re-specified - first with a per-resident wait interval, and if that is not enough, as most-to-least with fading.*

Function F9 (hand over to a care worker): the System requests care-worker assistance when prompting has been exhausted without progress, when the resident shows signs of distress, or when a safety-critical condition occurs. It passes only the resident's room, the activity, the step reached, and the reason for handover. The triggers, priorities, acknowledgement and re-escalation rules are in the *Escalation specification*. Serves OBJ-10, OBJ-12, OBJ-13, OBJ-15. Bounded by VT-07 - whose current resolution includes the safety-critical path - and by VT-08, VT-13. Enabled by TECH-12 (notification to the care worker's work phone), with TECH-10 telling the resident that someone is coming.

Function F10 (record the session for handover): the System records, per session, whether the activity was completed, the highest prompt level required, and whether a care worker was called. It retains no step-by-step detail of the activity itself. Serves OBJ-13, OBJ-15. Bounded by VT-08. Enabled by TECH-12 (session record), from TECH-11 data.

Note: F10 traces to OBJ-15, derived from the care worker's own Goal on SA-07 (care continuity at handover). Previously it had no Objective; see Open item O-12.

Escalation specification

Answers the midterm feedback to say in more detail how and when support escalates. It makes F7's stall rule, F8's ladder and F9's handover concrete enough to build and to test in EM-05. Every value below is a proposed starting value from group reasoning, not a finding: none has been tried with a resident or a care worker. Each one is configurable per resident, and the defaults are to be confirmed in the first EM-05 pilot and with care workers (Open item O-29).

Table 2.1: Timing parameters (UC04)

Parameter	Controls	Proposed default	Per-resident range	Basis
T-stall	Time without the next expected item event before F7 declares a stall (Action Sequence step 4)	20 s	15 to 45 s	SA-05 records 2 to 5 minutes per episode. About two minutes of that is brushing, so each of the other steps normally takes in the order of 10 to 30 s. Twenty seconds without the expected event is therefore long for a normal step, and short enough for the whole ladder to fit inside one episode.
T-pause	Pause in brushing motion that counts as brushing			

	having stopped, during SA-05 task 7	10 s	5 to 20 s	Short pauses to spit or rinse the brush are part of normal brushing and must not trigger a prompt (VT-11).
W	Silent wait after a detected stall, before the first prompt (step 4)	15 s	0 to 60 s	The wait is the design (OS-04). W is the first value to change per resident if UC04-F8-CL13 holds; W = 0 is the O-16 fallback for a resident who needs prompting at once.
R1, R2	Response window after an orienting (L1) or naming (L2) prompt, before the next level	20 s	15 to 40 s	Long enough to hear one short sentence and act on it. A shorter window would risk outpacing, which HFC-12 names as a detractor.
R3	Response window after the demonstration (L3)	30 s	20 to 60 s	The picture or clip takes longer to take in than a sentence.
N-repeat	Times the same item event repeats without the next expected event before it counts as a stall	3	2 to 5	SA-05 records performing a single step over and over as a breakdown. TECH-11 sees it as the same pick-up and put-down repeating.
B-target	Total brushing time that counts as SA-05 task 7 done	120 s	60 to 180 s, set with ST-12 or a dental professional	The commonly recommended two minutes. Seuntjens et al. (2025) meta-analysed single brushing exercises and found two minutes removed more plaque than one, with a medium effect for manual brushes (SMD 0.69) and a small one for powered brushes (SMD 0.47), and rated the recommendation moderate certainty. Their trials did not involve care-home residents with dementia, so per-resident targets are still agreed with ST-12 or a dental professional (Open item O-32).
T-safe	Time with no item event and no spoken or touch response, once the activity has started, before the safety-critical path (step 10)	180 s	120 to 300 s	DS-02 speaks of several minutes. T-safe only starts after the first item event of the session, because silence before that is normal. With the defaults the ladder runs out about 105 s after the last event, so T-safe comes after a routine handover that has had no response.

With the defaults, a resident who stops and does not respond hears L1 about 35 s after their last action, L2 at 55 s, sees L3 at 75 s, and the care worker is called at about 105 s.

Prompt levels

Level	**M-13 code**	**What the resident gets**	**Channel (TECH-10)**
L0 Wait	0	Nothing. Pepper stays silent with dimmed eye lights.	None
L1 Orienting	1	A sentence about the activity, without naming the action: "Mary, you are brushing your teeth this morning."	Speech
L2 Naming	2	The next action, phrased as an offer (step-specific wording in the next table).	Speech
L3 Demonstration	3	The L2 sentence once more, plus a picture or short clip of the step on the chest tablet. The picture stays on screen until the step is done or the care worker arrives.	Speech and tablet
Handover	4	F9 (see the handover rules below). Pepper tells the resident that someone is coming.	Speech; message to the care worker's phone (TECH-12)

Ladder rules:

1. One level at a time, each given once, with the response window between levels. This is what UC04-F8-PR8 verifies.
2. Any expected item event counts as resuming. Prompting stops, and the next stall, on whatever step, starts again at L1.
3. If the resident says no to two prompts in a row, or says stop, prompting stops and F9 sends a routine handover with the reason "declined". Speech the System does not understand counts as no reply, because F8 must not depend on speech recognition (TECH-10).
4. If the resident asks for help or says they are upset, the remaining levels are skipped (step 8-alt) and F9 sends a priority handover. TECH-11 cannot see distress, so this is the System's weakest detection. Distress the System misses is what the observers measure for UC04-F8-CL13 (M-14).

5. The early-stop case at SA-05 task 7 gets L1 and L2 only, once per session, and never leads to a handover on its own. A resident who stops brushing early and does not take up the offer moves on to rinsing. Calling a care worker over brushing time alone would cost HV-14 and HV-09 more than the extra brushing gains (VT-14).

Table 2.2: Ladder per step (UC04)

SA-05 step	Progress signal (TECH-11)	Stall rule	L2 naming cue (L3 shows the same step on the tablet)	Notes
1 to 3. Recognise it is time, go to the sink, locate the items	First lift on the toothbrush cup or toothpaste pad	Initiation path (step 2-alt): no lift when 10 minutes of the window remain	L1: "Good morning Mary, it is time for your teeth." L2: "Would you like to come and brush your teeth?"	Pepper stands at the bathroom doorway, so a resident elsewhere in the room may not hear it (O-30). No lift by the time the window closes: routine handover, reason "not started".
4. Pick up toothbrush	Toothbrush motion sensor: picked up	T-stall after the first lift, or N-repeat	"Would you like to pick up your toothbrush?"	
5 and 6. Open toothpaste, apply it	Toothpaste lifted from its pad and returned while the brush is held	T-stall after the brush is picked up	"Would you like some toothpaste on your brush?"	TECH-11 cannot confirm paste is on the brush. A lift and return is taken as done.
7. Brush all teeth	Brushing motion; total brushing time counted toward B-target	No brushing motion T-stall after the toothpaste is returned; or brushing stops for T-pause before B-target (early stop)	"You could brush your teeth now." Early stop: "Would you like to brush a little longer?"	Early stop follows ladder rule 5. Brushing time is used within the session only and is not recorded (F10, VT-14).
8. Rinse and spit	Tap flow sensor, or cup lifted	T-stall after brushing ends	"Would you like to rinse your mouth?"	
9. Put things away	Toothbrush back on the cup pad, toothpaste back on its pad	T-stall after rinsing	"Would you like to put the brush back in the cup?"	If the ladder runs out here, nobody is called: the record marks the session incomplete and the care worker sees it at handover.

Handover rules (F9)

Every message carries the same four items, the resident's room, the activity, the step reached and the reason, plus a priority. The room was added after the ER-01 walkthrough found that a care worker with several residents on the round could not tell where to go (WF-07, Open item O-34); it goes only to the care worker who is already responsible for that resident. The priority describes the message, so it adds nothing about the resident to what F10 or TECH-12 holds.

Code	Condition	Priority	Reason sent	Acknowledgement and re-escalation (TECH-12)
H1	Ladder exhausted on steps 4 to 8; initiation path ended with nothing started; resident declined twice	Routine	"no progress", "not started" or "declined"	Not acknowledged within 5 min: re-sent once to the same care worker. Not acknowledged within 10 min: sent to a second care worker.
H2	Resident asks for help or says they are upset	Priority	"asked for help"	Not acknowledged within 2 min: sent to a second care worker.
H3	T-safe elapsed with no item event and no response (step 10)	Urgent. Preempts any prompting.	"no response, possible risk"	Re-sent every 60 s. Not acknowledged within 60 s: sent to a second care worker as well, and through the home's nurse-call system where it can be connected (TECH-12 limitation).
D	The System cannot track the activity: sensor loss, Pepper unavailable, or an unrecognised sequence (exception path)	Information	"System not tracking"	Sent once per session, never re-sent. The care worker falls back to the SA-05 workarounds.

Rules that keep calls rare and meaningful, as HP-05 asks and HFC-14 explains:

- At most one routine handover per resident per session. After H1 the System stops prompting and waits for the care worker.
- No message for an early stop at SA-05 task 7, or for a ladder that runs out at task 9.

- If more than half of a resident's sessions in a week end in H1, the care team reviews that resident's W and response windows, or whether the System suits them at all. At that rate the System is adding calls, not taking them away (VT-13, UC04-F9-CL18).

The priorities and the escalation to a second care worker follow two of the alarm-management recommendations in Cvach (2012): alarm priority levels, and notification systems that can delay and escalate. Sending no message for an early stop at task 7 or a ladder that runs out at task 9 follows its advice to send an alert only when it calls for action.

One row per Premise. Verification only - does the Function operate as specified.

Table 2.3: Premises of UC04

Premise ID	Function	Evaluation Method(s) / Measure(s)	Verification Criteria	Step(s)
UC04-F7-PR7	F7	EM-05, M-11, M-18 (brushing-time component). No human subjects.	Accuracy and robustness. The System identifies the resident's current SA-05 step correctly for at least 90% of staged steps, and detects a stall within T-stall + 5 s of its onset (25 s with the defaults). Repeated item events are recognised as a stall at the N-repeat count. During SA-05 task 7 the summed brushing time is within 10 s of the ground-truth log per session. Proposed values, O-29. On sensor loss or an unrecognised sequence, a degraded state is reported to the care worker rather than the System failing silently.	1, 3, 3-alt, 4, exception path
UC04-F8-PR8	F8	EM-05, M-11 (timing component). No human subjects.	Timing and conformity to specified interaction behaviour. The full wait interval elapses before any prompt is issued. Prompting ascends exactly one level at a time, never skipping a level except on the distress path. Prompting stops within 2 s of the resident resuming. W and each response window are kept to within 2 s of their configured values. No prompt is repeated at the same level more than once. Spurious prompts, given when no stall occurred, stay at or below 1 per 10 staged sessions. The early-stop offer at SA-05 task 7 is given at most once per session and never leads to a handover. Proposed values, O-29. The false-prompt rate matters more than it looks: mistimed prompts were the failure mode observed in COACH (Mihailidis et al., 2008), and are the mechanism by which UC04-F8-CL13 would come true.	4, 6, 7, 8
UC04-F9-PR9	F9	EM-05, M-11. No human subjects.	Timing, correctness, and availability. A handover request reaches a care worker within 10 s of the triggering condition (H1, H2), and within 5 s for the safety-critical path (H3). The request carries only room, activity, step reached, and reason, with its priority. The safety-critical path pre-empts all pending prompting and does not queue behind it. In every staged case where nobody acknowledges, the re-send and second-care-worker escalation happen at the configured times in the handover rules. Proposed values, O-29.	9, 10, 8-alt
UC04-F10-PR10	F10	EM-05, M-16 (record-content component). No human subjects.	Correctness and conformity. The session record contains exactly the three specified items and no step-level trace of the activity. Records are not readable by roles outside the agreed set - ST-10 in particular is not a recipient.	11

Claims of UC04

One row per Claim. Validation - does the Function produce the intended effect. Premise and Claim numbers are drawn from project-wide counters, not reset per Use Case or per Function.

UC04-F8-CL10

Function	F8
Serves Objective	OBJ-11, OBJ-10
Effect	Positive. Increases the number of SA-05 steps the resident completes with no prompt at all, compared with the care worker's current prompt-on-hesitation pattern. Concerns ST-08, SA-05, PS-05.
Grounding	HFC-11; HV-09, HV-12. Precedent: Rogers et al. (1999) for the effect; Mihailidis et al. (2008) for the same outcome measure in a prompting system.

Evaluation Method(s) / Measure(s)	EM-06; M-12, M-14
Validation Criteria	M-12 mean above the same residents' pre-System baseline, with no increase in M-14.
Step(s)	3-8

UC04-F8-CL11

Function	F8
Serves Objective	OBJ-12, OBJ-10
Effect	Positive. Where the resident does stall, the graduated ladder resolves it at a lower level of assistance than current practice, which moves to naming or demonstrating at once. Concerns ST-08, ST-09, SA-05, HV-12.
Grounding	HFC-11. Precedent: Rogers et al. (1999) found significantly fewer physical assists under graduated prompting.
Evaluation Method(s) / Measure(s)	EM-06; M-13, M-12
Validation Criteria	M-13 distribution shifted toward lower levels than baseline, at equal or better completion on M-12.
Step(s)	6-8

UC04-F8-CL12

Function	F8
Serves Objective	OBJ-11
Effect	Positive. Increases the resident's felt sense of doing the activity themselves rather than being led through it. Concerns ST-08, HV-09.
Grounding	HFC-09; HV-09. Ryan & Deci (2000) for the mechanism; Peek et al. (2014) for acceptance factors in this population.
Evaluation Method(s) / Measure(s)	EM-06; M-15 (self-report and observer rating)
Validation Criteria	Both parts of M-15 are required. Holds if the observer's support-of-autonomy rating is higher than the pre-System baseline, and the self-report mean is no lower than baseline. The observer rating carries the Claim; the self-report alone cannot, because resident self-reports show documented ceiling effects (Custers et al., 2013; Open item O-17). If the two disagree, report both and treat the Claim as unresolved.
Step(s)	3-8

UC04-F8-CL13

Function	F8
Serves Objective	OBJ-11, OBJ-12
Effect	**Adverse.** For residents whose current ability falls below a step's demand, the deliberate wait before prompting increases observed frustration rather than producing independent completion. Concerns ST-08, HV-13.
Grounding	HFC-11; HV-13. This tests VT-09 directly. Precedent: mistimed and false-alarm prompts in COACH (Mihailidis et al., 2008) were observed to produce general frustration and confusion - the same mechanism CL10 relies on, running the wrong way.
Evaluation Method(s) / Measure(s)	EM-06; M-14, M-13
Validation Criteria	Holds if M-14 (Resistiveness to Care Scale) rises above baseline for the lower-ability subgroup while M-12 does not improve for them. If it holds, the wait interval must become per-resident rather than configured once.
Step(s)	4, 6

UC04-F8-CL14

Function	F8
Serves Objective	OBJ-14
Effect	**Adverse.** Being prompted through tooth brushing by a robot is experienced as infantilising, independently of whether it improves completion. Concerns ST-08, HV-10.
Grounding	HFC-12, HFC-13; HV-10. Kitwood (1997), who names infantilisation and outpacing among the detractors that erode personhood in dementia care. Robot-specific: Lee et al. (2016) on ageing stereotypes in assistive robot design; Coghlan et al. (2021), whose older participants (living at home, no dementia) called companion robots "patronizing" when they were toy-like or imitated human companionship; and Furlanis et al. (2026), whose "embodied treatment" dimension covers the gaze, timing and tone F8 controls - including a case of exaggerated gentleness toward older adults read as condescension.
Evaluation Method(s) / Measure(s)	EM-06, EM-07; M-14, M-16
Validation Criteria	Holds if residents (EM-06 post-session interview) or care staff (EM-07) describe the prompting as talking down to the resident or as child-like (M-16 code T), or if M-14 rises above baseline during prompting. Tests VT-10. Sensitive to §1.3: Pemberton et al. (2026) suggest an expressive humanoid form would raise this risk relative to an appliance-like one, so a negative result on one embodiment does not clear the other. The selected embodiment is Pepper (TECH-10), the higher-risk form; TECH-14 is the fallback if this Claim holds.
Step(s)	6-8

UC04-F9-CL15

Function	F9
Serves Objective	OBJ-10, OBJ-12
Effect	Positive. Reduces the share of sessions that end incomplete, by bringing a care worker in when prompting has run out instead of the activity being abandoned. Concerns ST-08, ST-09, SA-05, PS-05.
Grounding	HV-13; HFC-11
Evaluation Method(s) / Measure(s)	EM-06; M-12
Validation Criteria	Fewer incomplete sessions than baseline. (The time effect was split out into UC04-F9-CL18; Open item O-18.)
Step(s)	9, 10, 8-alt

UC04-F9-CL18

Function	F9
Serves Objective	OBJ-15
Effect	Positive. Reduces the care worker's prompting time per resident during the morning round, because the System carries the waiting and the lower prompt levels and the care worker is called only when needed. Concerns ST-09, SA-07, HV-14.
Grounding	HV-14; HFC-11, HFC-14. At risk: Rogers et al. (1999) found graduated prompting took considerably *more* caregiver time than usual care - absorbing that time is the point of the System, but it is an empirical question. Tests VT-13.
Evaluation Method(s) / Measure(s)	EM-06, EM-07; M-17
Validation Criteria	M-17 total care-worker prompting time per session below the pre-System baseline, with completion (M-12, CL15) no worse. Holds only if both are true: saving time by letting sessions go unfinished does not count.
Step(s)	9, 8-alt

UC04-F7-CL16

Function	F7
Serves Objective	OBJ-13
Effect	Positive. The resident does not experience the step tracking as being watched, because it requires nothing of them and observes the activity rather than the person. Concerns ST-08, HV-11.
Grounding	HFC-10, HFC-13; HV-11. Niemeijer et al. (2010, 2015) - residents can experience surveillance technology as supporting autonomy, but only within a person-centred approach. Pemberton et al. (2026) sharpens the risk: older adults assigned humanoid robots the "Warden" role - oversight without attachment - which is this Claim failing by design rather than by accident.
Evaluation Method(s) / Measure(s)	EM-06, EM-07; M-16
Validation Criteria	No resident expressions of being watched in M-16 coding (code W, EM-06 post-session interview), and care staff judge the retained record proportionate (EM-07). Sensitive to §1.3 on the same axis as CL14: Pepper's cameras are disabled and covered, but physically present.
Step(s)	1, 3

UC04-F10-CL17

Function	F10
Serves Objective	OBJ-13, OBJ-15
Effect	Positive. The three-item session record gives the care worker enough to adjust the support they offer, without exposing how the resident performed a personal-care activity step by step. Concerns ST-09, ST-08, HV-11.
Grounding	HV-11; VT-08 resolution; Niemeijer et al. (2010) on proportionality of what is recorded.
Evaluation Method(s) / Measure(s)	EM-07; M-16
Validation Criteria	Care staff report the record sufficient for handover and identify nothing in it they consider intrusive.
Step(s)	11

UC04-F8-CL19

Function	F8
Serves Objective	OBJ-10
Effect	Positive. The early-stop offer at SA-05 task 7 makes the resident brush for longer than at baseline, and so leaves less plaque. Concerns ST-08, SA-05, and PS-05's completion sub-problem (stopping brushing too early).
Grounding	HFC-11; HV-13, HV-12. Group reasoning from PS-05's completion sub-problem. Seuntjens et al. (2025) support the step from longer brushing to less plaque, in single brushing exercises outside this population. Nearest precedent: in MOUTh (Jablonski et al., 2018), a relationship-based intervention for nursing-home residents with dementia who resisted mouth care, care staff achieved longer mouth care than controls (d = 0.56) but only a small differential improvement in oral health (d = 0.18). Care staff delivered that mouth care, so no published precedent has been found for prompting the resident to brush longer themselves (Open item O-32). MOUTh's small oral-health effect is also why this Claim requires plaque to fall, and not brushing time alone.
Evaluation Method(s) / Measure(s)	EM-06; M-18, M-14, M-16
Validation Criteria	Holds if brushing time (M-18 part b) is above the same residents' baseline *and* plaque (M-18 part a) is below it, with no rise in M-14 and no T codes about the brushing offer in M-16. If brushing time rises and plaque does not fall, the Claim is unsupported, since time spent is only a proxy for how well the teeth were cleaned.

Step(s)	7, 3-alt

UC04-F9-CL20

Function	F9
Serves Objective	OBJ-10, OBJ-15
Effect	**Adverse.** Because the care worker is now absent for most of the routine, brushing quality falls even where independent step completion rises. The System counts SA-05 task 7 as done once brushing motion has occurred, while a care worker standing beside the resident would also have seen how well they brushed. Concerns ST-08, ST-09, HV-13.
Grounding	HV-13; VT-13, VT-14. Raised by the midterm feedback, which asked how well the teeth were actually brushed. M-12 cannot detect this effect, because it scores step completion only.
Evaluation Method(s) / Measure(s)	EM-06; M-18, M-12
Validation Criteria	Holds if plaque (M-18 part a) is higher than the same residents' baseline during the System phase. If it holds while UC04-F8-CL10 also holds, the System is trading oral health for independence. The first response is to raise B-target for those residents; the second is a periodic brushing check by the care worker, outside the System.
Step(s)	7, 9

UC05 Support the resident through the medication routine

Evidence basis	Would formalize SA-06 and PS-06. No primary data collected by this project.
Scope decision	SA-06 and SA-05 have different triggers and different Goals, so this is a separate Use Case rather than an alternate flow of UC04 - it is not a candidate for merging. It is deferred for this cycle rather than dropped. Reasons: • PS-06's out-of-scope field already narrows it heavily, including a care-home blacklist for residents on critical medication, which needs settling before Functions can be written. • PS-06 adds a fourth sub-problem, confirmation, that has no counterpart in PS-05 - the resident failing to confirm they have taken the medication, risking a double dose. That needs its own Function and its own Claim, and possibly a safety Premise, rather than reusing F7-F10. • Specifying it properly would roughly double the Premise and Claim count, and the medication activity will need Measures of its own on the §1.2 Measures page, numbered from M-25. F7-F10 are expected to transfer, since both activities are stepwise sequences with initiation, sequencing and completion breakdowns. Confirming that recurrence across two Use Cases is what would justify abstracting the TDP and IDP in Steps 12-13 - so UC05 should be written before those are finalized, not after. Group decision (Open item O-22): tooth brushing is this cycle's spine. Material already written for medication is kept for UC05 rather than discarded: DS-03 (§2.1), the tray or dispenser option TECH-15 (§1.3), and the medication-timing tension VT-12 (§1.2).

2.3.3 Open Items

These are gaps recorded as explicit decisions rather than left silent, per Step 14.

ID	Item	Blocks	Owner
O-10	Resolved. M-11…M-18 are in §1.2.4 and EM-05…EM-07 in §1.2.5. New Measures start at M-24; M-19 to M-23 belong to the main concept, and new Evaluation Methods start at EM-09 because EM-08 belongs to the main concept.	Nothing	Closed
O-11	Resolved (group decision: in scope). Notifying a care worker on immediate risk of serious harm stays in F9 and step 10. VT-07's non-negotiable floor now says "none identified", and the escalation rule has moved into its current resolution, since it is a design decision, not a regulation. PS-06's out-of-scope now excludes only emergency *treatment*. Because there is no camera, a possible emergency is inferred from no sensor change and no response (TECH-11 limitation, step 10).	Nothing	Closed

O-12	Resolved. SA-07 (the care worker's morning round) now records the care-continuity and handover task, and its Goal grounds the new OBJ-15, which F10 (and F9) serve.	Nothing	Closed
O-13	Resolved. UC04-F8-CL14 now grounds on HFC-12 (Personhood and malignant social psychology, Kitwood 1997) in §1.2, not on a Claim-level citation.	Nothing	Closed
O-14	Resolved. §1.3 now documents six Technology Options: Pepper (TECH-10), item sensors (TECH-11) and staff notification (TECH-12) selected; camera-based recognition (TECH-13) and a tool-like sink device (TECH-14) rejected; the medication tray (TECH-15) deferred. F7-F10's "Enabled by" clauses now cite them. Numeric thresholds are still open - see O-29.	Nothing	Closed
O-15	Resolved. The §1.1 Stakeholders page and the §2.2 Personas (HP-04, HP-05) now use the §1.2 Human Value numbering, and both Personas now elaborate all five values, including HV-12 Competence, which OBJ-11 and CL10-CL11 rest on.	Nothing	Closed
O-16	F8's direction is contested by the literature. F8 is least-to-most: allow the stall, prompt minimally, then escalate. The dementia ADL literature largely favours the opposite - errorless learning with vanishing cues, prompting early to prevent the error and fading support across sessions (de Werd et al., 2013). The largest trial of that approach, REDALI-DEM (Voigt-Radloff et al., 2017; n = 161, nine 1-hour sessions over eight weeks), found that structured relearning improved ADL performance and that improvements held for six months - but that errorless learning gave no additional effect over trial-and-error learning. Note that de Werd is an author on both the review and the trial that failed to confirm it. Two things follow. First, the evidence neither endorses F8's direction nor rules it out. Second, REDALI-DEM's participants had Alzheimer's or mixed-type dementia and were living at home, so even that null result transfers to care-home residents only cautiously. Either F8 is re-specified as most-to-least with fading, or it stays as a deliberate bet that UC04-F8-CL13 is designed to settle - which has to be stated as a bet rather than presented as established. Resolved (group decision: keep least-to-most, as a stated bet). F8's section now says so explicitly, names CL13 as the Claim that settles it, and states the fallback if CL13 holds: first a per-resident wait interval, then most-to-least with fading.	Nothing	Closed
O-17	Resolved. UC04-F8-CL12 now requires both parts of M-15: the observer's support-of-autonomy rating carries the Claim, the self-report must be no lower than baseline, and a disagreement between them leaves the Claim unresolved. M-15's items and scoring are now recorded from the source (Custers et al., 2013). Background: self-report of autonomy in this population is unreliable, and this is measured, not just suspected. Custers et al. (2013) found residents rated their own need fulfilment higher than trained observers did, with only weak-to-moderate agreement. Remaining caveat: that sample was physically frail residents (n = 36, mean age 80), not specifically residents with dementia.	Nothing	Closed
O-18	Resolved. UC04-F9-CL15 is now only the completion effect; the time effect is UC04-F9-CL18 (OBJ-15, HV-14, M-17), which holds only if care-worker time falls *and* completion is no worse.	Nothing	Closed
O-19	Resolved. The earlier TODO asked whether being prompted by a *robot* specifically, rather than a screen or a voice, changes the infantilisation effect in CL14. It does. Furlanis et al. (2026) argue that social robots are "socially legible bodies, interpreted in context through posture, timing, tone, and spatial behavior", and give three dimensions along which bias is enacted: embodied stereotypes (appearance), embodied constraints (affordances), and embodied treatment - bias arising in real-time interaction through gaze, timing and tone. F8's prompt pacing falls in the third. Their illustrative case includes a robot that exaggerated slowness and gentleness toward older adults, which was read as condescending rather than accommodating, which is CL14's mechanism. Note this paper is explicitly conceptual, so it frames the risk rather than measuring it. The empirical source they cite for this is Lee et al. (2016), "That robot is not for me: Addressing stereotypes of aging in assistive robot design", now read. Lee et al. interviewed 7 independently living older adults aged 57 to 72, who were shown photos and videos of assistive robots. All 7 said they did not identify with the image of older users the robots were designed around, and most saw the robots as useful for others but "not for me". The paper studies stereotyped representations in robot design and use scenarios. It does not study a robot's behaviour during an interaction, so it supports the stereotype side of CL14 but does not test the condescension mechanism itself. Coghlan et al. (2021) are closer to that mechanism. They interviewed 16 older adults (65+) living alone, who were shown videos of a talking assistant (ElliQ), a toy-like robot (Vector) and a robot dog. A number of participants called the robots "patronizing" or "condescending": Vector for being toy-like, and ElliQ because it imitated human conversation and companionship. Their participants were also cognitively intact and living at home, and the robots were companions, not care assistants. Form also changes the role the robot gets assigned. Pemberton et al. (2026) interviewed 13 older adults and found they sort socially assistive robots into four roles by autonomy and relationship tolerance - Carer, Pet, Warden, Tool. Humanoid robots were assigned the "Warden" role: oversight without attachment. Older adults preferred "tool" or "pet" framings over "friend" or "assistant" to		

	protect dignity and autonomy. Coghlan et al. (2021) report a related pattern: some participants who found the talking assistant controlling preferred the robot dog, because they would be in charge of it. Pemberton et al. also cite Prakash & Rogers (2015) for a preference for mechanical-looking robots in private contexts such as bathing. The original paper, now read, supports this only in part. In its rating task (32 younger and 32 older adults), a robotic face was rated more likeable than a human face for a personal-care task ($p = 0.01$), but it was not rated more trustworthy or more useful. In the interviews, participants were divided: some wanted a human-looking robot because of the "care" in personal care, and others wanted an "impersonal looking creature" because the task is private. Across all four tasks, more older adults preferred a human face than a robotic one, although 37% leaned toward the least human-looking faces. Tooth brushing is a personal-care task, so the finding applies, but it shows a split preference with a modest lean toward the mechanical face. So CL14 and CL16 are both sensitive to the §1.3 choice, and in the same direction: a humanoid, expressive form raises the infantilisation risk in CL14 *and* the being-watched risk in CL16. See Open item O-20. Transfer limits: Pemberton et al. studied cognitively intact older adults, in a VR home setup, ageing in place - not care-home residents with dementia.	CL14, CL16; feeds §1.3	Closed
O-20	Arising from O-19. The evidence points somewhere specific on the open embodiment question: for a personal-care activity, a less humanlike, appliance-like form that preserves emotional distance is better supported than an expressive humanoid one. Correction after reading the originals: this conclusion rests on Pemberton et al. and, more weakly, on Prakash & Rogers (2015), whose personal-care result is mixed (see O-19). An earlier version also cited Zsiga et al. (2013), via Pemberton et al., for older adults naming overreach in control and cognitive support as key barriers. The original does not report that. It is a brief focus-group report on a home companion robot, with 11 potential end users aged 77 to 85 plus caregivers, in Austria, France and Hungary. Its closest findings are one-line entries in its results table: French end users listed wanting to keep their autonomy as an obstacle, and French caregivers said the caregiver should not be bypassed and someone should keep control over the robot. It also reports that opinions differed on whether the robot should look humanoid, with the Hungarian older adults preferring a more humanoid robot. Two actions: (a) §1.3 should record this as the rationale for whichever Technology Option is selected, and as the reason for rejecting a humanoid one; (b) §1.2 should add an HFC on embodiment and perceived role, so CL14 grounds on a concept rather than on Claim-level citations. That would also close the remaining half of O-13. Resolved, with a recorded tension. (a) §1.3 records this evidence: the tool-like device (TECH-14) is best supported, but the group selected Pepper (TECH-10) as the project's platform. The tension is stated on TECH-10 with mitigations - tool framing, low expressivity, demonstration on the tablet, cameras disabled and covered - and TECH-14 is the named fallback if CL14 or CL16 hold. (b) Added HFC-13 (Embodiment and perceived role), which CL14 and CL16 now ground on.	Nothing	Closed
O-21	Resolved (group decision: add a resident interview). EM-06 now ends each System session with M-15's items and three short, verbally administered M-16 questions, asked only if the resident is willing and not distressed. CL14 and CL16 now say which part of M-16 comes from EM-06 and which from EM-07.	Nothing	Closed
O-22	Resolved (group decision: tooth brushing is the spine). New DS-02 in §2.1 reimagines PS-05 with the selected System; DS-03 (medication) is kept, labelled as UC05 material. HP-04 and HP-05 now centre on DS-02. RP-02 is rewritten as Pepper at the bathroom doorway, with tracking from item sensors (TECH-11) - so the Robot Profile can now perform F7. The tray device is kept as TECH-15 (deferred).	Nothing	Closed
O-23	Resolved. Added SA-07 (morning care round, performed by ST-09), HV-14 Attentive care (held by ST-09, ST-11, ST-10), VT-13 Competence vs Attentive care, and OBJ-15. ST-09's values are now its own, not the resident's list. CL18 now has a baseline to compare against (M-17).	Nothing	Closed
O-24	Resolved. Added VT-10 Dignity vs Competence (tested by CL14), VT-11 detection reliability (tested by PR8 and CL13), and VT-12 medication timing (time-criticality, deferred with UC05). VT-13 also covers time-criticality for UC04. VT-08's "nature of the conflict" now describes privacy vs well-being, and VT-09's domain area is corrected.	Nothing	Closed
O-25	Resolved. PS-05's out-of-scope field now gives four exclusions, each with a reason, and treats medication as deferred to UC05 rather than excluded.	Nothing	Closed
O-26	Resolved (group confirmed: no primary data yet). Every Evidence basis in §1.1 and §1.2 now says literature and group reasoning only, flagged provisional. Stakeholder relationships are written per Stakeholder, each Stakeholder lists only the values it holds, with a priority, and HV "Held by" fields match. Value sources now cite specific works (Ryan & Deci, 2000; Friedman et al., 2008; Kitwood, 1997; Niemeijer et al., 2010).	Nothing	Closed

O-27	Resolved. §1.4 was written from the Foundation pages as "d. Foundation Coherence" under 1. Foundation: edge inventory, open tensions with stance, completeness check. Writing it found 13 edges recorded on one page and missing on the other, and two orphaned Technology Options; all were fixed on the source pages except TECH-15, which waits for UC05.	Nothing	Closed
O-28	Resolved. M-14's scoring (duration × intensity per item, total 0-156) and M-15's items and scales are recorded from the sources. Each observer- or coder-rated Measure (M-12, M-13, M-16, M-17) now states a reliability plan: at least 20% double-coded, kappa or ICC ≥ .60 (Landis & Koch, 1977). M-16 has explicit codes (W, R, T, N; P/I).	Nothing	Closed
O-29	Premise thresholds and escalation timing could not be set from the literature for item sensors (TECH-11). Resolved provisionally. The *Escalation specification* now proposes every timing parameter (T-stall, T-pause, W, R1 to R3, N-repeat, B-target, T-safe) and the handover acknowledgement times, and PR7 to PR9 carry proposed thresholds. All of them are group reasoning, labelled as proposals. What remains: run the first EM-05 pilot at a mock sink, adjust the values from it, and review W, the response windows and the acknowledgement times with care workers (HP-05). The pilot script also includes two cases from ER-01: recognition of the words yes, no, stop and help (WF-05), and rinsing or spitting during brushing, to settle whether that ends SA-05 task 7 or is a pause inside it (WF-10).	Confirmation of PR7 to PR9 thresholds only	Group
O-30	Pepper (TECH-10) is about 1.2 m tall on a wheeled base and not waterproof. Whether it can stand at a care-home bathroom doorway within earshot of the sink, without blocking the resident's path, needs checking on site or with a mock-up. If it cannot, the fallback is TECH-14.	Pre-condition of UC04; RP-02's embodiment	Group
O-31	HP-05 needs the System's calls to be "rare and meaningful", but no Human Factors Concept covered alert fatigue, so TECH-12's alerting policy had nothing to ground on. Resolved. HFC-14 (Alert fatigue) is added in §1.2. The handover rules in the *Escalation specification* define four message types with priorities, acknowledgement times, one re-send and escalation to a second care worker, a limit of one routine call per resident per session, and a weekly review when calls become frequent. TECH-12 now points to these rules, and M-17 counts notifications. HFC-14's two sources have since been verified against PubMed (O-33); Cvach (2012) has been read in full and Sendelbach & Funk (2013) as an abstract.	Nothing	Closed
O-32	Arising from the midterm feedback, which asked how well the teeth were actually brushed. Brushing quality is now covered by M-18, UC04-F8-CL19, UC04-F9-CL20, VT-14 and TECH-16/TECH-17. Status of the three parts: (a) Source found: Seuntjens et al. (2025) support two minutes over one, in single brushing exercises. Still open: agree per-resident targets with ST-12 or a dental professional. (b) Resolved. CL19 has no direct precedent. The nearest is MOUTh (Jablonski et al., 2018), where care staff, not the resident, lengthened mouth care; CL19 now cites it. (c) Both papers read. Proposed: the Plaque Score of MPS (Henriksen et al., 1999), with its scoring scheme now on the Measures page. OHAT's plaque item agreed poorly with a dentist's Plaque Index (Chalmers et al., 2005). Still open: the group confirms the choice, and the calibration manual is obtained from the authors or rebuilt with a dental hygienist.	UC04-F8-CL19; M-18	Group
O-33	Resolved. The four sources added for O-31 and O-32 (Cvach, 2012; Sendelbach & Funk, 2013; Henriksen et al., 1999; Chalmers et al., 2005) have each been checked against PubMed: title, authors, journal, year, volume, issue and pages all match, and DOIs and PMIDs are added to the References. Cvach (2012), Henriksen et al. (1999) and Chalmers et al. (2005) have since been read in full and their metadata matches the PDFs. Sendelbach & Funk (2013) has been read as an abstract only, because the full text could not be obtained.	Nothing	Closed
O-34	Found by the ER-01 walkthrough (WF-07): the handover message said what and why, but not which resident or room, so a care worker with several residents on the round could not tell where to go. Resolved: the room is now a fourth item of every handover message, in F9, step 9, the handover rules, UC04-F9-PR9, DS-02, TDP-02 and TECH-12. Steps 9 and 10 should be walked again on the revised text.	Nothing	Closed
O-35	Resident-facing prompt script, from the ER-01 walkthrough. Five changes are proposed, none applied yet: (a) WF-01: no stall rule covers a resident who reaches the sink and lifts nothing. Proposed: a non-camera presence signal at the sink, for example a floor or door sensor added to TECH-11, so that T-stall starts on arrival. (b) WF-02: the L1 sentence "you are brushing your teeth this morning" is used at every step, and after brushing it can send the resident back to the brush. Proposed: a step-aware orienting sentence.		

	(c) WF-04: the L3 picture is on Pepper's tablet behind the resident, and the sentence does not point to it. Proposed: add "There is a picture on my screen", or show the picture where the resident is looking (depends on O-30); separate L3 pictures for opening and applying toothpaste. (d) WF-06: after "someone is coming", nothing more is said for up to 10 minutes. Proposed: repeat the reassurance once at the re-send time, and say when the care worker has acknowledged (needs O-37). (e) WF-12: DS-02 ends with "All done", which the specification lacks. Proposed: add a closing line without praise to the prompt levels.	AT-02 Build A prompt script; EV-01, EV-02	Group
O-36	How the System hears the resident, from ER-01 (WF-03, WF-05). Ladder rules 3 and 4 need it to recognise "no", "stop" and requests for help, while rule 3 says F8 must not depend on speech recognition, and RP-02's touch input cannot be discovered by the resident. Two quick "no"s can also end prompting when the resident would have gone on. Proposed: recognise a fixed set of four words (yes, no, stop, help) and treat everything else as no reply; count a "no" as declining only if no item event follows in the response window; drop touch as a help input or tell the resident about it. UC04-F8-PR8 would then add a recognition-rate check in EM-05 (O-29).	Ladder rules 3 and 4; UC04-F8-PR8	Group
O-37	From ER-01 (WF-08): the handover rules time the care worker's acknowledgement, but no acknowledgement action is specified. Proposed: a one-tap acknowledgement on the TECH-12 message, which also lets Pepper tell the resident that the care worker is on the way (O-35 d).	TECH-12; handover rules	Group
O-38	From ER-01 (WF-09), Evidence EV-05: the record's "highest prompt level" does not say at which step it was needed, so it may not be enough for the care worker to adjust support, which UC04-F10-CL17 claims. Adding the step would add detail about how a personal-care activity was performed (VT-08, HP-04's constraint). Group decision: keep three items and let EM-07 judge sufficiency, or add the step and re-check VT-08.	UC04-F10-CL17; F10; UC04-F10-PR10	Group
O-39	ER-01 is one evaluator's pass. The walkthrough procedure asks for 3 to 5 independent evaluators who compare their lists in a joint session; ideally one has care experience. Until then, EV-01 to EV-07 rest on a single reviewer's judgement.	Finalising ER-01 and EV-01 to EV-07	Group
O-40	Decisions ER-02 needs before an ethics application: (a) Ask the TU Delft HREC whether the study counts as research under the Dutch WMO, because it adds a plaque check and changes the care routine of adults who cannot consent. If it does, an accredited METC must review it too. (b) Confirm the non-inferiority margins proposed in ER-02: M-14 no more than 5 points higher, M-15 self-report no more than 0.5 points lower, and the incomplete share no more than 10 percentage points higher. (c) Confirm the target of 8 to 10 residents. With Holm's correction over six tests, fewer than 8 completing residents cannot give a significant confirmatory result.	ER-02 (study start)	Group

2.4 Team Design Pattern

TDP-02 Graduated activity support with human handover

Evidence basis

Literature and group reasoning only. No residents or care workers have been interviewed or observed by this project, and the pattern has not yet been evaluated.

Rogers et al. (1999) found that systematic graduated prompting during nursing-home morning care increased independent task performance and reduced physical assistance. The prompts in that study were delivered by care staff, so it supports the graduated-assistance structure but not the allocation of the lower prompt levels to a robot.

Mihailidis et al. (2008) evaluated COACH, an automated prompting system for a comparable stepwise activity, handwashing, with older adults with moderate-to-severe dementia. It supports assigning activity tracking and routine prompts to the System as a way to reduce direct caregiver involvement, while its mistimed prompts and tracking failures show why a care-worker fallback is required. It did not test Pepper, tooth brushing, item sensors, or the TECH-12 handover channel.

Peters et al. (2014) evaluated TEBRA, an automated system for supporting tooth brushing through task recognition and a hierarchy of prompts. This is a direct precedent for assigning routine tooth-brushing support and graduated prompting to the System. Its participants had moderate cognitive disabilities rather than dementia, and TEBRA did not use Pepper or the proposed item sensors and handover channel.

Wang et al. (2017) studied robot-delivered stepwise prompts for handwashing and tea making with ten older adults with early- to middle-stage Alzheimer's disease and their caregivers. The study makes robot-delivered activity support plausible, but participants' views of having a robot were mixed. It therefore supports investigating this allocation rather than assuming acceptance.

Braley et al. (2019) found that some people with dementia, especially participants with moderate dementia, showed confusion or frustration during repeated automated prompting and benefited from communication with the researchers. This supports a human handover when prompting becomes unresolved or stressful; it does not establish the proposed System's handover thresholds.

The project instantiation is derived from SA-05, SA-07, PS-05 and UC04. The resident retains the activity tasks, the System receives F7-F10, and the care worker retains reassurance, physical assistance and exception handling. This allocation remains provisional until it is evaluated through the Claims listed below.

Problem

A resident with dementia may lose their place in a familiar activity, while a care worker cannot continuously guide every step. If the System takes over too soon, it removes an opportunity to act independently; if it waits or prompts indefinitely, the resident may become distressed or the activity may remain incomplete. The team needs an explicit rule for what the resident attempts, what the System may prompt, and when a care worker assumes responsibility.

Solution Structure

1. Resident performs the activity.
2. System monitors task state (F7) and provides graduated assistance according to IDP-04 (F8).
3. System requests care-worker support (F9) when prompting fails, task-state uncertainty is high, stress or confusion appears, or physical or safety help is needed.
4. Care worker assesses the situation and provides the appropriate support.

If F7 cannot reliably determine progress, the System reports a degraded state and the care worker uses the existing SA-05 workarounds. If there is no sensor change and no response for the configured safety interval, the System follows UC04's agreed safety branch and calls the care worker. This is a conservative inference rather than direct emergency detection: TECH-11 cannot see distress or a collapse.

Solution Description

The resident remains the primary performer of the activity. Item sensors (TECH-11) track activity progress without filming the resident. The System provides graduated assistance according to IDP-04. The System hands responsibility to the care worker when prompting fails, task-state uncertainty is high, stress or confusion appears, or physical or safety help is needed. TECH-12 sends the handover and stores a minimal record for care continuity. TDP-02 defines who is responsible; IDP-04 defines how System prompting escalates.

Per-Actor allocation

Resident (ST-08):

1. Initiate and carry out the activity steps they can still perform (SA-05 tasks 1-9).
2. Continue after a prompt when able; no explicit response to the System is required for ordinary step tracking.

Care worker (ST-09):

1. Perform the wider morning-care round (SA-07) while the resident attempts SA-05.
2. Acknowledge an F9 message on the work phone; assess the resident and provide the next appropriate support, including reassurance or physical assistance when needed.
3. Review F10's three-item session record for care continuity and handover.
4. Use the existing care process if tracking, Pepper, or the notification channel fails.

System:

1. F7 through TECH-11: infer the current activity step and whether progress has stalled, limited to item events from this activity.
2. F8 through TECH-10: provide graduated assistance according to IDP-04.
3. F9 through TECH-12: request care-worker assistance when prompting fails, task-state uncertainty is high, stress or confusion appears, or physical or safety help is needed, passing only room, activity, step reached and reason.
4. F10 through TECH-12: store completion, highest prompt level, and whether a care worker was called; retain no step-by-step trace.

Consequences: benefits	Gives the resident a chance to complete steps independently; may resolve a stall without immediately requiring a care worker; provides a route to human assistance when System support is insufficient; may direct care-worker time toward residents who need it; provides a minimal handover record. These are expected effects, not observed outcomes (UC04-F8-CL10, UC04-F8-CL11, UC04-F9-CL15, UC04-F9-CL18, UC04-F10-CL17).
Consequences: risks	An incorrect item-sensor inference may interrupt a resident who was progressing. Waiting too long or escalating the wrong prompt may increase frustration (UC04-F8-CL13). Pepper's humanoid presence, and camera lenses that are covered but still visible, may feel supervisory (UC04-F7-CL16). A separate phone channel can create alert burden; the handover rules in UC04's Escalation specification limit it, grounded on HFC-14 (alert fatigue). Because the care worker is absent for most of the routine, brushing quality may fall even when steps are completed (UC04-F9-CL20). TECH-11 cannot directly see distress or danger. The escalation values and premise thresholds are proposals awaiting the EM-05 pilot (O-29), and Pepper may not fit safely at a bathroom doorway (O-30).
Related Patterns	IDP-04 Respectful graduated prompting shapes F8's interaction. The existing Human-Robot Group Game Management pattern concerns a different activity and does not validate this allocation.
Connection to Claims	Supports the design rationale for UC04-F8-CL10 (independent steps), UC04-F8-CL11 (lower assistance level), UC04-F9-CL15 (completion after handover), UC04-F9-CL18 (care-worker prompting time), and UC04-F10-CL17 (useful but proportionate record). UC04-F8-CL13 tests an adverse effect of waiting; UC04-F7-CL16 tests whether the tracking allocation preserves privacy as intended.
Validation path	Literature-grounded through HFC-11, Rogers et al. (1999), Mihailidis et al. (2008), Peters et al. (2014), Wang et al. (2017), and Braley et al. (2019). The linked Claims are operationalized by M-12, M-13, M-14, M-16 and M-17 through EM-06 and EM-07. Only one project instantiation is currently specified. UC05 will show whether the allocation recurs or whether medication confirmation and safety require a different pattern.

Table 2.4: Open items to check or research (TDP-02)

Item	What is needed	Why it matters
Premise thresholds (O-29)	Proposed values are now in UC04's Escalation specification and Premises. Pilot EM-05 at a mock sink and review the values with care workers.	Until the pilot confirms them, responsibility has been allocated to Functions whose dependable operating range is only estimated.
Pepper placement (O-30)	Test whether Pepper can stand within earshot and sight of the sink without entering the wet area or blocking the resident. Use TECH-14 if it cannot.	May change which device performs F8 and the resident-facing part of F9.
Alert policy (O-31)	HFC-14 and the handover rules are added, and HFC-14's sources are verified (O-33). Review acknowledgement times with care workers.	Needed to justify TECH-12 and to avoid moving workload from prompting to notification management.
Research: transfer and recurrence	Check how the COACH, TEBRA, Wang and Braley findings transfer to residents with dementia using Pepper and item sensors; complete UC05 far enough to compare actor responsibilities.	Tests the evidence transfer, acceptance assumptions and whether this is reusable beyond UC04.
Stakeholder validation	Review the allocation, fallback and record with residents and care staff if ethically feasible; otherwise retain the explicit literature-only limitation.	HP-04, HP-05, SA-07 and the alert assumptions are provisional.

2.5 Interaction Design Pattern

IDP-04 Respectful graduated prompting

Evidence basis

Literature and group reasoning only. No residents or care workers have been interviewed or observed by this project, and the interaction has not yet been evaluated.

The pattern is derived from SA-05's existing reminder and demonstration workarounds, PS-05, DS-02 and UC04 F8. HFC-09, HFC-11, HFC-12 and HFC-13 ground the principles concerning autonomy, adaptive assistance, personhood and robot embodiment. TECH-10 and RP-02 provide the selected modality: Pepper speech and tablet content

	with restrained movement. The cited literature does not establish that the proposed wording, timing or embodiment works for the intended residents.
Problem	In a stepwise activity, a resident may hesitate because they need time, have forgotten the next action, or are unable to continue. A prompt that arrives too early, repeats, or sounds childlike can remove control and dignity. A prompt that arrives too late or is too vague can leave the resident confused or distressed. The System needs a consistent way to offer help and to stop when help is no longer needed.
Principle	Match the specificity and timing of assistance to the person's present ability, while preserving a meaningful opportunity to act independently. Communicate with an adult respectfully and make assistance an offer, not a judgment about failure. The group selected wait-then-escalate, from least to most specific, as a stated design hypothesis. If UC04-F8-CL13 shows increased distress, first adapt the wait per resident and then consider earlier prompting with fading cues.
Solution	1. Remain silent while TECH-11 indicates that the resident is progressing. 2. After a reliably inferred stall, allow an individualized wait before prompting. UC04's Escalation specification proposes 15 s by default, adjusted per resident after the O-29 pilot. 3. Begin with one short, calm, low-information prompt. Phrase it as an offer in the second person and in an ordinary adult register; do not announce failure or use exaggerated praise. Confirm with residents whether using their name is welcome in this setting. 4. If needed, move to a more direct prompt and then a simple picture or short video on Pepper's tablet, one level at a time with a response window between them (UC04 Escalation specification). Avoid indefinite repetition; if there is no response, escalate, reassess, or hand over. If the resident declines twice, stop. Pepper does not mime the action with its arms. 5. Stop prompting as soon as the item sensors indicate that progress has resumed. 6. Hand over under F9 if the difficulty remains unresolved, stress or confusion appears, task-state uncertainty is high, or physical or safety help is needed. On unreliable sensor data, do not guess; report the degraded state. 7. Keep Pepper visibly tool-like during the activity: cameras disabled and covered, minimal head and arm motion, no gaze-following, soft eye lights rather than alarms, and no conversation beyond the routine. Position it at the doorway only if the O-30 mock-up shows that this is audible, visible and physically safe.
Research Link	HFC-09 links autonomy and perceived control to acceptance, citing Ryan & Deci (2000) and Peek et al. (2014). HFC-11 cites Rogers et al. (1999) for graduated assistance in nursing-home morning care and Mihailidis et al. (2008) for automated prompting during handwashing; it also records mistimed prompts as a possible source of confusion. Peters et al. (2014) provide a direct tooth-brushing precedent in TEBRA, including repeated prompts within a hierarchy of increasingly informative assistance. Wang et al. (2017) show that robot-delivered stepwise prompting is plausible for people with Alzheimer's disease while acceptance of a robot varies. Braley et al. (2019) support leaving automated prompting when confusion or frustration appears and restoring human communication. HFC-12 uses Kitwood (1997) for infantilisation and outpacing. HFC-13 covers how embodiment, gaze, timing and tone affect the role assigned to an assistive robot, drawing on Furlanis et al. (2026), Lee et al. (2016), and Pemberton et al. (2026). De Werd et al. (2013) and Voigt-Radloff et al. (2017) support the competing early-prompt/errorless-learning approach recorded in UC04 O-16. These works cover different activities and populations. The transfer limits recorded in HFC-11 and HFC-13 still apply; the selected interaction must be evaluated with the intended users rather than presented as established practice.
Connection to Claims	Shapes F8 at UC04 steps 4-8. Supports the design rationale for UC04-F8-CL10 (independent steps), UC04-F8-CL11 (lower prompt level), and UC04-F8-CL12 (felt autonomy). UC04-F8-CL13 (distress caused by waiting) and UC04-F8-CL14 (infantilising prompts) are adverse claims that test whether this pattern actually works as intended. The pattern does not by itself establish any of these effects.

Table 2.5: Open items to check or research (IDP-04)

Item	What is needed	Why it matters

Prompt thresholds (O-29)	Proposed timing is in UC04's Escalation specification. Use the EM-05 pilot to confirm it, then configure the wait per resident with care-worker input.	Needed to verify UC04-F8-PR8 and reduce the risk tested by UC04-F8-CL13.
Pepper placement (O-30)	Test doorway audibility, tablet visibility, water clearance and obstruction with the actual robot or a full-size mock-up.	Determines whether strategy 7 and tablet demonstration are feasible; TECH-14 is the fallback.
Exact interaction content	Co-design or review the name use, wording, volume, images, acknowledgement and touch input with residents and care staff where ethically feasible.	HP-04 and RP-02 are still provisional and the interaction has not been tested with intended users.
Research and evaluation	Check how the COACH, TEBRA, Wang and Braley findings transfer to residents with dementia using Pepper and item sensors, then apply M-12 to M-16 through EM-06 and EM-07, including both parts of M-15.	Tests independent performance, assistance level, distress, autonomy, privacy and infantilisation.
Reuse beyond UC04	Compare the pattern with UC05 after its medication-specific confirmation, reassurance and safety rules are specified.	Shows which parts transfer and which require a separate IDP.

Chapter 3

Evaluation

3.1 Artifact

AT-02 Pepper tooth-brushing aid

Scope

Instantiates UC04 (§2.3) and implements its four Functions: F7 step tracking from item sensors (TECH-11), F8 graduated prompting through Pepper's speech and chest tablet (TECH-10), F9 handover to the care worker's phone (TECH-12), and F10 the three-item session record (TECH-12). The prompt wording, timing and handover rules are those of UC04's *Escalation specification*.

Deliberately out of scope: UC05 (medication, deferred by O-22), any use of Pepper's cameras (TECH-13 rejected), physical assistance by the robot (left to the care worker through F9), and any judgement of how clean the teeth are (PS-05 out of scope; M-18 is a research measure, not a System function).

Simulation notice

Paper build: nothing is simulated by software. The specification stands in for the interface. Evaluators read the prompt script and the timings, and imagine the tablet picture, Pepper's position at the doorway and the care worker's phone message.

Build A: not applicable until it is built. If Pepper cannot be placed at the doorway (O-30), the planned fallback is to run F8 through a tool-like sink device (TECH-14), which would be recorded here as a separate build.

Premises verifiable

Paper build: none. It has no timing, no sensors and no message channel, so UC04-F7-PR7 to UC04-F10-PR10 can only be checked for whether they are specified coherently, which ER-01 did along the way.

Build A: all four Premises, through EM-05 at a mock sink (ground-truth log, no human subjects). UC04-F10-PR10's check on who can read the record is only partly verifiable at a mock sink, because it depends on the care home's own access settings.

Builds

Build	What changed
Spec-1 (paper)	First evaluated build. The UC04 specification as of this cycle: Action Sequence, Functions F7 to F10, Escalation specification, Premises and Claims tables, DS-02, HP-04, HP-05 and RP-02. Walked through in ER-01.
Build A (planned)	Working system: Pepper with the F8 prompt script and tablet pictures, item sensors on the toothbrush, toothpaste and cup (TECH-11), and the phone message and session record (TECH-12). Planned to include the changes ER-01 recommends (§2.3 Open items O-35 to O-38), on top of O-34, which is already in the specification. First evaluated in the EM-05 pilot (O-29); then used by ER-02.

3.2 Evaluation Reports

ER-01 Cognitive walkthrough of UC04 with Mary and Myra

Mode

Formative

Artifact used

AT-02, build Spec-1 (paper): the UC04 specification, DS-02, HP-04, HP-05 and RP-02, as they stood when the walkthrough was done.

Evaluation Method(s) applied

Ad hoc, not one of EM-05 to EM-07. A cognitive walkthrough is an inspection method with no end users involved. It addressed Claims, since it asks whether Mary and Myra can plausibly get the intended effect from the Functions. It also touched the Premises, but only for whether they are specified coherently: a paper build cannot verify timing or accuracy.

Variant: the baseline procedure from usabilitybok.org, with its four questions asked at each step. (Q1) Will the user try to achieve the right effect? (Q2) Will the user notice

that the correct action is available? (Q3) Will the user associate the correct action with the effect they are trying to achieve? (Q4) If the correct action is performed, will the user see that progress is being made?

Ethics No ethics review was required. The walkthrough involved no participants and no personal data: the evaluator worked from the specification and the Personas only.

Introduction Addresses the Claims attached to the human-facing steps of UC04: UC04-F8-CL10, UC04-F8-CL13, UC04-F8-CL14, UC04-F9-CL15, UC04-F7-CL16, UC04-F10-CL17 and UC04-F8-CL19. CL11, CL12, CL18 and CL20 depend on outcomes (prompt levels reached, felt autonomy, time saved, plaque) that a walkthrough cannot estimate, so they are left to ER-02.

Why now: the *Escalation specification* was added to answer the midterm feedback on how and when support escalates. A walkthrough is the cheapest check of whether that ladder makes sense to the people it is meant for, before a build is made. SCE supplies the walkthrough material: the Action Sequence, the Personas, and Premises and Claims tied to steps.

Method **What was done.** One evaluator walked UC04's Action Sequence twice: as Mary (HP-04) for the resident-facing steps, and as Myra (HP-05) for the handover and record steps. Steps that only the System performs (step 1, the start of tracking, and the inside of F7) were not walked, because nobody has to act on them. For each step that was walked, the evaluator tried to tell a credible story for each of the four questions, using what HP-04 and HP-05 say the person can and cannot do. Each failure story became a finding (WF-nn), with a design suggestion where one was obvious.

Steps walked. 2-alt (not started), 2 and 3 (arriving and starting), 4 to 8 at each SA-05 step from the *Ladder per step*, 3-alt (early stop during brushing), 8-alt (distress), the ladder rule on declining, 9 (handover, from both sides), 10 (safety path, Myra's side), and 11 (session record, Myra's side).

Limits. This is one pass by one evaluator. The procedure asks for 3 to 5 independent evaluators and a joint session, so the findings below are a first list, not an aggregate (O-39). The evaluator is not a person with dementia, and simulating one is the known weak point of the method; HP-04 is itself provisional. A walkthrough also says nothing about how often a problem would occur or how severe it would be for real residents.

Finding(s) See *Walkthrough findings* below: a step-by-step table, then findings WF-01 to WF-12, grouped by the Claim they bear on.

Implications See *Implications* below. One finding, WF-07, was a clear specification defect and has been fixed in §2.3 (O-34). WF-10 is added to the EM-05 pilot (O-29). The others are recorded as §2.3 Open items O-35 to O-38 for a group decision before Build A.

Walkthrough findings

Answers are Yes, Partly or No; a No or Partly points to the finding that explains it. UC04 steps 4 to 8 are the prompting ladder, walked once at each SA-05 step; those rows name the SA-05 step as well. Step numbers in the findings below (for example "steps 8 and 9" in WF-02) are SA-05 steps.

Table 3.1: Step by step (ER-01)

UC04 step	Persona	Q1 right effect	Q2 notices action	Q3 links action and effect	Q4 sees progress	Story
2-alt Not started	Mary	Partly (WF-01)	Partly	Partly	Yes	At 10 minutes before the window closes, Pepper says "Good morning Mary, it is time for your teeth." If Mary is in the bathroom or near it, this is a credible success. If she is elsewhere in the room, she may not hear a voice from the bathroom doorway (O-30).
2 and 3 Arrives, has not started	Mary	Yes	No (WF-01)	n/a	No	Mary stands at the sink and does not pick anything up. No item event happens, so no stall rule runs: step 4's T-stall starts only after the first lift, and the initiation path only acts at 10 minutes before the window closes. She can stand there unsupported for most of the window.
4 to 8 at SA-05 step 4, pick up toothbrush	Mary	Yes	Yes	Yes	Yes	She has lifted the cup and stalls. Pepper waits 15 s, then gives L1 and L2 ("Would you like to pick up your tooth-

						credible success. The silent wait fits HP-04's need for time before being prompted.
4 to 8 at SA-05 steps 5 and 6, toothpaste	Mary	Yes	Partly (WF-04)	Yes	Partly	L2 is credible. If the blocker is opening the tube, L2 and L3 must show opening, which the merged step does not say. TECH-11 counts a lift and return as done, so she gets no sign if no paste reached the brush; the cost of that is small.
4 to 8 at SA-05 step 7, brush	Mary	Yes	Partly (WF-04)	Yes	Yes	L2 "You could brush your teeth now." is credible. At L3 the picture is on Pepper's chest tablet, at the doorway behind her, while she faces the sink and mirror. The L3 sentence repeats L2 and does not tell her to look at the screen.
3-alt Early stop	Mary	Partly	Yes	Yes	Partly (WF-10)	"Would you like to brush a little longer?" is a single offer, with no call if she declines. Credible success. Unclear case: spitting or rinsing mid-brushing produces a tap or cup event, and the ladder does not say whether that is a pause in step 7 or the start of step 8.
4 to 8 at SA-05 step 8, rinse	Mary	Partly (WF-02)	Yes	No (WF-02)	Yes	L1 at this step is still "Mary, you are brushing your teeth this morning." After she has brushed, that can be heard as an instruction to brush again. If she picks up the brush, that is not the expected tap or cup event, so the ladder carries on as if she had not responded.
4 to 8 at SA-05 step 9, put away	Mary	Yes	Yes	Yes	Partly (WF-12)	L2 is credible. If the ladder runs out here nobody is called, which suits HV-14. The specification has no closing line; DS-02 has Pepper say "All done", the only point where Mary would hear that she has finished.
Declining (ladder rule 3)	Mary	n/a	n/a	Partly (WF-03)	n/a	Every L2 is a yes/no question. Two quick "no"s, which a person may give before thinking about the question, end prompting with a routine call, reason "declined", even if she would have gone on.
8-alt Distress	Mary	Yes	No (WF-05)	Partly	Partly	To get help she must say something the System recognises as asking for help, or touch Pepper's head or hand. Nothing tells her that touching the robot does anything, and the rule that F8 must not depend on speech recognition conflicts with recognising "help", "no" or "stop".
9 Handover, resident side	Mary	n/a	n/a	n/a	No (WF-06)	Pepper says that someone is coming. Under H1 a routine call can wait 5 minutes for a re-send and 10 for a second care worker. Nothing is specified for Mary during that time, and she is not told when someone has answered.
9 Handover, care-worker side	Myra	Yes	No (WF-07)	Partly (WF-08)	No (WF-08)	Myra is dressing another resident. The message gives the activity, step and reason, with a priority, but not which resident or which room. With several residents on the round (SA-07) she cannot tell where to go. The specification also times her acknowledgement without saying how she acknowledges.
10 Safety path	Myra	Yes	No (WF-07)	Yes	Partly	An urgent message is distinct from a routine one and is re-sent every 60 s, which is credible. The missing room matters most here.
11 Session record	Myra	Yes	Yes	No (WF-09)	n/a	The record says completed, highest level "demonstration", no call. Myra wants to know what support Mary needs (HP-05 task 3), but the record does not say at which step the demonstration was needed, so she cannot target her help next time.

Table 3.2: Findings by Claim (ER-01)

ID	Finding	Suggestion	Bears on
WF-01	No stall rule covers a resident who has reached the sink but not lifted anything. The System cannot tell this apart from not having come in at all.	A presence signal at the sink that is not a camera, for example a floor or door sensor added to TECH-11, so that T-stall can start on arrival. Interacts with O-30.	UC04-F8-CL10, UC04-F8-CL13
WF-02	The L1 orienting sentence is the same at every step. At steps 8 and 9 it can be heard as an instruction to brush again, and at step 4 it describes brushing that has not started.	Make L1 step-aware while keeping it at the orienting level, for example "Mary, you have nearly finished your teeth" at steps 8 and 9.	UC04-F8-CL10, UC04-F8-CL14
WF-03	Phrasing every L2 as a yes/no offer makes "no" an easy answer, and two of them end prompting. Offers serve OBJ-14, so this is a trade-off with OBJ-10.	Count a "no" as declining only if no item event follows within the response window; keep L2 as an offer.	UC04-F8-CL10, UC04-F8-CL14
WF-04	At L3 the picture is behind the resident, and the spoken part does not point to it. The toothpaste		

	step merges opening and applying, so the picture may show the wrong part.	Add a pointer to the L3 sentence ("There is a picture on my screen"), or show the picture where the resident is looking, depending on O-30. Give steps 5 and 6 their own L3 pictures.	UC04-F8-CL10
WF-05	Ladder rules 3 and 4 need the System to recognise "no", "stop" and requests for help, which conflicts with the rule that F8 must not depend on speech recognition. The touch input in RP-02 cannot be discovered by the resident.	Recognise a small fixed set of words only (yes, no, stop, help), treat everything else as no reply, and verify the recognition rate in EM-05. Drop touch as a help input, or tell the resident about it.	UC04-F8-CL13, UC04-F8-PR8
WF-06	After "someone is coming" the resident hears nothing more, for up to 10 minutes on a routine call.	Pepper says once more that someone is coming at the re-send time, and tells the resident when the care worker has acknowledged.	UC04-F8-CL13, UC04-F9-CL15
WF-07	The handover message does not say which resident or which room.	Fixed: the room is now a fourth item of every handover message (O-34).	UC04-F9-CL15, UC04-F9-PR9
WF-08	The handover rules time an acknowledgement, but no acknowledgement action is specified, and the resident is not told when one happens.	One-tap acknowledgement on the TECH-12 message, which also triggers WF-06's "Myra is on her way".	UC04-F9-CL15, UC04-F9-CL18
WF-09	The three-item record does not say at which step the highest prompt level was needed, so it may not be enough for the care worker to adjust support. Adding the step would add detail about how a personal-care activity was performed.	Group decision between privacy (VT-08, HP-04's constraint) and usefulness at handover (HP-05).	UC04-F10-CL17
WF-10	It is not specified whether a tap or cup event during brushing ends step 7 or is a pause inside it.	Add rinsing mid-brushing as a scripted case in the EM-05 pilot and settle the rule from it.	UC04-F8-CL19, UC04-F7-PR7
WF-11	Prompts arrive just when the resident stops and name the step they are on, which shows that something follows what they do at the sink, even with Pepper's cameras covered.	Explain the item sensors when consent is sought, and in RP-02's framing. Whether residents remember this is unknown.	UC04-F7-CL16
WF-12	Unexpected, outside the Claims: DS-02 ends with "All done", which the specification does not contain. Without it, the end of the routine gives the resident no sign of having finished.	Add a closing line to the prompt levels, in an ordinary adult register and without praise.	none (no Evidence)

Implications of ER-01

- **Applied.** WF-07: the room is added to every handover message in F9, the handover rules, UC04-F9-PR9, DS-02 and TDP-02 (§2.3 O-34). The walkthrough of steps 9 and 10 from Myra's side should be repeated on the revised text.
- **For a group decision before Build A.** WF-01, WF-02, WF-04, WF-06 and WF-12 change the resident-facing prompt script (O-35). WF-03 and WF-05 change how the System hears the resident (O-36). WF-08 adds an acknowledgement action to TECH-12 (O-37). WF-09 is a trade-off between the record's privacy and its usefulness (O-38). WF-11 changes what residents are told about the item sensors at consent and in RP-02, and goes with the prompt-script changes (O-35).
- **For EM-05.** WF-05's word recognition and WF-10's rinse-or-pause rule become scripted cases in the first pilot (O-29).
- **For ER-02.** The walkthrough found ways UC04-F8-CL13's adverse effect could arise that do not depend on the wait W: standing unsupported at the sink (WF-01), missing the picture (WF-04), and waiting after a handover (WF-06). ER-02 therefore codes M-14 per step and records which ladder rule fired, so that a rise in distress can be traced to its step.
- **For the method.** Repeat the walkthrough with 3 to 5 independent evaluators, at least one with care experience if the group can find one, and compare their lists before merging them (O-39).

ER-02 Within-resident baseline comparison of the tooth-brushing aid (study design)

Mode Summative

Artifact used	AT-02, Build A (planned). The study may only start once EM-05 has verified UC04-F7-PR7 to UC04-F10-PR10 on that build. Any Premise that is only partly met has to be named in the Results of every Claim that rests on it.
Evaluation Method(s) applied	EM-06 (Within-resident baseline comparison), addressing Claims only. EM-07 (care-staff interview) runs alongside on the same ward but is a separate report, because it has different participants and a different consent basis. UC04-F10-CL17, and the care-staff halves of CL14, CL16 and CL18, belong to that report.
Ethics	**Review.** Residents with dementia are vulnerable participants who mostly cannot give informed consent themselves. On the TU Delft HREC checklist that, and the collection of health data, rule out the light procedure: the group needs a full HREC application with a Data Management Plan. The study also adds a plaque examination to residents' care and changes their routine. The group has not yet checked whether that makes it research under the Dutch WMO (Medical Research Involving Human Subjects Act). If it does, an accredited medical research ethics committee (METC) has to review it as well, and research with adults who cannot consent is then allowed only under strict conditions. Asking the HREC this is the first step (O-40). **Consent.** Written informed consent from each resident's legal representative, after an information meeting and a plain-language leaflet with pictures. On every study morning the care worker also checks the resident's own assent. Any sign of objection, in words or behaviour, means no observation that morning, and the resident can leave the study at any time without any effect on their care. Care staff give their own written consent for being observed (M-17) and interviewed (EM-07). **Protections.** No video or audio recording, in line with HV-11 and RP-02. Observers code on paper, from the doorway. Data are pseudonymised, and health data are handled as special-category personal data under the GDPR. The brushing-time log (M-18 part b) exists only for the study and is deleted after analysis. Care workers' timing data (M-17) are not shared with the care home (ST-11) for any judgement of their work. Residents, representatives and staff are told that the robot is an automated system that follows a script. The plaque check is skipped if the resident is unwilling or distressed on the day.
Introduction	Addresses the EM-06 Claims for the resident: UC04-F8-CL10, UC04-F8-CL11, UC04-F8-CL12, UC04-F8-CL13, UC04-F8-CL14 (resident interview part), UC04-F9-CL15, UC04-F7-CL16 (resident interview part), UC04-F9-CL18 (observed time part), UC04-F8-CL19 and UC04-F9-CL20. Why a formative check is not enough: ER-01 found where the design could fail, but a walkthrough cannot say how often a problem occurs or how bad it is for residents. Its evaluator also could not simulate a person with dementia reliably. Every Claim here is about an effect on residents or care workers in the real routine, so only observing that routine can support, qualify or challenge it.
Method	**Participants.** Residents of one psychogeriatric ward in a Dutch care home, recruited through the care team and their legal representatives. Inclusion: a diagnosis of dementia; currently brushing their own teeth with verbal prompting from care staff (SA-05's existing workarounds); able to stand or sit at their own sink; own teeth or dentures. Exclusion: needs full physical help to brush; hearing that a hearing aid cannot correct to understand Pepper's speech; an oral condition for which the dentist advises against a plaque check; previous strong distress around robots or new devices. Target: 8 to 10 residents, so that at least 8 complete (see the analysis plan for why 8). This is close to the six residents in COACH (Mihailidis et al., 2008). Also the care workers on that ward's morning round, typically 4 to 8. **Experimental design.** Within-subject single-case A-B-A-B design per resident, following the alternating baseline-intervention structure of COACH. A is usual care: the care worker prompts as they do now, and Pepper stands at the doorway switched to standby. B is the System: F7 to F10 are active, and the care worker continues the round and answers F9 messages. Starts are staggered by one week across residents, so that a change in all residents at the same calendar week can be told apart from the System's effect. One acclimatisation week comes first, with Pepper present and silent and nothing scored. Keeping Pepper present in A means that A and B differ only in whether the Functions run, which is what the Claims are about. It also hides the phase from the plaque assessor (see Procedure).

Each phase lasts at least five scored mornings and until the resident's M-12 baseline shows no clear trend, planned as two weeks of weekday mornings (about 10 sessions). With acclimatisation, that is about nine weeks on the ward per resident.

Variables. Independent variable: condition, nominal with two levels (A usual care, B System), manipulated by phase. Grouping variable for UC04-F8-CL13, not manipulated: ability, coded per resident and step as lower-ability where the resident's baseline M-13 at that step is 3 (demonstration) or higher. Dependent variables are the Measures in *Hypotheses* below, with their measurement levels.

Tasks. The resident's own morning tooth brushing in their own bathroom, SA-05 steps 1 to 9, at their usual time. Nothing is added to the task.

Measures. M-12, M-13, M-14, M-15, M-16 (resident interview), M-17 and M-18, as defined on the §1.2 Measures page, plus two counts from the System log in B: which ladder rule ended each episode, and the F9 messages by type. M-15 is asked after sessions in both phases, because UC04-F8-CL12 compares it with baseline. M-16's three questions are about the robot and are asked after B sessions only.

Procedure. One study morning, in both phases unless marked:

1. The care worker checks that the resident is well and assents. Otherwise there is no session that morning.
2. The care worker gives the usual reminder that it is time to brush, as at the start of DS-02.
3. A: the care worker prompts as usual. B: the System runs; the care worker goes on with the round and answers F9 messages.
4. One observer stands at the bathroom doorway next to Pepper and codes each SA-05 step live: unprompted completion (M-12), highest prompt level (M-13), and Resistiveness to Care behaviours with duration and intensity (M-14). In at least 20% of sessions a second observer codes independently. A third person time-samples the care worker's prompting episodes (M-17).
5. Within 10 minutes after the session, if the resident is willing and calm, the observer asks the three M-15 items, and in B also the three M-16 questions, taking no more than five minutes. Answers are written down, not recorded.
6. Twice a week, on fixed days, a trained assessor scores plaque (M-18 part a) within 30 minutes of brushing. The assessor is not told the phase and does not see the session; since Pepper is present in both phases, its presence does not reveal the phase.
7. Stopping rules: an H3 message, an observed M-14 behaviour at extreme intensity, or any objection ends the session and the care worker takes over. If two B sessions in a row end in distress, that resident leaves B, and the care team reviews W and the response windows, which is the fallback UC04-F8-CL13 names.

Material. AT-02 Build A: Pepper with the prompt script and tablet pictures, item sensors on the toothbrush, toothpaste and cup (TECH-11), the care worker's phone app with the handover message and session record (TECH-12). Paper coding sheets for M-12, M-13 and M-14, and an M-17 time-sampling sheet. A large-print card with the M-15 items and the M-16 questions. The MPS kit (two dental mirrors, a head torch) and its calibration manual (O-32). Leaflets and consent forms.

Result(s) Not run. The study is designed only, so there are no results and no Evidence yet.

Discussion Not run.

Conclusions Not run.

Hypotheses and analysis plan of ER-02

Each Claim becomes a yes/no question about the same residents in B compared with A. Unless the row says otherwise, the test uses one value per resident per phase type: the mean over that resident's A sessions and the mean over their B sessions. The confirmatory test is then a Wilcoxon signed-rank test across residents on those pairs, one-sided in the predicted direction, at $\alpha = .05$. Holm's correction is applied across the six confirmatory tests marked with an asterisk. Effect sizes are reported with every test. Each resident's sessions are also plotted by phase, and judged for level, trend and overlap, as single-case designs are read.

Claim	Question (yes/no)	Dependent variable and level	Analysis and decision rule

UC04-F8-CL10 *	Do residents complete more SA-05 steps unprompted in B than in A, without more distress?	M-12, count of steps 0 to 9 per session (ratio). M-14, total 0 to 156 (treated as interval).	M-12: Wilcoxon, B greater than A. M-14 must not rise: the upper end of the confidence interval for the B minus A difference stays below 5 points, a margin the group has to confirm (O-40).
UC04-F8-CL11 *	On steps where the resident stalls, is the highest prompt level lower in B than in A, with completion no worse?	M-13 per stalled step, 0 to 4 (ordinal). M-12 as above.	Per resident, the median M-13 over stalled steps; Wilcoxon, B lower than A. M-12 no worse, as in CL10.
UC04-F8-CL12 *	Do observers rate support of autonomy higher in B, while residents rate their own autonomy no lower?	M-15 observer, 7-point (ordinal). M-15 self-report, mean of three 5-point items (ordinal, treated as interval).	Observer: Wilcoxon, B greater than A. Self-report: lower end of the confidence interval above −0.5 points, a margin to confirm (O-40). If the two disagree the Claim stays unresolved, as CL12 already states.
UC04-F8-CL13	For lower-ability steps, does distress rise in B while unprompted completion does not improve?	M-14 and M-12, restricted to lower-ability steps; plus which ladder rule fired.	Too few residents are expected in this group for a test, so it is judged per resident from the plots. It holds for a resident if their M-14 on lower-ability steps is higher in both B phases than in both A phases, and their M-12 on those steps is not. The ladder log shows whether the rise follows the wait W or WF-01, WF-04 or WF-06 from ER-01.
UC04-F8-CL14 (resident part)	Do residents say the robot talks down to them?	M-16 code T (nominal counts), from B interviews.	No test. Holds if any resident's answer is coded T, reported with the quote, or if M-14 during prompting rises above baseline (from CL10's data). Read together with EM-07.
UC04-F9-CL15 *	Do fewer sessions end incomplete in B than in A?	Session outcome, complete or incomplete (nominal); per resident, the proportion incomplete (ratio).	Wilcoxon on the per-resident proportions, B lower than A.
UC04-F7-CL16 (resident part)	Do residents say they feel watched?	M-16 code W (nominal counts), from B interviews.	No test. The Claim is supported for residents if no answer is coded W; every W is reported with its quote.
UC04-F9-CL18 * (observed part)	Is the care worker's prompting time per session lower in B, with completion no worse?	M-17, seconds of prompting per session (ratio); notifications per round by type (count).	Wilcoxon, B lower than A. Completion no worse: the B minus A difference in the proportion incomplete stays below 10 percentage points, a margin to confirm (O-40). Notification counts are reported alongside, so that time saved on prompting is not just moved to messages (HFC-14).
UC04-F8-CL19 * and UC04-F9-CL20	Do residents brush longer in B, and is there less plaque in B, or more?	M-18 part b, seconds of brushing per session (ratio), and the share of sessions reaching B-target. M-18 part a, Plaque Score 1 to 4 (ordinal).	Brushing time: Wilcoxon, B greater than A. Plaque: Wilcoxon on the per-resident mean Plaque Score, two-sided, because the two Claims predict opposite directions. Lower in B with longer brushing supports CL19. Higher in B means CL20 holds. Reported across residents only, since MPS was built for groups.

Why 8 residents. With Holm's correction over six tests, the smallest p-value must fall below .05 / 6 ≈ .008. A one-sided Wilcoxon test on n residents can reach at best p = 1/2^n: .016 for 6 residents, .008 for 7 and .004 for 8. With 6 or fewer no confirmatory test could succeed, whatever the data. With 7 a test succeeds only if every resident moves in the predicted direction. With 8 it can still succeed if the resident with the smallest change moves the other way, which is why the target is 8. Even then, a test only succeeds if almost all residents move in the predicted direction. The single-case plots therefore carry as much weight as the tests, and a non-significant result is reported as "not shown", never as "no effect".

Reliability. Inter-observer agreement on the 20% double-coded sessions as each Measure's page states: Cohen's kappa for M-12, weighted kappa for M-13 and M-15 observer, ICC for M-14 and M-17 durations, with .60 as the minimum (Landis & Koch, 1977). Observers train on acclimatisation-week sessions until they reach it.

Preregistration. Hypotheses, margins, tests and stopping rules are registered (for example on OSF) before the first A phase. Anything decided after seeing data is reported as exploratory.

Table 3.3: Quality and logistics (ER-02)

Concern	How this design deals with it	What remains
Reliability	Double coding of 20% of sessions, agreement thresholds per Measure, observer training before scoring.	M-13 and the tooth-brushing version of M-15 are untested scales.
Validity	The real routine in the resident's own bathroom, with their usual care worker, so ecological validity is high. Pepper is present in both phases, so the comparison isolates the Functions.	The observer at the door is a presence the resident does not normally have, in both phases. M-18 part b measures time spent, not which teeth were cleaned.
Bias	The plaque assessor does not know the phase. Return to A after B shows whether an effect lasts beyond the first exposure. Staggered starts separate the System from calendar events. Preregistration fixes the analysis in advance.	Observers cannot be blind, because Pepper speaks only in B. Novelty is reduced by the acclimatisation week, not removed.
Scope	Inclusion criteria match HP-04.	One care home, one robot, residents who can still brush their own teeth with verbal cues. Results do not transfer to TECH-14 or to UC05.
Budget and equipment	Pepper on loan, item sensors, a phone app, coding sheets, the MPS kit.	A dental hygienist's time for calibration and twice-weekly plaque checks needs funding.
Time and location	About nine weeks per resident on one ward, after an EM-05 pilot, with staggered starts adding a few weeks.	Approval time for the HREC (and a METC, if needed) comes before all of this.

3.3 Evidence

EV-01 Walkthrough on independent step completion

Bears on UC04-F8-CL10

Source ER-01 (Cognitive walkthrough of UC04 with Mary and Myra)

Assessed against criteria Not assessed. CL10's criterion needs M-12 and M-14 from residents.

Result observed At steps 4 to 7 the wait and the L1 and L2 prompts gave credible success stories for Mary. Four places gave failure stories: a resident at the sink who has not lifted anything gets no support (WF-01); the L1 sentence can send her back to brushing at steps 8 and 9 (WF-02); two quick "no"s end prompting (WF-03); and the L3 picture is behind her (WF-04).

Verdict Qualifies

Basis for verdict Reviewer judgement from one walkthrough pass. The Claim's mechanism is credible on the steps where the resident faces the cue and has already started; the four findings name conditions under which it would not hold as specified.

Limitations One evaluator, who is not a person with dementia, on a provisional Persona. A walkthrough cannot say how often these failures would happen. The verdict applies to Spec-1; WF-01 to WF-04 are not yet fixed (O-35, O-36).

EV-02 Walkthrough on frustration from waiting

Bears on UC04-F8-CL13 (adverse)

Source ER-01 (Cognitive walkthrough of UC04 with Mary and Myra)

Assessed against criteria Not assessed. CL13's criterion needs M-14 and M-12 from lower-ability residents.

Result observed The 15 s wait itself gave a credible success story, since HP-04 needs time before being prompted. Three other waits gave failure stories, none of them caused by W: standing at the sink with no stall rule running (WF-01), waiting after "someone is coming" with no further word for up to 10 minutes (WF-06), and having no discoverable way to ask for help (WF-05).

Verdict Qualifies

Basis for verdict Reviewer judgement. The walkthrough widens the conditions under which CL13's adverse effect is plausible beyond the configured wait. If CL13 holds in ER-02, changing W

	alone may therefore not be enough, which is why ER-02 records which ladder rule fired.
Limitations	As EV-01. Frustration is exactly what an evaluator without dementia is least able to simulate.

EV-03 Walkthrough on infantilising prompts

Bears on	UC04-F8-CL14 (adverse)
Source	ER-01 (Cognitive walkthrough of UC04 with Mary and Myra)
Assessed against criteria	Not assessed. CL14's criterion needs M-16 codes from residents or staff and M-14.
Result observed	The L2 and L3 wording is in an ordinary adult register, phrased as offers, with no praise, and the closing line in DS-02 avoids praise as well. Two points could read as talking down: an L1 sentence that narrates brushing the resident is not doing (WF-02), and a help input that consists of touching the robot's head (WF-05).
Verdict	Qualifies
Basis for verdict	Reviewer judgement on the wording only. No failure story suggested the prompt script as a whole is infantilising; two specific lines could be.
Limitations	As EV-01. Whether a prompt feels condescending depends on voice, timing and the relationship, which a paper build leaves out (HFC-13).

EV-04 Walkthrough on completion after handover

Bears on	UC04-F9-CL15
Source	ER-01 (Cognitive walkthrough of UC04 with Mary and Myra)
Assessed against criteria	Not assessed. CL15's criterion needs session outcomes from residents.
Result observed	From Myra's side, the handover message did not say which resident or room it was about (WF-07), and no acknowledgement action was specified (WF-08). From Mary's side, nothing was said to her after "someone is coming" (WF-06).
Verdict	Challenges
Basis for verdict	Reviewer judgement. As specified in Spec-1, a care worker with several residents on the round could not reliably get to the right resident, so the handover could not deliver the completion effect CL15 claims. WF-07 has since been fixed (O-34); WF-06 and WF-08 are open (O-35, O-37).
Limitations	Speaks to Spec-1, not to the revised specification. The revised steps 9 and 10 need walking again.

EV-05 Walkthrough on the session record

Bears on	UC04-F10-CL17
Source	ER-01 (Cognitive walkthrough of UC04 with Mary and Myra)
Assessed against criteria	Not assessed. CL17's criterion needs care staff's judgement in EM-07.
Result observed	Myra can read the three-item record, but "highest level: demonstration" does not say at which step, so it does not tell her where Mary needs help next time (WF-09).
Verdict	Challenges
Basis for verdict	Reviewer judgement against HP-05's third task demand. The privacy half of CL17 is met by design, while the sufficiency half is in doubt.
Limitations	Care workers may already know a resident well enough that the record only needs to flag that something changed. Only EM-07 can settle this. A fix would trade against VT-08 (O-38).

EV-06 Walkthrough on the early-stop offer

Bears on	UC04-F8-CL19
Source	ER-01 (Cognitive walkthrough of UC04 with Mary and Myra)

Assessed against criteria	Not assessed. CL19's criterion needs M-18 brushing time and plaque from residents.
Result observed	A single offer to keep brushing, with no call if declined, gave a credible success story. The rule is unclear when the resident rinses or spits during brushing, which could skip the offer or make it at the wrong moment (WF-10).
Verdict	Qualifies
Basis for verdict	Reviewer judgement. The interaction is credible; the claimed effect on plaque cannot be judged by a walkthrough.
Limitations	As EV-01. Whether a resident with dementia takes up the offer, and whether longer brushing leaves less plaque in this population, are outside what a walkthrough can show (O-32).

EV-07 Walkthrough on feeling watched

Bears on	UC04-F7-CL16
Source	ER-01 (Cognitive walkthrough of UC04 with Mary and Myra)
Assessed against criteria	Not assessed. CL16's criterion needs M-16 codes from residents and staff.
Result observed	Pepper stays silent with dimmed lights while Mary manages, and its cameras are covered. But each prompt arrives just as she stops and names the step she is on, which shows her that something follows what she does at the sink (WF-11).
Verdict	Qualifies
Basis for verdict	Reviewer judgement. Item sensors avoid filming, but a well-timed prompt makes tracking evident to the person tracked. Whether that is felt as being watched or as being helped is the open question CL16 states.
Limitations	As EV-01. Explaining the sensors at consent may change this, but whether residents remember the explanation is unknown.

3.4 Verification and Validation Summary

Where each UC04 Premise and Claim stands. "Planned" means the Evaluation Method is designed but has not produced Evidence.

Premise or Claim	Planned method	Report so far	Evidence so far	Verdict so far
UC04-F7-PR7	EM-05	none (needs Build A)	none	none
UC04-F8-PR8	EM-05	none (needs Build A)	none	none
UC04-F9-PR9	EM-05	none (needs Build A)	none	none
UC04-F10-PR10	EM-05	none (needs Build A)	none	none
UC04-F8-CL10	EM-06	ER-01; ER-02 designed	EV-01	Qualifies (walkthrough only)
UC04-F8-CL11	EM-06	ER-02 designed	none	none
UC04-F8-CL12	EM-06	ER-02 designed	none	none
UC04-F8-CL13 (adverse)	EM-06	ER-01; ER-02 designed	EV-02	Qualifies (walkthrough only)
UC04-F8-CL14 (adverse)	EM-06, EM-07	ER-01; ER-02 designed	EV-03	Qualifies (walkthrough only)
UC04-F9-CL15	EM-06	ER-01; ER-02 designed	EV-04	Challenges, as specified in Spec-1; WF-07 is fixed and needs walking again, WF-06 and WF-08 are open (O-35, O-37)
UC04-F7-CL16	EM-06, EM-07	ER-01; ER-02 designed	EV-07	Qualifies (walkthrough only)

UC04-F10-CL17	EM-07	ER-01	EV-05	Challenges (walkthrough only; O-38)
UC04-F9-CL18	EM-06, EM-07	ER-02 designed	none	none
UC04-F8-CL19	EM-06	ER-01; ER-02 designed	EV-06	Qualifies (walkthrough only)
UC04-F9-CL20 (adverse)	EM-06	ER-02 designed	none	none

No Claim has yet been tested on a verified Premise, because EM-05 has not run.

ER-01 judged Claims on a paper build (Spec-1), on which no Premise can be verified. Its verdicts therefore say whether the mechanism each Claim relies on is credible as specified; they cannot say whether the System produces the effect. Two verdicts are Challenges (CL15, CL17): as specified, the handover and the session record could not deliver what those Claims state. Five are Qualifies: the mechanism is credible at some steps and fails under conditions the findings name. None is Supports, which one evaluator's walk-through could not justify. In the next iteration EM-05 runs on Build A first, so that ER-02 tests Claims on verified Premises and a Claim that fails can be told apart from a System that did not work. Interpretation across both concepts is in §3.5 of the Meaningful Connections report.

References

Braley, R., Fritz, R., Van Son, C. R., & Schmitter-Edgecombe, M. (2019). Prompting Technology and Persons With Dementia: The Significance of Context and Communication. *The Gerontologist, 59*(1), 101-111. doi:10.1093/geront/gny071.

Chalmers, J. M., King, P. L., Spencer, A. J., Wright, F. A. C., & Carter, K. D. (2005). The oral health assessment tool: validity and reliability. *Australian Dental Journal*, 50(3), 191-199. doi:10.1111/j.1834-7819.2005.tb00360.x.

Coghlan, S., Waycott, J., Lazar, A., & Barbosa Neves, B. (2021). Dignity, autonomy, and style of company: dimensions older adults consider for robot companions. *Proceedings of the ACM on Human-Computer Interaction*, 5(CSCW1). doi:10.1145/3449178.

Custers, A. F. J., Westerhof, G. J., Kuin, Y., Gerritsen, D. L., & Riksen-Walraven, J. M. (2013). Need fulfillment in the nursing home: resident and observer perspectives in relation to resident well-being. *European Journal of Ageing*, 10(3), 201-209. doi:10.1007/s10433-013-0263-y.

Cvach, M. (2012). Monitor alarm fatigue: an integrative review. *Biomedical Instrumentation & Technology*, 46(4), 268-277. doi:10.2345/0899-8205-46.4.268.

de Werd, M. M. E., Boelen, D., Olde Rikkert, M. G. M., & Kessels, R. P. C. (2013). Errorless learning of everyday tasks in people with dementia. *Clinical Interventions in Aging*, 8, 1177-1190. doi:10.2147/CIA.S46809.

Friedman, B., Kahn, P. H., Jr., & Borning, A. (2008). Value Sensitive Design and Information Systems. In K. E. Himma & H. T. Tavani (Eds.), *The Handbook of Information and Computer Ethics*. doi:10.1002/9780470281819.ch4.

Furlanis, T., Brščić, D., & Kanda, T. (2026). Robotic Bias and Its Dimensions. In *Proceedings of the 21st ACM/IEEE International Conference on Human-Robot Interaction (HRI '26)*, Edinburgh, UK, 562-571. doi:10.1145/3757279.3785613.

Galik, E., Resnick, B., Vigne, E., Holmes, S. D., & Nalls, V. (2017). Reliability and validity of the Resistiveness to Care Scale among cognitively impaired older adults. *Journal of the American Medical Directors Association*, 18(1), 59-64. doi:10.1016/j.jamda.2016.08.008.

Heerink, M., Kröse, B., Evers, V., & Wielinga, B. (2010). Assessing acceptance of assistive social agent technology by older adults: the Almere model. *International Journal of Social Robotics*, 2, 361-375. doi:10.1007/s12369-010-0068-5.

Henriksen, B. M., Ambjørnsen, E., & Axéll, T. E. (1999). Evaluation of a mucosal-plaque index (MPS) designed to assess oral care in groups of elderly. *Special Care in Dentistry*, 19(4), 154-157. doi:10.1111/j.1754-4505.1999.tb01378.x.

Jablonski, R. A., Kolanowski, A. M., Azuero, A., Winstead, V., Jones-Townsend, C., & Geisinger, M. L. (2018). Randomised clinical trial: Efficacy of strategies to provide oral hygiene activities to nursing home residents with dementia who resist mouth care. *Gerodontology*, 35(4), 365-375. doi:10.1111/ger.12357.

Kitwood, T. (1997). *Dementia Reconsidered: The Person Comes First*. Buckingham: Open University Press.

Landis, J. R. & Koch, G. G. (1977). The measurement of observer agreement for categorical data. *Biometrics*, 33(1), 159-174. doi:10.2307/2529310.

Lawton, M. P., Van Haitsma, K., & Klapper, J. (1996). Observed affect in nursing home residents with Alzheimer's disease. *Journal of Gerontology B: Psychological Sciences and Social Sciences*, 51(1), P3-P14. doi:10.1093/geronb/51b.1.p3.

Lawton, M. P., Van Haitsma, K., Perkinson, M., & Ruckdeschel, K. (1999). Observed affect and quality of life in dementia: Further affirmations and problems. *Journal of Mental Health and Aging*, 5(1), 69-81.

Lee, H. R., Tan, H., & Šabanović, S. (2016). That robot is not for me: Addressing stereotypes of aging in assistive robot design. In *25th IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN '16)*, 312-317. doi:10.1109/ROMAN.2016.7745148.

Mahoney, E. K., Hurley, A. C., Volicer, L., Bell, M., Gianotis, P., Hartshorn, M., Lane, P., Lesperance, R., MacDonald, S., Novakoff, L., Rheaume, Y., et al. (1999). Development and testing of the Resistiveness to Care Scale. *Research in Nursing & Health*, 22(1), 27-38. doi:10.1002/(SICI)1098-240X(199902)22:1<27::AID-NUR4>3.0.CO;2-T.

Mihailidis, A., Boger, J. N., Craig, T., & Hoey, J. (2008). The COACH prompting system to assist older adults with dementia through handwashing: an efficacy study. *BMC Geriatrics*, 8, 28. doi:10.1186/1471-2318-8-28.

Niemeijer, A. R., Depla, M. F. I. A., Frederiks, B. J. M., & Hertogh, C. M. P. M. (2015). The experiences of people with dementia and intellectual disabilities with surveillance technologies in residential care. *Nursing Ethics*, 22(3), 307-320.

Niemeijer, A. R., Frederiks, B. J. M., Riphagen, I. I., Legemaate, J., Eefsting, J. A., & Hertogh, C. M. P. M. (2010). Ethical and practical concerns of surveillance technologies in residential care for people with dementia or intellectual disabilities: an overview of the literature. *International Psychogeriatrics*, 22(7), 1129-1142. doi:10.1017/S1041610210000037.

Peek, S. T. M., Wouters, E. J. M., van Hoof, J., Luijkx, K. G., Boeije, H. R., & Vrijhoef, H. J. M. (2014). Factors influencing acceptance of technology for aging in place: a systematic review. *International Journal of Medical Informatics*, 83(4), 235-248.

Pemberton, L., Mehandjiev, N., & Carter, M. (2026). The Tool, the Pet, the Carer & the Warden: A Conceptual Framework of How Older Adults Perceive Socially Assistive Robots. *Information Systems Frontiers*, 28, 1039-1059. doi:10.1007/s10796-026-10699-2.

Peters, C., Hermann, T., Wachsmuth, S., & Hoey, J. (2014). Automatic Task Assistance for People with Cognitive Disabilities in Brushing Teeth: A User Study with the TEBRA System. *ACM Transactions on Accessible Computing, 5*(4), 1-34. doi:10.1145/2579700.

Pino, M., Boulay, M., Jouen, F., & Rigaud, A.-S. (2015). "Are we ready for robots that care for us?" Attitudes and opinions of older adults toward socially assistive robots. *Frontiers in Aging Neuroscience*, 7, 141. doi:10.3389/fnagi.2015.00141.

Prakash, A., & Rogers, W. A. (2015). Why some humanoid faces are perceived more positively than others: effects of human-likeness and task. *International Journal of Social Robotics*, 7(2), 309-331. doi:10.1007/s12369-014-0269-4.

Ramos, M. D., Harris, J., Manning, J., & Barrier, K. (2026). Strategies, challenges, and regulatory implications for safe medication management in Alzheimer's disease: a scoping review. *Journal of Nursing Regulation*, 17, 57-67. doi:10.1016/j.jnr.2026.01.005.

Rogers, J. C., Holm, M. B., Burgio, L. D., Granieri, E., Hsu, C., Hardin, J. M., & McDowell, B. J. (1999). Improving morning care routines of nursing home residents with dementia. *Journal of the American Geriatrics Society*, 47(9), 1049-1057.

Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist*, 55(1), 68-78. doi:10.1037/0003-066X.55.1.68.

Sendelbach, S., & Funk, M. (2013). Alarm fatigue: a patient safety concern. *AACN Advanced Critical Care*, 24(4), 378-386. doi:10.1097/NCI.0b013e3182a903f9.

Seuntjens, M. T., Thomassen, T. M. J. A., Van der Weijden, F. (G. A.), & Slot, D. E. (2025). Plaque scores after 1 or 2 minutes of toothbrushing: a systematic review and meta-analysis. *International Journal of Dental Hygiene*, 23(3), 614-624. doi:10.1111/idh.12840.

Vandemeulebroucke, T., Dierckx de Casterlé, B., & Gastmans, C. (2020). Ethics of socially assistive robots in aged-care settings: a socio-historical contextualisation. *Journal of Medical Ethics*, 46(2). doi:10.1136/medethics-2019-105615.

Voigt-Radloff, S., de Werd, M. M. E., Leonhart, R., Boelen, D. H. E., Olde Rikkert, M. G. M., Fliessbach, K., Klöppel, S., Heimbach, B., Fellgiebel, A., Dodel, R., Eschweiler, G. W., Hausner, L., Kessels, R. P. C., & Hüll, M. (2017). Structured relearning of activities of daily living in dementia: the randomized controlled REDALI-DEM trial on errorless learning. *Alzheimer's Research & Therapy*, 9, 22. doi:10.1186/s13195-017-0247-9.

Wang, R. H., Sudhama, A., Begum, M., Huq, R., & Mihailidis, A. (2017). Robots to assist daily activities: Views of older adults with Alzheimer's disease and their caregivers. *International Psychogeriatrics, 29*(1), 67-79. doi:10.1017/S1041610216001435.

Zsiga, K., Edelmayer, G., Rumeau, P., Péter, O., Tóth, A., & Fazekas, G. (2013). Home care robot for socially supporting the elderly: focus group studies in three European countries to screen user attitudes and requirements. *International Journal of Rehabilitation Research*, 36(4), 375-378. doi:10.1097/MRR.0b013e3283643d26.